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SpaceCDF — Course Workbook

Worksheets and exercises for the 40-hour Concurrent Design Facility intensive

SpaceCDF

2026

Faculty of Engineering · School of Engineering Design and Teaching Innovation (SEDTI)

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Classification: Internal

Note to the Learner

This workbook is the companion volume to the *SpaceCDF Textbook*. It is designed to be used by you, alone or in a study team, at your own pace — no facilitator or instructor is required. Each chapter contains worksheets, calculations to perform, and reflections to write. Hints direct you back to the textbook chapter that introduces the underlying theory.

When the workbook mentions the SpaceCDF tool, you should have a working installation in front of you (see the *SpaceCDF Installation Guide*). The tool runs entirely on your own computer; instructions for navigation are written so that you can perform the steps without supervision.

Spaces marked with a thin pencil rule are intended for your own notes. There are no marking rubrics in this volume.

Acknowledgement — Generative AI (AIG)

This workbook was produced with the assistance of generative AI as part of the SpaceCDF Concurrent Design Facility workflow. The worksheet content, calculation prompts, equation sheets and embedded figures are generated by the SpaceCDF toolchain (Python · matplotlib · python-docx) guided by ECSS, NASA SEH and SMAD-4 references. Editorial framing and pedagogical commentary remain owned by the SpaceCDF teaching team.

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Any course deliverable that you produce based on this workbook — or any document exported by SpaceCDF — must carry the AIG badge and a short note describing how generative AI was used. Refusing to use AI where it is permitted is fine; what is not acceptable is using it without saying so.

Worksheet 1.1: Introduction to Space Mission Design

Name:	_____
Date:	_____
Team:	_____
Session:	1.1 -- Introduction to Space Mission Design

Part A: The 17 SE Processes -- SpaceCDF Mapping

For each NASA SE process (NPR 7123.1D), identify which SpaceCDF feature implements it. Write the tab name, button, or panel. If the process requires human judgment that the tool cannot automate, mark "Manual" in the last column.

Hint: Navigate through SpaceCDF's workflow steps (Step 1 Mission Need · Step 2 Concept · Step 3 Requirements · Step 4 Design) and explore each tab. Many processes map to more than one feature.

#	SE Process	SpaceCDF Feature(s)	Manual?
1	Stakeholder Expectations Definition	_____	_____
2	Technical Requirements Definition	_____	_____
3	Logical Decomposition	_____	_____
4	Design Solution Definition	_____	_____
5	Product Implementation	_____	_____
6	Product Integration	_____	_____
7	Product Verification	_____	_____
8	Product Validation	_____	_____
9	Product Transition	_____	_____
10	Technical Planning	_____	_____
11	Requirements Management	_____	_____
12	Interface Management	_____	_____
13	Technical Risk Management	_____	_____
14	Configuration Management	_____	_____
15	Technical Data Management	_____	_____
16	Technical Assessment	_____	_____
17	Decision Analysis	_____	_____

Reflection: Which processes are well-supported by the tool? Which ones fundamentally require human judgment and cannot be fully automated? Write your answer below:

Part B: Lifecycle Phase Identification

For each activity below, identify which NASA lifecycle phase (Pre-A through F) it belongs to and which review gate marks the end of that phase:

Activity	NASA Phase	Review Gate
"We need to monitor sea ice thickness in the Canadian Arctic"	_____	_____
Writing system-level "shall" requirements with verification methods	_____	_____
Selecting a specific reaction wheel vendor after evaluating 3 options	_____	_____
Running thermal-vacuum qualification tests on the flight model	_____	_____
Operating the satellite and producing daily data products	_____	_____
Decommissioning the satellite and performing passivation	_____	_____
Preliminary orbit trade study comparing SSO, ISS-orbit, and polar	_____	_____
Detailed finite element analysis of the primary structure	_____	_____

Hint: Refer to the lifecycle phase table in Session 1.1, Section 4.3. Consider: What level of design detail does this activity require?

Part C: CDF Position Interaction Analysis

For each design conflict below, identify which engineering positions must interact to resolve it. Then briefly describe how you would approach the resolution.

Conflict 1: The payload requires 50 W but the solar array can only provide 30 W.

Positions involved: _____

Resolution approach: _____

Conflict 2: The reaction wheel vibration exceeds the payload's pointing stability requirement.

Positions involved: _____

Resolution approach: _____

Conflict 3: The selected orbit (700 km) requires a propulsive deorbit for debris compliance, but there is no mass budget remaining for a propulsion system.

Positions involved: _____

Resolution approach: _____

Conflict 4: The X-band downlink antenna pattern interferes with the payload's sensitive optical receiver.

Positions involved: _____

Resolution approach: _____

Part D: SpaceCDF Tool Exploration

Navigate through the SpaceCDF tool and answer each question:

1. How many KPI cards are shown on the Design Dashboard? List them by name:

1. In the Gate Review tab, what are the MCR exit criteria? Which are auto-evaluated?

1. What engineering positions are available when starting a collaborative session?

1. What workflow steps does SpaceCDF guide you through (from left to right)?

1. In the Positions tab, find your assigned position. What key questions does it list?

Part E: V-Model Sketch

In the space below, sketch the System-V model from memory. Label each level on both the left side (decomposition) and right side (integration). Draw the horizontal traceability arrows and label them.

Use this space for your sketch:

Notes & Reflections

Record key points below, questions you want to follow up on, and connections to your mission concept:

Worksheet 1.2: Canadian Space Ecosystem & Regulatory Environment

Name:	_____
Date:	_____
Team:	_____
Session:	1.2 -- Canadian Space Ecosystem & Regulatory Environment

Part A: Canadian Space Industry Mapping

For your team's mission concept, identify potential Canadian industry partners at each tier. If no Canadian supplier exists for a capability, note what country you would source from and why.

Tier	Capability Needed	Canadian Candidate(s)	International Alternative	Justification
Tier 1 (Prime/ Integration)	___	___	___	___
Tier 2 (Subsystem)	___	___	___	___
Tier 2 (Subsystem)	___	___	___	___
Tier 3 (Component/ Service)	___	___	___	___
Tier 4 (R&D/Academic)	___	___	___	___

Question: What CSA funding program(s) might support your mission? Why?

Part B: RSSSA Compliance Assessment

Answer the following questions for your team's mission concept:

1. Does your mission involve any remote sensing of Earth (imaging, SAR, radiometry, etc.)?

1. If yes, will the satellite be operated from Canada?

1. What RSSSA conditions might apply? (Check all that apply)

Condition	Applies?	Notes
Priority access for Government of Canada	_____	___
Shutter control provisions	_____	___
Data disposition / handling plan	_____	___
System disposal plan	_____	___
Raw data protection	_____	___
Interoperability arrangements (foreign access)	_____	___

1. If your mission does NOT involve remote sensing, what other regulatory requirements apply? (e.g., spectrum licensing, export control)

Part C: Spectrum Licensing Plan

Identify the frequency bands your mission will use and the licensing path for each:

Link	Band	Frequency (GHz)	Data Rate	Use (TT&C / Payload / ISL)	Licensing Path
Uplink	_____	_____	_____	_____	_____
Downlink (TT&C)	_____	_____	_____	_____	_____
Downlink (Payload)	_____	_____	_____	_____	_____
Other	_____	_____	_____	_____	_____

Hint: For a typical CubeSat, TT&C uses S-band (2.0--2.3 GHz) and payload data uses X-band (8.025--8.4 GHz). The licensing path for each is: ISED application -> ISED technical review -> ITU filing (API) -> ITU coordination -> ISED licence issuance.

Question: How early in your project should you begin the spectrum licensing process? Why?

Part D: Export Control Classification

List the 3 most critical components in your preliminary mission concept. For each, identify the likely export control classification:

Component	Country of Origin	Likely Classification	Licence Required?
___	_____	ECCN: _ / ***USML Cat:*** _	_____
___	_____	ECCN: _ / ***USML Cat:*** _	_____
___	_____	ECCN: _ / ***USML Cat:*** _	_____

Hint: Most commercial-off-the-shelf (COTS) space components from US manufacturers are classified under EAR (ECCN 9A515 for spacecraft parts). Military-grade or specifically designed items may fall under ITAR (USML Category XV). Canadian-origin components are subject to EIPA if they appear on the Export Control List.

Question: If your mission includes a US-origin star tracker and you plan to launch on an Indian PSLV, what export control issues arise?

Part E: Regulatory Timeline

Based on your mission concept, estimate when each regulatory activity should begin and how long it will take. Mark the activities on the timeline below:

Regulatory Activity	Start Phase	Duration	Critical Path?
RSSSA licence application	_____	_____	_____
ISED spectrum application	_____	_____	_____
ITU coordination	_____	_____	_____
Export permit(s)	_____	_____	_____
Launch provider interface agreement	_____	_____	_____

Question: Which regulatory activity is most likely to be on the critical path for your mission? Why?

Notes & Reflections

Record key points below, regulatory questions you need to research further, and implications for your mission design:

Worksheet 1.3: International Standards for Space Systems

Name:	_____
Date:	_____
Team:	_____
Session:	1.3 -- International Standards for Space Systems

Part A: Standard Applicability Matrix

For your team's mission concept, determine which ECSS and NASA standards apply. For each standard, decide whether it is fully applicable, partially applicable (with tailoring), or not applicable. Justify your decision.

Standard	Branch	Applicability	Tailoring Notes / Justification
ECSS-M-ST-10C (Project Management)	M	_____	_____
ECSS-M-ST-40C (Configuration Management)	M	_____	_____
ECSS-M-ST-80C (Risk Management)	M	_____	_____
ECSS-E-ST-10C (Systems Engineering)	E	_____	_____
ECSS-E-ST-20C (Electrical & Electronic)	E	_____	_____
ECSS-E-ST-32C (Structures)	E	_____	_____
ECSS-E-ST-40C (Software)	E	_____	_____
ECSS-E-ST-50C (Communications)	E	_____	_____
ECSS-Q-ST-10C (Product Assurance)	Q	_____	_____
ECSS-U-AS-10C (Debris Mitigation)	U	_____	_____
ISO 24113 (Debris Mitigation)	--	_____	_____
NPR 7123.1D (SE Processes)	--	_____	_____

Hint: A university CubeSat mission would typically invoke ECSS-E-ST-10C with heavy tailoring, ECSS-U-AS-10C fully, and might waive ECSS-Q-ST-10C (product assurance) due to cost constraints. A CSA-funded mission would typically require more complete ECSS compliance.

Question: What is the minimum set of standards your mission MUST comply with regardless of mission class?

Part B: CDS Rev 14 Compliance Checklist

If your mission uses a CubeSat form factor, complete this compliance checklist. If not, skip to Part C.

CDS Requirement	Your Design Value	Compliant?	Notes
Unit dimensions (100 x 100 x 113.5 mm per U)	_____	_____	_____
Mass per U ($\leq$ 2.0 kg)	_____	_____	_____
Total mass (for your form factor)	_____	_____	_____
Rail material (hard-anodised Al)	_____	_____	_____
CG within 2 cm of geometric centre	_____	_____	_____
No protrusions beyond envelope during launch	_____	_____	_____
RF silence for 30 min after deployment	_____	_____	_____
Deployment switches on all deployables	_____	_____	_____
Propulsion inhibits (3 independent, if applicable)	_____	_____	_____

Hint: Use SpaceCDF's Compliance panel to check CDS compliance. The tool will flag non-compliant parameters automatically based on your mass and dimensions entries.

Part C: Debris Mitigation Compliance

Complete the following debris mitigation assessment for your mission orbit:

1. **Planned orbit altitude:** _____ km
2. **Estimated natural orbital lifetime** (without propulsion):

Use the rule of thumb: $\tau_{\text{years}} \approx \frac{\mathrm{h} - 200}{30} \times \frac{\mathrm{m/A}}{50}$ *where* h *is altitude (km),* m *is mass (kg),* A *is cross-section area (m* ^{2} *)*

h = _ ***km,*** m = _ kg, A = _____ m ^{2}

$\mathrm{m/A}$ = _____ kg/m ^{2}

τ = _____ years (approximate)

1. **Compliance assessment:**

Requirement	Limit	Your Mission	Compliant?
IADC post-mission lifetime	$\leq$ 25 years	_____	_____
FCC post-mission lifetime (2024+)	$\leq$ 5 years	_____	_____
ISO 24113 disposal reliability	$\geq$ 90%	_____	_____

1. **If non-compliant with the 5-year FCC rule, what options exist?**

1. **Passivation plan** (list all stored energy sources and how you will deplete them at end of life):

Energy Source	Passivation Method
Battery	------
Pressure vessels (if any)	------
Reaction wheels	------
RF transmitter	------
Other: ____	------

Part D: Structural Design Margins

If your mission has preliminary structural design information, calculate the margin of safety:

Use the formula: $\text{MoS} = \frac{\sigma_{\text{allowable}}}{\text{FoS} \times \sigma_{\text{applied}}} - 1$

Worked example: For Al 7075-T6 (σ_{yield} = 503 MPa), applied stress = 200 MPa, FoS (yield) = 1.25:

$$\text{MoS} = \frac{503}{1.25 \times 200} - 1 = \frac{503}{250} - 1 = 1.012$$

This is a positive margin (compliant).

Your calculation:

Material: _ , $\sigma_{\text{allowable}}$ = _ MPa

σ_{applied} = _ ***MPa*, *FoS* =** _

MoS = ____ (show work below)

Part E: SpaceCDF Compliance Panel

Navigate to the Compliance panel in SpaceCDF and answer:

1. What is the debris mitigation compliance status for your mission?

1. What orbital altitude would be needed to comply with the FCC 5-year rule WITHOUT propulsion? (Use the tool's orbital lifetime calculator)

1. What standards has the tool automatically suggested for your mission type?

1. Are there any compliance flags (warnings or errors) on your current design?

Notes & Reflections

Record key points about standards applicability, compliance challenges, and design implications for your mission:

Worksheet 1.4: Mission Needs, Trade Studies & Concept of Operations

Name:	_____
Date:	_____
Team:	_____
Session:	1.4 -- Mission Needs, Trade Studies & ConOps

Part A: Stakeholder Analysis Matrix

Identify at least 4 stakeholders for your team's mission. For each, capture their needs, constraints, priority, and influence level.

#	Stakeholder	Role	Key Needs	Constraints Imposed	Priority (P/S/T)	Influence (H/M/L)
1	_____	_____	_____	_____	_____	_____
2	_____	_____	_____	_____	_____	_____
3	_____	_____	_____	_____	_____	_____
4	_____	_____	_____	_____	_____	_____
5	_____	_____	_____	_____	_____	_____
6	_____	_____	_____	_____	_____	_____

P = Primary (must satisfy), S = Secondary (should satisfy), T = Tertiary (nice to have) H = High influence, M = Medium, L = Low

Question: Which stakeholder has the most influence over your mission? Do any stakeholders have conflicting needs? If so, how might you resolve the conflict?

Part B: Problem Statement

Write your team's mission problem statement. It must answer all four questions: What is the problem? Who is affected? What is the impact? What constraints exist?

Problem Statement:

Peer Review Checklist (exchange with another team):

Criterion	Pass?	Feedback
Describes WHAT is needed, not HOW to achieve it?	_____	_____
Identifies the capability gap?	_____	_____
Names specific affected stakeholders?	_____	_____
Quantifies the impact of not solving it?	_____	_____
States constraints (budget, schedule, regulatory)?	_____	_____
Free of solution language? (No specific satellite type, orbit, component named)	_____	_____

Part C: Mission Objectives

Write 2--3 mission objectives. Score each against the quality checklist.

Objective 1: _____

Quality Criterion	Yes/No	Notes
States WHAT, not HOW?	_____	_____
Has measurable criterion with number + unit?	_____	_____
Achievable with current technology (TRL $\geq$ 4)?	_____	_____
Relevant to the problem statement?	_____	_____
Traceable to a specific stakeholder need?	_____	_____
Classified as primary or secondary?	_____	_____

Objective 2: _____

Quality Criterion	Yes/No	Notes
States WHAT, not HOW?	_____	_____
Has measurable criterion with number + unit?	_____	_____
Achievable with current technology (TRL $\geq$ 4)?	_____	_____
Relevant to the problem statement?	_____	_____
Traceable to a specific stakeholder need?	_____	_____
Classified as primary or secondary?	_____	_____

Objective 3 (optional): _____

Part D: MoE / MoP / TPM Trace

For your primary objective, trace the measurement hierarchy:

Objective: _____

MoE (Measure of Effectiveness -- operational effectiveness from user perspective):

MoP (Measure of Performance -- technical performance of the system):

TPM (Technical Performance Measure -- design parameter tracked during development):

Hint: If your objective is "Provide 10 m GSD multispectral imagery...", then: - MoE might be "% of target area with usable (<30% cloud) imagery within 5 days" - MoP might be "GSD $\leq$ 10 m at nadir, SNR $\geq$ 100 in all bands" - TPM might be "current estimate of aperture diameter (target: 15 cm), imager mass (target: 1.5 kg)"

Part E: Trade Study -- Space vs Non-Space

Complete a trade study evaluating whether a dedicated satellite is the right solution for your mission need.

Decision statement: _____

Alternatives:

#	Alternative	Description
1	_____	_____
2	_____	_____
3	_____	_____
4	Do nothing / status quo	_____

Criteria and Weights (must sum to 1.0):

Use pairwise comparison or direct assignment. Show your weight derivation below.

Criterion	Weight	Direction (higher/lower is better)
_____	_____	____
_____	_____	____

Criterion	Weight	Direction (higher/lower is better)
_____	_____	_____
_____	_____	_____
_____	_____	_____
Total	**1.00**	

Weight derivation (pairwise comparison table or notes):

Scoring Matrix (normalised 0--1):

Alternative	Criterion 1	Criterion 2	Criterion 3	Criterion 4	Criterion 5	Weighted Total
1: _____	_____	_____	_____	_____	_____	_____
2: _____	_____	_____	_____	_____	_____	_____
3: _____	_____	_____	_____	_____	_____	_____
4: Do nothing	_____	_____	_____	_____	_____	_____

Recall: Weighted Total = $\sum \mathrm{w_c} \times \mathrm{s(a, c)}$

Result: The highest-scoring alternative is: _____

Decision statement (1 paragraph -- include top 2 alternatives and why one was selected over the other):

Sensitivity check: If you increase the weight of your second-most-important criterion by 20% (and re-normalise), does the ranking change?

Part F: Concept of Operations Outline

Document your preliminary ConOps following the NASA SEH Appendix S structure:

1. Mission Architecture:

Space segment: _____

Ground segment: _____

User segment: _____

2. Orbit-Average Power Budget:

Calculate using: $\mathrm{P_{avg}} = \sum \mathrm{P_{mode,\,i}} \times \mathrm{DC_i}$

Mode	Power (W)	Duty Cycle (fraction of orbit)	P x DC (W)
Idle / Housekeeping	_____	_____	_____
Science / Imaging	_____	_____	_____
Downlink	_____	_____	_____
Eclipse	_____	_____	_____
		Orbit-average power:	**_____ W**

Solar array requirement: $\mathrm{P_{SA}}$ = _____ W

Show your calculation:

3. Data Budget:

Daily data generation: _____ GB/day

Downlink capacity per pass: _____ GB/pass

Passes per day: _____

Daily downlink capacity: _____ GB/day

Balance (downlink $\geq$ generation)? _____

Part G: SpaceCDF Tool Entry

In SpaceCDF, enter your mission need (Step 1) and concept (Step 2). Then answer:

1. In Step 1, did the tool auto-detect your mission type from the objective text? What type did it suggest?

1. In Step 2, run the mission trade analysis. Does the tool's recommendation match your manual trade study result from Part E?

1. In the ConOps tab, build your architecture diagram. How many systems did you define?

1. In the Trade Studies tab, run an orbit selection trade with at least 3 orbit options. What orbit did it recommend?

Notes & Reflections

Record key decisions made today, open questions, and what you want to investigate further:

Worksheet 2.1: The System-V and Requirements Engineering

Name:	_____
Date:	_____
Team / Mission:	_____
Role / Position:	_____

SpaceCDF Tab: Requirements

Key Equations Reference

Traceability Coverage:

$$\text{Coverage} = \frac{\text{N}_{\text{verified requirements}}}{\text{N}_{\text{total requirements}}} \times 100\%$$

Target: 100% -- every requirement needs a parent trace, a verification method, and a responsible subsystem.

Requirement Quality (SMART): **S**pecific -- **M**easurable -- **A**chievable -- **R**elevant -- **T**raceable

Verification Methods (ATID): **A**nalysis -- **T**est -- **I**nspection -- **D**emonstration

Part A: System-V Mapping

For your mission, identify one element at each level of the V-model:

V-Level	Left Branch (Decomposition)	Right Branch (Verification)
Stakeholder Need		
Mission Requirement		
System Requirement		
Subsystem Requirement		

Part B: Requirement Quality Assessment

Evaluate each requirement using the SMART checklist and NASA Appendix C criteria. Mark Pass (P) or Fail (F). Rewrite any failures.

#	Requirement	S	M	A	R	T	WHAT or HOW?	Rewrite (if needed)
1	"The spacecraft shall operate at 500 km altitude"							
2	"The system shall achieve GSD <= 10 m at nadir"							
3	"The EPS shall use triple-junction GaAs cells"							
4	"The system shall have a design lifetime of at least 3 years"							
5	"The structure shall be made of aluminium 7075"							

Part C: Requirement Hierarchy and Traceability

For your mission, write one requirement at each level. Draw the traceability chain.

Stakeholder Objective (traces FROM): _____

Mission Requirement (MR-___): _____

System Requirement (SR-___), derived from MR above: _____

Subsystem Requirement (SSR-___), derived from SR above: ___

Traceability diagram (draw arrows showing parent-child relationships):

Part D: Splitting Compound Requirements

Split this compound requirement into individually testable statements:

> "The system shall provide 10 m GSD multispectral imagery with 5-day revisit, 24-hour data latency, and positive power margin in all operational modes including eclipse."

#	Split Requirement (one concern each)	Type (F/P/I/C/E)	Verification (A/T/I/D)	Phase
1				
2				
3				
4				
5				

Part E: SpaceCDF Requirements Generation

1. In SpaceCDF, click **"Generate from Objectives."** How many requirements were generated? ____
2. Use the Level filter -- count at each level: - Mission: _ ***System:*** _ Subsystem: ____
3. Classify 5 requirements by type:

Req ID	Requirement Text (abbreviated)	Type (Functional / Performance / Interface / Constraint / Environmental)

1. Find one requirement that says HOW (implementation) not WHAT. Write it and your correction:

- **Original:** _____
- **Corrected:** _____

1. Accept at least 5 requirements and reject at least 1. Record your decisions:

Req ID	Action (Accept/Edit/Reject)	Rationale

1. For 5 key requirements, assign verification method and phase:

Req ID	Verification Method (A/T/I/D)	Phase (B/C/D/E)	Rationale for Method

Notes & Reflections

Worksheet 2.2: Functional Decomposition and Allocation

Name:	_____
Date:	_____
Team / Mission:	_____
Role / Position:	_____

SpaceCDF Tab: Functions

Key Equations Reference

Function Types: Observe, Communicate, Navigate, Point, Power, Protect, Store, Process, Support

Universal Functions (every spacecraft): 1. Generate electrical power 2. Maintain thermal environment 3. Survive launch environment 4. Communicate with ground (TTC) 5. Dispose at end of life

Coverage Rule: Every leaf function must trace to at least one requirement. No gaps.

Allocation Rule: Multi-allocated functions create interfaces that must be explicitly managed.

Part A: Mission Function Tree

Draw your mission's complete function tree. Use minimum 3 levels of decomposition. Include both mission-specific and universal functions.

Mission-specific functions:

```
F-001: _____
  +-- F-002: _____
      +-- F-002a: _____
      +-- F-002b: _____
  +-- F-003: _____
      +-- F-003a: _____
  +-- F-004: _____
  +-- F-005: _____
```

Universal functions:

```
F-010: Generate electrical power
  +-- F-011: _____
  +-- F-012: _____
F-020: Maintain thermal environment
  +-- F-021: _____
```

```
F-030: Survive launch environment
F-040: Communicate with ground (TTC)
F-050: Dispose at end of life
```

Part B: Function-to-Requirement Traceability

For each leaf function, write one derived requirement with a measurable threshold:

Function ID	Function Name	Allocated To (Subsystem)	Derived Requirement	Type

Part C: Multi-Allocation Analysis

Identify at least one function that is allocated to more than one subsystem:

Function: _____

Subsystem A (primary): _____

Subsystem B (contributor): _____

Interface created between A and B: _____

Who "owns" this function? _____

What interface requirements are needed? _____

Part D: Performance Criteria

For 4 key functions, define quantitative performance criteria:

Function	Performance Criterion	Value	Unit	Source

Part E: SpaceCDF Coverage Check

After entering functions in SpaceCDF:

1. How many total functions in your tree? ____
2. How many leaf functions? ____
3. How many leaf functions have NO linked requirements (amber badge)? ____
4. List the uncovered functions and derive requirements for them:

Function ID	Function Name	Derived Requirement

1. Are there any functions the tool generated that are NOT appropriate for your mission? List and explain why you removed them:

Notes & Reflections

Worksheet 2.3: Orbit Selection and Mission Architecture

Name:	_____
Date:	_____
Team / Mission:	_____
Role / Position:	_____

SpaceCDF Tabs: Mission Architecture, Orbit Trade Advisor

Key Equations Reference

Orbital period: $T = 2\pi\sqrt{a^3/\mu}$ where $a = R_E + h$, $\mu = 3.986 \times 10^{14}$ m^3/s^2, $R_E = 6371$ km

Orbital velocity: $v = \sqrt{\mu/a}$

Eclipse fraction: $f_{ecl} = \frac{1}{\pi}\arccos\left(\frac{\sqrt{a^2 - R_E^2}}{a}\right)$

Sun-synchronous inclination: $\cos(i) = -\frac{2\dot{\Omega}a^{7/2}}{3R_E^2 J_2\sqrt{\mu}}$ where $J_2 = 1.0826 \times 10^{-3}$

Hohmann ΔV: $\Delta V_1 = \sqrt{\mu/r_1}\left(\sqrt{2r_2/(r_1 + r_2)} - 1\right)$

Swath width: $W = 2h\tan(\theta)$

Part A: Orbital Mechanics Calculations

Your selected orbit: Altitude h = _ km, **Inclination** i = ***deg***, ***Type:*** _

Show all working for each calculation:

1. Semi-major axis:

$a = R_E + h = 6371 +$ _ = ***km*** = __ m

2. Orbital period:

$T = 2\pi\sqrt{a^3/\mu} = 2\pi\sqrt{____^3/\ 3.986 \times 10^{14}} =$ _ **s** = ___ min

3. Orbital velocity:

$v = \sqrt{\mu/a} = \sqrt{3.986 \times 10^{14}/\ _} =$ ___ m/s

4. Eclipse fraction (maximum):

$$f = \frac{1}{\pi}\arccos\left(\frac{\sqrt{a^2 - R_E^2}}{a}\right) = \frac{1}{\pi}\arccos(_) = ____$$

5. Sunlight and eclipse time per orbit:

$t_{sun} = T \times (1 - f) =$ _ × = __ min

$t_{ecl} = T \times f =$ _ × = __ min

6. Sun-synchronous inclination (if applicable):

$\cos(i) =$ _ $\Rightarrow i =$ ___ deg

Does this match your selected inclination? Y / N

Part B: Orbit Trade Matrix

From SpaceCDF's Orbit Trade Advisor, record the top 3 candidates:

Rank	Orbit Type	Alt (km)	Inc (deg)	GSD (m)	Revisit (d)	Natural Lifetime (yr)	FCC 5-yr?	Score
1								
2								
3								

Selected orbit: __ ***Rationale:*** _____

Part C: Debris Compliance

For your selected orbit:

Parameter	Value
Natural orbital lifetime	_____ years
FCC 5-year compliant?	Y / N
IADC 25-year compliant?	Y / N
Propulsion needed for deorbit?	Y / N
If yes, estimated deorbit ΔV	_____ m/s
Deorbit method (if needed)	_____

Show ΔV calculation if propulsion needed:

Part D: Downstream Cascade Analysis

How does your orbit selection affect each subsystem? Fill in the impact:

Subsystem	Orbit Parameter That Matters	Impact on Your Design
Payload (GSD)	Altitude = _____ km	GSD = _____ m
EPS (eclipse)	Eclipse fraction = _____	Eclipse duration = _____ min
Comms (link)	Slant range at 10 deg elev = _____ km	FSPL increase = _____ dB
Thermal	Eclipse/sunlight cycling	Hot case / Cold case concern: _____
AOCS	Orbit rate = _____ deg/min	Nadir tracking rate needed
Propulsion	Lifetime = _____ yr	Propulsion needed? Y / N
Radiation	TID estimate = _____ krad/yr	Electronics class: _____

Part E: Discussion

What would happen to your mission if you moved the orbit **100 km higher**?

What would happen if you moved it **100 km lower**?

Notes & Reflections

Worksheet 2.4: Mission Architecture -- Segments, Interfaces, and Budgets

Name:	_____
Date:	_____
Team / Mission:	_____
Role / Position:	_____

SpaceCDF Tabs: System Architecture, Interfaces, Dashboard, Budget Breakdown

Key Equations Reference

Mass margin: $\text{Margin}_{\%} = \frac{\text{M}_{\text{alloc}} - \text{M}_{\text{MEV}}}{\text{M}_{\text{alloc}}} \times 100\%$ Green: > 20% | Amber: 10--20% | Red: < 10%

MEV: $\text{M}_{\text{MEV}} = \text{M}_{\text{CBE}} + \text{equipment margin} + \text{system margin}$

Orbit-average power: $\text{P}_{\text{avg}} = \sum(\text{P}_{\text{mode}} \times \text{f}_{\text{duty}})$

SA power: $\text{P}_{\text{SA}} = \text{P}_{\text{peak}} + \frac{\text{P}_{\text{ecl}} \times \text{t}_{\text{ecl}}}{\text{t}_{\text{sun}} \times \eta_{\text{charge}}}$

ECSS Phase A margins: Equipment 20% + System 20% = ~44% compound

Part A: Mass Budget

Complete from your SpaceCDF design (or estimate if pre-run):

Subsystem	CBE Mass (kg)	% of Dry	Equip. Margin (20%)	MEV (kg)
Payload				
EPS				
AOCS				
Comms (TTC)				
OBC				
Thermal				
Structure				
Harness				
Propulsion				
Dry Total (CBE)		100%		

Subsystem	CBE Mass (kg)	% of Dry	Equip. Margin (20%)	MEV (kg)
System margin (20%)				
Dry MEV				
Propellant				
Wet Mass				
Launcher Allocation				
Mass Margin				
Mass Margin %			G / A / R	

Show margin calculation:

Part B: Power Budget by Mode

Subsystem	Safe (W)	Idle (W)	Imaging (W)	Downlink (W)	Eclipse (W)
OBC					
AOCS					
Payload					
Comms (TX)					
Thermal					
Total					
Duty cycle (%)					

Orbit-average power calculation (show working):

$$P_{avg} = _ \times + __ \times _ + \times __ + _ \times = __ W$$

SA power required:

$$P_{recharge} = \frac{P_{ecl} \times t_{ecl}}{t_{sun} \times \eta_{charge}} = \frac{\times}{\times 0.9} = _____ W$$

$$P_{SA,EOL} = P_{peak} + P_{recharge} = _ + = __ W$$

$$P_{SA,BOL} = P_{SA,EOL}/(1-0.025)^{n} = _ / = __ W$$

Part C: Interface Identification

List the 5 most critical interfaces in your design:

#	Subsystem A	Subsystem B	Interface Types (M/E/T/D/R/O)	Key Concern
1				
2				
3				
4				
5				

Write 2 formal interface requirements for your most critical pair:

Interface: __ **<->** ____

IR-001: _____

IR-002: _____

Verification method for IR-001: _ **IR-002:** ___

Part D: Conflict Resolution

From the SpaceCDF Interface Matrix, identify one conflict (red border):

Conflict description: _____

Severity: Critical / Major / Minor

Resolution options considered:

#	Option	Pros	Cons
1			
2			
3			

Selected resolution: _____

Rationale: _____

Part E: Budget Health Check

From the SpaceCDF Dashboard, record all KPIs:

Budget	Value	Margin	Status (G/A/R)
Mass	_ **kg wet vs** ___ kg alloc	_____%	

Budget	Value	Margin	Status (G/A/R)
Power (worst mode)	_ **W vs** ___ W SA	_____%	
Link	_____ dB margin	>= 3 dB?	
Cost	_ **MEUR vs** ___ ceiling	_____%	
Pointing	_ **deg vs** ___ deg req	_____%	

Which budget is tightest? _____

What single design change would improve it? ____

Notes & Reflections

Worksheet 3.1: Power System and Thermal Control Design

Name:	_____
Date:	_____
Team / Mission:	_____
Role / Position:	_____

SpaceCDF Tabs: Dashboard (Power KPI), Engineering Budgets, Timing Budget, Parametric

Quick Reference: Power and Thermal Concepts

What is a Solar Array?

A solar array converts sunlight into electrical power using photovoltaic cells. In space, the solar flux (called the solar constant) is $\mathrm{S} = 1361$ W/m^{2} at 1 AU from the Sun -- about 40% more intense than at Earth's surface because there is no atmosphere absorbing part of the spectrum.

Solar cell types and efficiency (AM0, space spectrum):

Cell Technology	Typical Efficiency	Degradation Rate	Cost Relative	Notes
Silicon (Si)	16--18%	~3.5%/yr	Low	Legacy; rarely used in new missions
Gallium Arsenide (GaAs) single-junction	22--24%	~2.5%/yr	Medium	Good radiation tolerance
Triple-junction GaAs (InGaP/GaAs/Ge)	28--30%	~2.5%/yr	High	CubeSat standard today
Multi-junction (4J, 5J)	32--35%	~2%/yr	Very high	Emerging; used on flagship missions

For CubeSat design, assume **triple-junction GaAs cells at 29.5% efficiency** ($\eta = 0.295$). Cells are mounted on panels with a **packing factor** of $\mathrm{f}_{\mathrm{pack}} = 0.80$ -- 0.85, meaning 80--85% of the panel area is actually covered by active cells (the rest is gaps, wiring, and structural margin).

Degradation: Solar cells lose efficiency over time due to radiation damage (proton and electron bombardment in the Van Allen belts). At a rate of $\delta = 2.5\%$ per year, after n years the remaining efficiency is $(1 - 0.025)^{\mathrm{n}}$. This means a 3-year mission loses about 7.3% of its beginning-of-life (BOL) power, and a 5-year mission loses about 11.9%.

Body-Mounted vs Deployable Solar Arrays

Configuration	Advantages	Disadvantages	Typical Power
Body-mounted	No moving parts, no deployment risk, no stowed volume penalty	Limited area (only the sunlit faces), power varies with attitude	1U: ~2 W, 3U: ~7 W, 6U: ~12 W
Single deployable	More area, moderate complexity	One mechanism, partially blocks FOV	1U: ~4 W, 3U: ~15 W, 6U: ~30 W
Dual deployable	Maximum area for the form factor	Two mechanisms, larger stowed volume, deployment risk	3U: ~25 W, 6U: ~48 W

Rule of thumb: If your total power demand (including battery recharge) exceeds what body-mounted panels can provide, you need deployable arrays. Compare your $P_{SA,BOL}$ to the reference power values above.

How Batteries Work in Space

Batteries store electrical energy to power the spacecraft during eclipse (when the satellite is in Earth's shadow and receives no sunlight). In LEO, eclipses occur every orbit -- typically 30--36 minutes out of a ~95-minute orbit period.

Key battery parameters:

- **Depth of Discharge (DOD):** The fraction of total battery capacity used each cycle. Lower DOD dramatically extends cycle life but requires a larger (heavier) battery.
- **Cycle life:** The number of charge/discharge cycles before the battery degrades below 80% of its original capacity.
- **Specific energy:** Energy stored per unit mass (Wh/kg). For space-grade Li-ion 18650 cells: 150--200 Wh/kg.

DOD	Typical Cycle Life (Li-ion 18650)	Suitable Mission Duration
80%	~500 cycles	Short missions (< 1 month)
50%	~2,000 cycles	Medium missions (< 6 months)
30%	~10,000 cycles	Multi-year LEO missions
20%	~30,000 cycles	Long-life LEO (> 5 years)

Design guidance: For a multi-year LEO mission (15 orbits/day), use DOD = 30%. For a 6-month technology demonstration, DOD = 50% is acceptable.

What Does the EPS Do?

The Electrical Power System (EPS) is the spacecraft's power utility. A typical CubeSat EPS board (e.g., GomSpace P31u) includes:

- **MPPT (Maximum Power Point Tracking):** A DC-DC converter that continuously adjusts the operating voltage of the solar array to extract maximum power. Efficiency: 90--95%.
- **Battery charge controller:** Manages battery charging with voltage and current limits to protect battery life.

- **Regulated voltage rails:** Provides fixed voltages (typically 3.3 V, 5 V, and unregulated battery voltage) to all subsystems.
- **Switched outputs:** Allows the OBC to turn subsystems on/off for power management and safe mode.
- **Overcurrent protection:** Protects against short circuits in any subsystem.

Thermal Control Basics

In space, there is no air for convection. Heat transfer occurs only by **radiation** and **conduction**. A spacecraft reaches thermal equilibrium when absorbed heat (solar flux, Earth albedo, Earth IR, internal electronics waste heat) equals radiated heat (to deep space at ~3 K).

ECSS thermal margins (ECSS-E-ST-31C): Operating ± 5 degC, Acceptance ± 10 degC, Qualification ± 15 degC. If the predicted temperature of any component is within 5 degC of its operating limit, you must add thermal control (heater, coating change, or radiator).

Key Equations Reference

SA EOL power: $\mathrm{P_{SA,EOL}} = \mathrm{P_{peak}} + \frac{\mathrm{P_{ecl}} \times \mathrm{t_{ecl}}}{\mathrm{t_{sun}} \times \eta_{\mathrm{charge}}}$

SA BOL power: $\mathrm{P_{SA,BOL}} = \frac{\mathrm{P_{SA,EOL}}}{(1-\delta)^{\mathrm{n}}}$ ($\delta = 0.025$/yr for GaAs)

SA area: $\mathrm{A} = \frac{\mathrm{P_{BOL}}}{\eta_{\mathrm{cell}} \times \mathrm{S} \times \cos\theta \times \mathrm{f_{pack}}}$ ($\eta = 0.295$, $\mathrm{S} = 1361$ W/m^2, $\mathrm{f_{pack}} = 0.85$)

Battery capacity: $\mathrm{C} = \frac{\mathrm{P_{ecl}} \times \mathrm{t_{ecl}}}{\mathrm{DOD} \times \eta_{\mathrm{discharge}}}$ ($\mathrm{DOD} = 0.30$, $\eta = 0.95$)

Thermal equilibrium: $\mathrm{T} = \left(\frac{\mathrm{Q_{abs}} + \mathrm{Q_{int}}}{\varepsilon \sigma \mathrm{A_{rad}}}\right)^{1/4}$ ($\sigma = 5.67 \times 10^{-8}$ W/m^2/K^4)

ECSS thermal margins: Operating +/-5 degC, Acceptance +/-10 degC, Qualification +/-15 degC

Worked Example: UniSat-1 (1U) Power and Thermal

Solar Array Sizing (Body-Mounted)

UniSat-1 uses body-mounted solar cells on 5 faces (the 6th face is the deployment interface). ISS orbit: 400 km altitude, 92.4 min period, 56 min sunlight, 36 min eclipse.

Given: $\mathrm{P_{peak,sunlight}} = 1.2$ W, $\mathrm{P_{eclipse}} = 0.5$ W (OBC only), mission lifetime = 6 months.

Step 1 -- Effective illuminated area (passive magnetic attitude, ~1.5 faces average):

$$\mathrm{A_{eff}} = 1.5 \times (0.10 \times 0.10) = 0.015 \text{ m}^2$$

Step 2 -- SA BOL power:

$$\mathrm{P_{SA,BOL}} = \eta \times \mathrm{S} \times \mathrm{A_{eff}} \times \mathrm{f_{pack}} = 0.295 \times 1361 \times 0.015 \times 0.80 = 4.82 \text{ W (illuminated peak)}$$

Step 3 -- Orbit-average power available:

$$\mathrm{P_{avg}} = \mathrm{P_{SA,BOL}} \times \frac{\mathrm{t_{sun}}}{\mathrm{T_{orbit}}} \times \eta_{\mathrm{EPS}} = 4.82 \times \frac{56}{92.4} \times 0.85 = 2.48 \text{ W}$$

Step 4 -- After 6-month degradation:

$(1-0.025)^{0.5} = 0.987$, so $P_{avg,EOL} = 2.48 \times 0.987 = 2.45$ W

Step 5 -- Power demand: $P_{avg,demand} = 0.68$ W. Margin = $2.45 - 0.68 = 1.77$ W (72%). **Comfortable.**

Battery Sizing

Given: $P_{ecl} = 0.5$ W, $t_{ecl} = 36$ min $= 0.60$ h, $DOD = 0.50$ (acceptable for 6-month mission), $\eta = 0.95$.

$$C_{bat} = \frac{0.5 \times 0.60}{0.50 \times 0.95} = \frac{0.30}{0.475} = 0.63 \text{ Wh}$$

Specify minimum **10 Wh** (standard GomSpace NanoPower P31u battery). Actual DOD per eclipse = 0.63/10 = 6.3%. Cycle count: 6 months at 15 orbits/day = 2,740 eclipses. At 6.3% actual DOD, battery life is not a concern. **Pass.**

Thermal Check (Passive Only)

UniSat-1 uses no heaters or MLI. At 400 km, strong Earth IR (~240 W/m^2) provides a warm floor. With COTS electronics rated -20 to +60 degC and a 400 km orbit, the thermal environment stays within approximately -10 to +45 degC for a 1U with aluminium/anodised surfaces. A simplified cold-case equilibrium calculation gives a very cold steady-state temperature, but the thermal mass of a 1 kg CubeSat limits actual cooling during 36-minute eclipses. **No heaters required for a 6-month mission at 400 km.**

Part A: Solar Array Sizing

Show all calculations step by step.

1. Peak sunlight power demand: $P_{peak} =$ _ **W (from mode: _____)**

2. Eclipse power demand: $P_{ecl} =$ _____ W

3. Eclipse and sunlight times (from Worksheet 2.3):

$t_{ecl} =$ _ **min** $t_{sun} =$ ___ min

4. Recharge power:

$$P_{recharge} = \frac{P_{ecl} \times t_{ecl}}{t_{sun} \times \eta_{charge}} = \frac{\quad \times \quad}{\quad \times 0.9} = \text{_____ W}$$

---

5. SA EOL required:

$$P_{SA,EOL} = P_{peak} + P_{recharge} = \text{_} + \quad = \text{__ W}$$

6. EOL degradation factor (mission lifetime = _____ years, $\delta = 0.025$/yr):

$$(1-0.025) \quad = \text{_____}$$

7. SA BOL required:

$$P_{SA,BOL} = \frac{P_{SA,EOL}}{(1-\delta)^n} = \text{__} = \text{_____ W}$$

---

8. SA area:

$$A = \frac{P_{BOL}}{0.295 \times 1361 \times \cos(\quad) \times 0.85} = \text{__} = \text{_____ m}^2$$

---

9. SA type needed (compare to CubeSat reference values):

- [ ] Body-mounted only (1U: ~2 W, 3U: ~7 W, 6U: ~12 W)
- [ ] Single deployable (1U: ~4 W, 3U: ~15 W, 6U: ~30 W)
- [ ] Dual deployable (3U: ~25 W, 6U: ~48 W)

10. SA mass:

$m_{SA} = A \times \sigma =$ _ $\times$ $=$ __ kg

(body-mounted: $\sigma = 2.5$ kg/m^2; deployable: $\sigma = 1.5$ kg/m^2)

Part B: Battery Sizing

Show all calculations:

1. Eclipse energy:

$E = P_{ecl} \times \frac{t_{ecl}}{60} =$ _ $\times$ $=$ __ Wh

2. Battery capacity required:

$C = \frac{E}{DOD \times \eta} = \frac{}{0.30 \times 0.95} = \frac{}{0.285} =$ _____ Wh

3. With 20% margin: $C_{spec} =$ _ $\times 1.2 =$ ___ Wh

---

4. Battery mass:

$m_{bat} = \frac{C_{spec}}{E_{specific}} = \frac{}{150} =$ _____ kg

5. Cycle life check:

Number of eclipses in mission = _ **orbits/day** $\times$ **365** $\times$ ***years*** $=$ __ cycles

At DOD = 30%, Li-ion provides ~10,000 cycles. Is this sufficient? Y / N

Part C: Thermal Analysis

1. Identify hot case and cold case for your orbit:

Parameter	Hot Case	Cold Case
Solar illumination		
Operational mode		
Internal dissipation		
Coating condition	BOL / EOL	BOL / EOL
Key concern		

2. From SpaceCDF Dashboard, record thermal predictions:

Component	Min Predicted (degC)	Max Predicted (degC)	Operating Min (degC)	Operating Max (degC)
Payload / sensor				
Battery				
OBC				
Transponder				

3. Margin check (need >= 5 degC):

Component	Hot margin = Max_op - Max_pred	Cold margin = Min_pred - Min_op	Pass?

4. If any margin fails, what thermal control action would you take?

Part D: SpaceCDF Verification

Compare your hand calculations to SpaceCDF values:

Parameter	Hand Calculation	SpaceCDF Value	Difference
SA power required (BOL)	_____ W	_____ W	_____ W
Battery capacity	_____ Wh	_____ Wh	_____ Wh
Orbit-average power	_____ W	_____ W	_____ W
Eclipse duration	_____ min	_____ min	_____ min

If there are significant differences, explain why:

Decision Justification

Explain WHY you chose each element of your power and thermal design. Consider the alternatives you rejected and the trade-offs involved.

Solar array configuration choice and rationale:

Battery sizing rationale (why this DOD? why this capacity margin?):

Thermal control approach (passive only? heaters? coatings?):

Notes & Reflections

Worksheet 3.2: Attitude and Orbit Control System (AOCS)

Name:	_____
Date:	_____
Team / Mission:	_____
Role / Position:	_____

SpaceCDF Tabs: Dashboard (AOCS KPI), Pointing Budget, Architecture (AOCS)

Quick Reference: AOCS Concepts

What is AOCS?

The Attitude and Orbit Control System performs two distinct jobs:

- **Attitude Determination (AD):** Figuring out which way the spacecraft is pointing, using sensors.
- **Attitude Control (AC):** Changing or holding the spacecraft's orientation, using actuators.

These are separate problems. You can know your orientation perfectly (great sensors) but still wobble if your actuators are weak. Or you can have powerful actuators but poor knowledge of where you are pointing.

Sensors: How Do We Know Where We Are Pointing?

Sensor	What It Measures	Accuracy	Mass	Power	Cost	When You Use It
Magnetometer	Direction of Earth's magnetic field	~2--5 deg	~5 g	~0.1 W	~1 kEUR	Coarse determination; always on for B-field knowledge
Coarse sun sensor	Direction to the Sun (1--2 axes)	~2--5 deg	~5 g	~0.01 W	~0.5 kEUR	Safe mode; initial acquisition
Fine sun sensor	Direction to the Sun (2 axes, high precision)	~0.1--0.5 deg	~10 g	~0.05 W	~2 kEUR	Sun-pointing missions; backup to star tracker
Star tracker	Full 3-axis attitude from star patterns	~0.001--0.01 deg (3--30 arcsec)	50--350 g	~0.5--1.5 W	~20--35 kEUR	Precision pointing for imaging, comms, science
			~20 g	~0.3 W	~3 kEUR	

Sensor	What It Measures	Accuracy	Mass	Power	Cost	When You Use It
MEMS gyroscope	Angular rate (rotation speed)	Drift: ~1--10 deg/hr				Fills gaps during star tracker blinding; slew control
Earth sensor	Nadir direction (local vertical)	~0.1--0.5 deg	~50 g	~0.5 W	~5 kEUR	Nadir-pointing missions (less common on CubeSats)

Key insight: Star trackers are by far the most accurate sensor. They work by photographing the sky, matching the observed star pattern to an onboard catalogue, and computing the spacecraft's orientation to within a few arcseconds. However, they are blinded by the Sun, Moon, or Earth limb entering their field of view, so sun sensors or gyroscopes provide backup during these periods.

Actuators: How Do We Control Our Orientation?

Actuator	How It Works	Torque	Precision	Mass	Power	When You Use It
Permanent magnet	Aligns spacecraft with Earth's B-field like a compass needle	~10^{-5} N m (restoring)	~10--15 deg	~20 g	0 W	Passive stabilisation for missions with no pointing need
Hysteresis rods	Dissipate rotational energy through magnetic hysteresis; damp tumbling	N/A (damping only)	N/A	~10 g each	0 W	Always paired with permanent magnet for passive stabilisation
Magnetorquers (MTQ)	Electromagnetic coils that interact with Earth's magnetic field to produce torque	~10^{-6} -- 10^{-4} N m	~2--5 deg (as primary); N/A (as desaturation)	~10--30 g each	0.1--0.2 W	**Primary control** for low-precision missions; **desaturation** of reaction wheels for all RW-based systems
Reaction wheels (RW)	Spinning flywheels that exchange angular momentum with the spacecraft	~10^{-4} -- 10^{-2} N m	~0.001--0.01 deg	30--120 g each	0.3--1.5 W	Precision pointing -- any mission requiring < 2 deg accuracy

Actuator	How It Works	Torque	Precision	Mass	Power	When You Use It
Thrusters	Expel mass (cold gas, ions) to produce torque via offset force	Variable	~0.1--1 deg	System-dependent	High	Large slews; orbit adjust; when MTQs are insufficient (deep space)

Magnetorquers -- two critical roles:

1. **As primary actuators** (missions needing only 2--5 deg pointing): MTQs alone provide 3-axis control, but only in LEO where Earth's magnetic field is strong. They cannot produce torque parallel to the local field vector, so control authority varies around the orbit.
2. **As desaturation actuators** (all RW-based missions): Reaction wheels absorb disturbance torques by spinning up. Over time, they accumulate angular momentum and reach saturation (maximum spin speed). MTQs "dump" this stored momentum by generating opposing torque against Earth's magnetic field. This is done approximately once per orbit. Because the available magnetic torque is small and drops to zero along the local field direction, a full dump typically takes **tens of minutes (order 20--50 min)**, not a few minutes -- budget an imaging inhibit for the whole of it.

Reaction wheels -- why 4 wheels? Three wheels (one per axis) provide minimum 3-axis control. A 4th wheel on a skew axis provides single-fault tolerance: if any one wheel fails, the remaining three still provide full 3-axis control.

AOCS Architecture Decision Table

Use this table to select your architecture based on your pointing requirement:

Pointing Requirement	Architecture	Sensors	Actuators	Mass	Power	Cost
> 5 deg (no pointing need)	Passive magnetic	Magnetometer	Permanent magnet + hysteresis rods	~50 g	0 W	~2 kEUR
2--5 deg	Magnetorquer-based	Coarse sun sensors + magnetometer	3-axis MTQ	~100 g	0.2 W	~8 kEUR
0.1--2 deg	Reaction wheel system	Fine sun sensors + magnetometer	3--4 RW + 3 MTQ (desaturation)	~500 g	2--4 W	~35 kEUR
< 0.1 deg	Fine pointing	Star tracker + fine sun sensors	4 RW + 3 MTQ	~800 g	3--5 W	~55 kEUR
< 0.01 deg	Very fine pointing	Star tracker + MEMS gyro + sun sensors	4 RW + 3 MTQ	~1200 g	4--6 W	~80 kEUR

How to use this table: Find your mission's pointing requirement (from your requirements worksheet). Read across to find the architecture, sensors, and actuators you need. If your mission is an Earth-observation

camera needing 0.05 deg pointing, you need fine pointing (star tracker + 4 RW + 3 MTQ). If your mission is an IoT relay needing only 5 deg pointing, magnetorquers alone suffice.

Key Equations Reference

Gravity gradient torque: $T_{gg} = \frac{3\mu}{2a^3}|I_z - I_x|\sin(2\theta)$

Aerodynamic torque: $T_{aero} = \frac{1}{2}\rho v^2 C_D A \cdot d_{cp-cm}$

SRP torque: $T_{SRP} = \frac{S}{c} A_s (1+q) \cdot d_{sp-cm}$

Magnetic torque: $T_{mag} = M \times B$

Momentum storage: $H = T_{dist} \times t_{desat}/2$

Pointing error (RSS): $\theta_{total} = \sqrt{\sum \theta_i^2}$

MTQ torque: $T_{MTQ} = m_{dipole} \times B \times \sin(\alpha)$

Worked Example: UniSat-1 (1U) AOCS

Architecture Selection

UniSat-1 carries a MEMS magnetometer payload that does not require accurate pointing. In fact, a slowly rotating/tumbling state provides magnetic field measurements across multiple directions, improving data quality.

Architecture: Passive magnetic stabilisation (permanent magnet + hysteresis rods). No pointing error budget is needed because there is no pointing requirement.

Parameter	Value
Pointing accuracy	~10--15 deg (to local magnetic field)
Mass	~30--50 g
Power	0 W
Cost	~1--2 kEUR

Disturbance Torques at 400 km

Even though UniSat-1 has no active AOCS, understanding the disturbance environment confirms that the passive magnet provides sufficient restoring torque.

Gravity gradient (1U is nearly cubic, so $I_z \approx I_x$, making this very small):

$$T_{gg} \approx \frac{3 \times 3.986 \times 10^{14}}{2 \times (6.771 \times 10^6)^3} \times |I_z - I_x| \approx 1 \times 10^{-8} \text{ N m}$$

Aerodynamic (ρ at 400 km $\approx 4 \times 10^{-12}$ kg/m^3, $v \approx 7670$ m/s):

$$T_{aero} = 0.5 \times 4 \times 10^{-12} \times 7670^2 \times 2.2 \times 0.01 \times 0.01 \approx 3 \times 10^{-8} \text{ N m}$$

Permanent magnet restoring torque: $\sim 1 \times 10^{-5}$ N m -- three orders of magnitude larger than all disturbances. **The satellite stays field-aligned.**

Part A: AOCS Architecture Selection

Pointing requirement from your mission: _____ deg (3-sigma)

AOCS architecture selected (tick one):

- [ ] Passive magnetic (> 5 deg)
- [ ] Magnetorquers only (2--5 deg)
- [ ] Reaction wheels + MTQ (0.1--2 deg)
- [ ] Fine pointing: RW + star tracker + MTQ (< 0.1 deg)
- [ ] Very fine: RW + ST + gyro + MTQ (< 0.01 deg)

Justification for selection:

Estimated AOCS mass: _ kg **Estimated AOCS power: ___ W**

Part B: Disturbance Torque Calculations

Calculate at least 2 disturbance torques for your spacecraft and orbit. Show all working.

Spacecraft properties:

Parameter	Value	Unit
Mass		kg
Form factor		U
I_z (largest MOI)		kg m^2
I_x (smallest MOI)		kg m^2
Cross-section area (largest face)		m^2
d_{cp-cm} (CP-CM offset estimate)		m
Orbit altitude		km

1. Gravity gradient torque:

$$T_{gg} = \frac{3 \times 3.986 \times 10^{14}}{2 \times (\quad)^3} \times |I_z - I_x| =$$

$T_{gg} =$ _____ N m

2. Aerodynamic torque (ρ at **_ km** $\approx$ ___ kg/m^3):

$T_{aero} = \frac{1}{2} \times _ \times _^2 \times 2.2 \times __ \times _____ =$

$T_{aero} =$ _____ N m

3. (Optional) Solar radiation pressure torque:

$T_{SRP} =$ _____ N m

4. (Optional) Residual magnetic dipole torque ($M \approx$ _____ A m^2, $B \approx 3 \times 10^{-5}$ T):

$T_{mag} = _ \times _ =$ __ N m

Summary:

Source	Torque (N m)
Gravity gradient	
Aerodynamic	
SRP	
Magnetic dipole	
Total (RSS or sum)	

Which disturbance dominates? _____

Part C: Reaction Wheel Sizing

Skip this section if your mission uses passive magnetic or MTQ-only architecture.

1. Torque requirement ($\geq 2 \times$ total disturbance):

$T_{RW,req} = 2 \times _ =$ ***N m*** = __ mN m

2. Momentum storage (desaturation interval = 1 orbit = _____ s):

$H = T_{dist} \times \frac{t_{desat}}{2} = _ \times \frac{}{2} =$ ___ mN m s

3. Selected reaction wheel:

Parameter	Required	Selected Product
Torque	_____ mN m	_____ mN m
Momentum	_____ mN m s	_____ mN m s
Mass	--	_____ g
Power	--	_____ W
Quantity	4 (3+1 redundancy)	

Part D: Pointing Error Budget

Skip this section if your mission uses passive magnetic stabilisation (no pointing requirement).

Complete the RSS pointing error budget:

Error Source	Value (deg)	Value ^{2}
Sensor accuracy		
Actuator resolution		
Alignment knowledge		
Thermal distortion		
Jitter (RW)		
Orbit knowledge		
Timing error		
Sum of squares		
RSS Total = $\sqrt{\text{sum}}$	**= _____ deg**	

Requirement: _____ deg

Margin: _ deg (___ %)

Does the pointing budget close? Y / N

Which error source dominates? _____

What action would most improve the budget? _____

Part E: SpaceCDF Verification

Compare your hand calculations to SpaceCDF's Pointing Budget display:

Parameter	Hand Calculation	SpaceCDF	Match?
RSS pointing error	_____ deg	_____ deg	
Dominant contributor	_____	_____	
Margin to requirement	_____ %	_____ %	

If there are differences, explain:

Decision Justification

Explain WHY you chose your AOCS architecture. Consider the alternatives and what drove your decision.

Why this architecture and not a simpler one?

Why this architecture and not a more capable one?

What is the single biggest risk in your AOCS design?

Notes & Reflections

Worksheet 3.3: Communications and Link Budget Design

Name:	_____
Date:	_____
Team / Mission:	_____
Role / Position:	_____

SpaceCDF Tabs: Link Budget, Spectrum Selector, Equipment Browser (Comms)

Quick Reference: Communications Concepts

What is a Link Budget?

A link budget is an accounting statement for your communication link. It adds up every gain (transmitter power, antenna gain) and subtracts every loss (distance, atmosphere, pointing error) to determine whether the signal arriving at the receiver is strong enough to decode. Everything is computed in **decibels (dB)** so that multiplication and division become addition and subtraction.

Decibel refresher:

Conversion	Formula	Common Values
Watts to dBW	$\mathrm{P_{dBW}} = 10\log_{10}(\mathrm{P_W})$	2 W = +3.0 dBW; 1 W = 0 dBW; 0.5 W = -3.0 dBW
Kelvin to dBK	$\mathrm{T_{dBK}} = 10\log_{10}(\mathrm{T_K})$	150 K = 21.8 dBK; 600 K = 27.8 dBK
Ratio to dB	$\mathrm{G_{dB}} = 10\log_{10}(\mathrm{G})$	x2 = +3 dB; x10 = +10 dB; x100 = +20 dB

What Each Term in the Link Budget Means

Term	What It Is	Analogy
TX Power ($\mathrm{P_{TX}}$)	Electrical power fed to the transmitter amplifier	How loud you shout
TX Antenna Gain ($\mathrm{G_{TX}}$)	How well the antenna focuses the signal into a beam (vs radiating equally in all directions)	Using a megaphone vs shouting in all directions
TX Line Losses ($\mathrm{L_{TX}}$)	Power lost in cables and filters between amplifier and antenna	Sound absorbed by the megaphone walls
EIRP	Effective Isotropic Radiated Power = $\mathrm{P_{TX}} + \mathrm{G_{TX}} - \mathrm{L_{TX}}$. The total "signal strength leaving the spacecraft"	Total loudness heard from a distance
FSPL		Voice getting quieter the farther away you walk

Term	What It Is	Analogy
	Free Space Path Loss -- signal weakens as it spreads over a larger sphere. Depends on distance and frequency	
Atmospheric Loss	Signal absorbed/scattered by Earth's atmosphere (rain, oxygen, water vapour)	Fog muffling sound
Pointing Loss	Signal loss because the antenna beam is not perfectly aimed	Megaphone pointed slightly off-target
Polarisation Loss	Loss when TX and RX antenna polarisations do not match	Holding the megaphone sideways when the listener expects vertical
RX Antenna Gain ($\mathrm{G_{RX}}$)	Ground station antenna ability to collect signal. Larger dish = more gain	Listener using a large ear trumpet
System Noise Temp ($\mathrm{T_{sys}}$)	Combined noise from antenna, sky, and receiver electronics. Lower = better	Background noise in the room
G/T	Receiver figure of merit = $\mathrm{G_{RX}} - \mathrm{T_{sys}}$ (in dB). Captures how "good" the ground station is	Ear trumpet size minus room noise
$\mathrm{C/N_0}$	Carrier-to-noise-density ratio -- the received signal strength relative to noise	Signal-to-noise ratio before decoding
$\mathrm{E_b/N_0}$	Energy per bit to noise density -- the fundamental measure of link quality. Divides $\mathrm{C/N_0}$ by data rate	How much energy is available per bit of information
Link Margin	How much $\mathrm{E_b/N_0}$ you have above the minimum needed. Must be >= 3 dB	Safety factor for your communication

Frequency Band Comparison

Parameter	UHF (Amateur)	S-band	X-band	Ka-band
Frequency	435--438 MHz	2200--2290 MHz	8025--8400 MHz	25.5--27.0 GHz
Bandwidth available	20 kHz	5 MHz	375 MHz	1.5 GHz
Maximum data rate	< 20 kbps	0.1--10 Mbps	10--400 Mbps	100--2000 Mbps
FSPL at 500 km (nadir)	139 dB	153 dB	165 dB	175 dB
FSPL at 1300 km (10 deg elev)	148 dB	162 dB	173 dB	183 dB
Atmospheric loss	~0.1 dB	~0.5 dB	~1.0 dB	~3--10 dB (rain!)
Typical spacecraft antenna	Monopole (0 dBi)	Patch (6 dBi)	Horn/patch array (10--15 dBi)	Small dish/array (20+ dBi)
Ground antenna	Yagi (10--13 dBi)	3 m dish (35 dBi)	3 m dish (45 dBi)	1 m dish (45 dBi)

Parameter	UHF (Amateur)	S-band	X-band	Ka-band
License type	IARU coordination (free)	ISED/FCC ($30--45K)	ISED/FCC + ITU (higher)	Complex ITU filing
License timeline	~6 months	6--12 months	12+ months	12--18 months
Equipment availability	Many COTS	Many COTS	Growing COTS	Emerging
Best for	TT&C, telemetry, simple missions	Medium data rate, standard EO	High data rate, EO/SAR	Very high data rate

How to choose: Start with your data rate requirement. If you need < 20 kbps and have non-commercial data, UHF amateur is free and simple. If you need up to 10 Mbps, S-band is the CubeSat workhorse. For high-resolution imaging producing GBs per day, you need X-band or Ka-band.

Antenna Types

Antenna Type	Gain	Beamwidth	Pointing Need	Mass	Typical Band	Use Case
Monopole/ dipole	~0 dBi	Omnidirectional	None	5--20 g	UHF	TT&C, simple telemetry
Patch (single)	~5--8 dBi	~70--90 deg	Low (~10 deg)	10--50 g	S-band	Standard CubeSat downlink
Patch array (2x2)	~12 dBi	~30--40 deg	Medium (~5 deg)	30--100 g	S/X-band	Higher-rate downlink
Horn	~10--15 dBi	~20--30 deg	Medium (~5 deg)	50--200 g	X-band	High-rate downlink
Parabolic reflector	~20--35 dBi	~2--10 deg	Tight (< 1 deg)	200--500 g	X/Ka-band	Very high-rate, DTE
Phased array	~15--25 dBi	Electronically steered	Electronic (fast)	100--300 g	S/X/Ka-band	Agile beam, multiple targets

Trade-off: Higher gain antennas provide stronger signals (enabling higher data rates or lower TX power), but they have narrower beams, which means the spacecraft must point more accurately. An omnidirectional antenna works regardless of spacecraft orientation but provides very low gain.

Modulation and Coding

The modulation scheme and forward error correction (FEC) code determine the minimum $\mathrm{E_b/N_0}$ required for a target bit error rate (BER):

Modulation + Coding	$\mathrm{E_b/N_0}$ Required (BER 10^{-6})	Spectral Efficiency	Best For
BPSK uncoded	10.5 dB	1.0 bps/Hz	Legacy telecommand
QPSK uncoded	10.5 dB	2.0 bps/Hz	Simple telemetry

Modulation + Coding	E_b/N_0 Required (BER 10^{-6})	Spectral Efficiency	Best For
QPSK + convolutional (r=1/2)	5.0 dB	1.0 bps/Hz	Standard CCSDS TM
QPSK + LDPC (r=1/2)	2.0 dB	1.0 bps/Hz	High-efficiency downlink
QPSK + LDPC (r=3/4)	4.0 dB	1.5 bps/Hz	Balanced (CubeSat standard)
8PSK + LDPC (r=3/4)	6.5 dB	2.25 bps/Hz	High-rate downlink

Design guidance: For most CubeSat missions, **QPSK + LDPC (r=3/4)** at $E_b/N_0 = 4.0$ dB is the standard choice, offering a good balance of coding gain and implementation simplicity.

Key Equations Reference

EIRP: $EIRP = P_{TX} + G_{TX} - L_{TX}$ (dBW)

FSPL: $FSPL = 20\log_{10}\left(\frac{4\pi df}{c}\right)$ (dB)

G/T: $G/T = G_{RX} - 10\log_{10}(T_{sys})$ (dB/K)

C/N$_0$: $C/N_0 = EIRP - FSPL - L_{losses} + G/T + 228.6$ (dBHz)

E_b/N_0: $E_b/N_0 = C/N_0 - 10\log_{10}(R_b)$ (dB)

Link margin: $Margin = E_b/N_{0,\,avail} - E_b/N_{0,\,req} - L_{impl} \geq 3$ dB

Antenna gain: $G = \eta_a(\pi D/\lambda)^2$

dB conversion: $P_{dBW} = 10\log_{10}(P_W)$; 2 W = +3.0 dBW; 1 W = 0 dBW

Worked Example: UniSat-1 (1U) UHF Link Budget

UniSat-1 uses UHF amateur band at 437 MHz with 9600 bps downlink to an amateur ground station.

Step-by-Step Link Budget

Scenario: 400 km orbit, 437 MHz, 9600 bps, 10 deg minimum elevation (slant range ~1150 km), ground station with 13 dBi cross-Yagi (RHCP).

Line	Parameter	Calculation	Value	Unit
1	TX Power	0.5 W = $10\log_{10}(0.5)$	-3.0	dBW
2	TX Antenna Gain	Monopole	0.0	dBi
3	TX Line Losses	Short cable run	-0.5	dB
4	**EIRP**	$-3.0 + 0.0 - 0.5$	**-3.5**	dBW
5	FSPL	$21.98 + 20\log_{10}(1.15 \times 10^6) + 20\log_{10}(437 \times 10^6) - 169.54$	-146.5	dB

Line	Parameter	Calculation	Value	Unit
6	Atmospheric Loss	Minimal at UHF	-0.3	dB
7	Pointing Loss	Omni antenna, minimal	-0.5	dB
8	Polarisation Loss	RHCP-to-RHCP (cross-Yagi)	-0.5	dB
9	RX Antenna Gain	13 dBi cross-Yagi	+13.0	dBi
10	System Noise Temp	600 K (amateur + LNA) = $10\log_{10}(600)$	27.8	dBK
11	**G/T**	$13.0 - 27.8$	**-14.8**	dB/K
12	Boltzmann Constant		+228.6	dBW/K/Hz
13	**C/N** $_0$	$-3.5 - 146.5 - 0.3 - 0.5 - 0.5 + (-14.8) + 228.6$	**+62.5**	dBHz
14	Data Rate	9600 bps = $10\log_{10}(9600)$	39.8	dBbps
15	E_b/N_0 **available**	$62.5 - 39.8$	**+22.7**	dB
16	E_b/N_0 required	GMSK + conv. code r=1/2 (K=7)	5.0	dB
17	Implementation Loss		2.0	dB
18	**LINK MARGIN**	$22.7 - 5.0 - 2.0$	**+15.7**	dB

Result: Link closes with 15.7 dB margin (> 3 dB). **Pass.**

FSPL Calculation Detail

$$\mathrm{FSPL} = 21.98 + 20\log_{10}(1.15 \times 10^6) + 20\log_{10}(437 \times 10^6) - 169.54$$

$$20\log_{10}(1.15 \times 10^6) = 20 \times 6.0607 = 121.2 \text{ dB}$$

$$20\log_{10}(437 \times 10^6) = 20 \times 8.640 = 172.8 \text{ dB}$$

$$\mathrm{FSPL} = 21.98 + 121.2 + 172.8 - 169.54 = 146.5 \text{ dB}$$

Check with the single-formula form, slant range in metres (1.15×10^6 m) and frequency in Hz (437×10^6 Hz):

$$20\log_{10}(4\pi \times 1.15 \times 10^6 \times 437 \times 10^6 / 3 \times 10^8) = 146.5 \text{ dB}$$

And with the engineering form (range in km, frequency in MHz):

$$\mathrm{FSPL} = 32.44 + 20\log_{10}(1150) + 20\log_{10}(437) = 32.44 + 61.2 + 52.8 = 146.5 \text{ dB}$$

All three forms agree. **Common error:** using 41.44 instead of 32.44 in the km/MHz form adds a spurious 9 dB to every link budget. If your FSPL looks about 9 dB too high, check that constant first.

Data Throughput

At 4800 bps useful throughput (9600 bps channel rate with r=1/2 FEC), a 7-minute pass delivers:

$$V_{\text{pass}} = 4800 \times 420 \times 0.85 = 1.71 \text{ Mbit} = 214 \text{ kB per pass}$$

With 4 passes/day: ~0.86 MB/day. The magnetometer generates ~1.13 MB/day -- marginal. May need a second ground station or data prioritisation.

Part A: Frequency Band Selection

Mission data rate requirement: _____ Mbps

License type selected: Amateur / Experimental / Commercial

Band selected: ___ **Centre frequency:** ___ MHz

Rationale for band selection:

Licensing/filing required: _____

Estimated licensing cost: _ **Timeline:** ___ months

Part B: Complete Link Budget

Fill in ALL rows. Show computation for EIRP, FSPL, G/T, C/N_0, and margin.

Line	Parameter	Formula / Source	Value	Unit
1	TX Power	_____ W =		dBW
2	TX Antenna Gain			dBi
3	TX Line Losses			dB
4	**EIRP** = Line 1 + 2 + 3		_____	dBW
5	Free Space Path Loss	$20\log_{10}(4\pi \times __ \times __ / c)$		dB
6	Atmospheric Loss			dB
7	Pointing Loss			dB
8	Polarisation Loss			dB
9	RX Antenna Gain			dBi
10	System Noise Temp	_____ K =		dBK
11	**G/T** = Line 9 - 10		_____	dB/K
12	Boltzmann Constant		+228.6	dBW/K/Hz
13	**C/N_0** = 4 + 5 + 6 + 7 + 8 + 11 + 12		_____	dBHz
14	Data Rate	_ **bps =** $10\log_{10}(_)$		dBbps
15	E_b/N_0 **available** = 13 - 14		_____	dB
16	E_b/N_0 required (mod+code: __)			dB

Line	Parameter	Formula / Source	Value	Unit
17	Implementation Loss			dB
18	**LINK MARGIN** = 15 - 16 - 17		_____	dB

Show FSPL calculation:

$$\mathrm{FSPL} = 20\log_{10}(4\pi) + 20\log_{10}(\mathrm{d}) + 20\log_{10}(\mathrm{f}) - 20\log_{10}(\mathrm{c})$$

$$= 21.98 + 20\log_{10}(\ _\ ) + 20\log_{10}(\ \ ) - 169.54 = \ __\ \text{dB}$$

Does the link close ($\geq$ 3 dB)? Y / N

If margin is excessive, maximum achievable data rate at 3 dB margin:

$$\mathrm{R_{b,\,max}} = 10^{(\mathrm{E_b/N_{0,\,avail}} - \mathrm{E_b/N_{0,\,req}} - \mathrm{L_{impl}} - 3)/10} \times \mathrm{R_b} = \ _____\ \text{Mbps}$$

Part C: Data Budget

Daily data generation:

$$\mathrm{V_{gen}} = \mathrm{R_{payload}} \times \mathrm{t_{imaging}} \times \mathrm{N_{orbits}} \times \mathrm{f_{compression}}$$

$$= \ _\ \textbf{Mbps} \times \textbf{\textit{s}} \times __ \times _ \ = \ \textbf{\textit{MB/day}} \ = \ __\ \text{GB/day}$$

Daily downlink capacity:

$$\mathrm{V_{DL}} = \mathrm{R_{DL}} \times \mathrm{t_{contact}} \times \mathrm{N_{passes}} \times \eta_{\mathrm{protocol}}$$

$$= \ _\ \textbf{Mbps} \times \textbf{\textit{s}} \times __ \times _ \ = \ \textbf{\textit{MB/day}} \ = \ __\ \text{GB/day}$$

Data budget closes? ($\mathrm{V_{DL}} \geq \mathrm{V_{gen}}$): Y / N

If no, what is your mitigation plan?

Part D: SpaceCDF Comparison

Open the Link Budget tab, enter your parameters, and compare:

Parameter	Hand Calculation	SpaceCDF	Difference
EIRP (dBW)			
FSPL (dB)			
C/N $_0$ (dBHz)			
Link Margin (dB)			

If there are differences, explain:

Part E: Modulation and Coding Selection

Selected modulation: _____

Selected FEC code: _____

Required $\mathrm{E_b/N_0}$ at BER 10^{-6}: _____ dB

Spectral efficiency: _____ bps/Hz

Rationale:

Decision Justification

Explain WHY you chose your communications architecture. Address each decision explicitly.

Why this frequency band and not another?

Why this antenna type? What pointing does it require from AOCS?

Why this modulation/coding scheme? What trade-off did you make between data rate and link robustness?

Notes & Reflections

Worksheet 3.4: Structure, Propulsion, and Equipment Selection

Name:	_____
Date:	_____
Team / Mission:	_____
Role / Position:	_____

SpaceCDF Tabs: Equipment Browser, Dashboard, Trade Studies, Budget Breakdown

Quick Reference: Structure and Propulsion Concepts

CubeSat Design Specification (CDS Rev 14) Key Dimensions

The CDS defines standard CubeSat form factors. All CubeSats must comply with these dimensions to fit inside standard deployers (e.g., ISIPOD, NRCSD, Exolaunch). The unit "U" represents a 10 x 10 x 11.35 cm cube.

Form Factor	Dimensions (mm)	Max Mass (kg)	Internal Volume (cm^3)	Rails	Typical Use
1U	100 x 100 x 113.5	2.0	~1,000	4	Technology demo, simple sensors
1.5U	100 x 100 x 170.2	3.0	~1,500	4	Extended 1U with more volume
2U	100 x 100 x 227.0	4.0	~2,000	4	IoT, AIS, simple payloads
3U	100 x 100 x 340.5	6.0	~3,000	4	Standard EO, comms, science
6U	100 x 226.3 x 340.5	12.0	~6,000	4	High-performance EO, SAR, comms
12U	226.3 x 226.3 x 340.5	24.0	~12,000	8	Advanced payloads, propulsion
16U	226.3 x 226.3 x 454.0	32.0	~16,000	8	Microsatellite-class missions

Key CDS requirements (all form factors):

Requirement	Specification	Why It Matters
Rail material	Hard anodised Al 7075-T6 or 6061-T6	Prevents cold welding in vacuum; provides structural load path

Requirement	Specification	Why It Matters
Rail cross-section	8.5 x 8.5 mm minimum contact	Transmits launch loads through deployer rails
Deployment switches	Minimum 1 (on a rail standoff)	Ensures spacecraft is powered off until deployed
RBF (Remove Before Flight) pin	Required	Physically disconnects batteries during ground handling
No protrusions beyond rail envelope (stowed)	Required	Prevents jamming inside the deployer
CG offset from geometric centre	<= 2 cm	Prevents uncontrolled tumbling at deployment; deployer balance
Fundamental frequency	> 40 Hz (first mode)	Avoids resonance with launch vehicle modes

Structural Loads and Margins

During launch, the spacecraft experiences:

- **Quasi-static acceleration:** 6--9 g axial (along rocket axis), 2--4 g lateral (sideways). Think of your 5 kg CubeSat feeling like it weighs 45 kg.
- **Random vibration:** Broadband shaking at 20--2000 Hz. Critical for PCB solder joints and connectors.
- **Shock:** Brief, high-g pulses (500--2000 g) at separation events. Critical for brittle components.

Margin of Safety (MoS): The ratio of what the structure can withstand to what it must withstand, minus 1. Must be >= 0. If MoS is negative, the structure fails under the design load.

Material properties (Al 7075-T6): Yield strength $\sigma_{\mathrm{y}} = 503$ MPa, Ultimate strength $\sigma_{\mathrm{u}} = 572$ MPa. Factor of Safety: 1.25 (yield), 1.5 (ultimate).

Propulsion Decision Tree

Not every CubeSat needs propulsion. Use this decision tree:

```
START: Does my mission need propulsion?
  |
  +-- Orbit altitude < 500 km?
  |     YES --> Natural atmospheric drag deorbits within 5 years (FCC compliant)
  |             Do I need orbit maintenance to extend lifetime?
  |               NO  --> NO PROPULSION NEEDED
  |               YES --> Need drag compensation (5-15 m/s per year)
  |
  +-- Orbit altitude 500-700 km?
  |     Natural lifetime is 5-25+ years
  |     FCC 5-year rule: likely NON-COMPLIANT without active deorbit
  |     --> PROPULSION LIKELY REQUIRED for deorbit (50-150 m/s)
  |
  +-- Orbit altitude > 700 km?
  |     Natural lifetime is decades to centuries
  |     IADC 25-year guideline: NON-COMPLIANT
  |     --> PROPULSION REQUIRED for deorbit (high Delta-V)
```

```
|
+-- Constellation phasing needed?
|     YES --> Add 10-50 m/s for orbit plane/phase adjustment
|
+-- Collision avoidance needed?
      YES --> Add 1-5 m/s reserve per manoeuvre
```

Key rule: Below ~500 km, you almost certainly do not need propulsion. Above ~600 km, you almost certainly do. The 500--600 km range depends on your specific ballistic coefficient and mission lifetime.

Propulsion Technologies Comparison

Type	How It Works	I_{sp} (s)	Thrust	Propellant Mass (100 m/s, 5 kg S/C)	System Dry Mass	Total System Mass	Burn Time	Cost
Cold gas	Pressurised gas expelled through nozzle	40--80	10--100 mN	0.87 kg	0.3 kg	**1.17 kg**	Minutes	~15 kEUR
Resistojet	Gas heated before expulsion	80--150	10--50 mN	0.34 kg	0.3 kg	**0.64 kg**	Minutes	~25 kEUR
Electrospray (FEEP)	Liquid metal ionised and accelerated by electric field	500--1500	0.01--1 mN	0.04 kg	0.9 kg	**0.94 kg**	Months	~50 kEUR
Hall effect	Xenon ionised and accelerated by magnetic/ electric fields	800--1500	1--10 mN	0.05 kg	1.5 kg	**1.55 kg**	Weeks	~80 kEUR
Green monoprop	Catalytic decomposition of non-toxic propellant	200--250	0.1--1 N	0.21 kg	1.0 kg	**1.21 kg**	Seconds	~60 kEUR

The fundamental trade-off:

- **High I_{sp} (electric propulsion):** Uses very little propellant but has very low thrust, so manoeuvres take weeks or months. The thruster hardware is also heavier and more expensive.
- **Low I_{sp} (chemical/cold gas):** Uses much more propellant but provides high thrust for quick manoeuvres. Hardware is simpler, lighter, and cheaper.

Choose based on: How much ΔV do you need? How fast must the manoeuvre happen? How much mass/ volume/cost can you afford?

Key Equations Reference

Structural Margin of Safety: $\text{MoS} = \frac{\sigma_{\text{allow}}}{\sigma_{\text{design}} \times \text{FoS}} - 1 \geq 0$ FoS: Yield 1.25, Ultimate 1.5 (metallic)

Tsiolkovsky rocket equation: $\Delta V = I_{sp} \cdot g_0 \cdot \ln(m_0/m_f)$

Propellant mass: $m_{prop} = m_{dry} \times (e^{\Delta V/(I_{sp} \cdot g_0)} - 1)$ $g_0 = 9.80665$ m/s^2

Data storage: $S = V_{daily} \times N_{days} \times f_{safety}$

Fundamental frequency: $f_1 > 40$ Hz (deployer requirement)

Worked Example: UniSat-1 (1U) Structure and Propulsion

CDS Compliance

Parameter	1U Specification	UniSat-1 Design	Compliant?
Dimensions	100 x 100 x 113.5 mm	100 x 100 x 113.5 mm (ISIS 1U frame)	Y
Maximum mass	2.0 kg (CDS) / 1.33 kg (NanoRacks)	1.0 kg target	Y
Rail material	Hard anodised Al 7075-T6	ISIS standard frame	Y
Deployment switches	Min 1	2 (ISIS standard)	Y
RBF pin	Required	Included	Y
CG offset	<= 2 cm	< 1 cm (symmetric layout)	Y
Protrusions (stowed)	None beyond rail envelope	UHF antenna stowed along rail	Y

Propulsion Decision

Orbit altitude: 400 km. **Natural lifetime:** ~8--14 months (ballistic coefficient $BC = m/(C_D A) = 1.0/(2.2 \times 0.01) = 45.5$ kg/m^2).

FCC 5-year rule satisfied without propulsion. No orbit maintenance needed (6-month mission). No constellation phasing. **Decision: No propulsion.**

Structural Margin (Axial Load)

Given: Mass = 1.0 kg, axial load = 9 g, 4 rails.

$$F_{axial} = \frac{1.0 \times 9 \times 9.81}{4} = \frac{88.3}{4} = 22.1 \text{ N per rail}$$

$$\sigma = \frac{22.1}{8.5 \times 10^{-3} \times 8.5 \times 10^{-3}} = \frac{22.1}{7.225 \times 10^{-5}} = 0.31 \text{ MPa}$$

$$\text{MoS}_y = \frac{503}{0.31 \times 1.25} - 1 = \frac{503}{0.3875} - 1 = 1297 \gg 0$$ **Pass** (by a very large margin)

Key insight: Quasi-static axial stress is never the critical case for CubeSats. The real structural risks are: (a) PCB solder joint fatigue from random vibration, (b) deployment mechanism reliability, (c) stiffness (fundamental frequency > 40 Hz).

Equipment List

#	Category	Component	Mass (g)	Power (W)	Cost (kEUR)
1	Structure	ISIS 1U frame	200	--	4.0
2	EPS + Battery	GomSpace P31us	200	0.3	12.0
3	Solar cells	Body-mounted GaAs (5 faces)	50	--	8.0
4	OBC	Custom MSP430/Cortex-M	30	0.3	3.0
5	Comms	NanoCom AX100 (UHF)	60	0.5	8.0
6	Antenna	UHF monopole (deployable)	20	--	2.0
7	Payload	MEMS magnetometer PCB	50	0.2	5.0
8	AOCS	Permanent magnet + hysteresis rods	30	0	1.0
9	Harness	Internal cables, connectors	50	--	1.0
	TOTAL		**690**	**~1.3 (peak)**	**~44**

Mass margin: 1330 - 690 = 640 g (48%). With 20% equipment margin + 20% system margin (MEV): 1330 - 994 = 336 g (25%). **Green.**

Part A: CDS Compliance Check

CDS Requirement	Your Design Value	Specification	Compliant?
Form factor: _____ U	Dimensions: _____ mm	_____ mm	Y / N
Maximum mass	_____ kg (wet)	_____ kg	Y / N
Rail material		Hard anodised Al	Y / N
Deployment switches	_____ (quantity)	Min 1	Y / N
RBF pin provided	Y / N	Required	Y / N
No protrusions beyond rails (stowed)	Y / N	Required	Y / N
CG within 2 cm of geometric centre	Estimated: _____ cm offset	<= 2 cm	Y / N
Fundamental frequency	Estimated: _____ Hz	> 40 Hz	Y / N

Non-compliance items and corrective actions:

Part B: Structural Margin Calculation

Axial launch load: _ g Lateral launch load: ___ g

Load per rail (4 rails, equal sharing):

$F_{axial} = \frac{m \times a_{axial} \times g_0}{4} = \frac{\quad \times \quad \times 9.81}{4} =$ _____ N

Stress in rail (cross-section 8.5 x 8.5 mm = 7.225×10^{-5} m^2):

$\sigma = \frac{F}{A} = \frac{\quad}{7.225 \times 10^{-5}} =$ _____ MPa

Margin of safety (yield, Al 7075-T6, $\sigma_y = 503$ MPa):

$MoS = \frac{503}{\quad \times 1.25} - 1 =$ _____

Pass? Y / N

Part C: Propulsion Decision

Does your mission need propulsion?

Question	Answer
Orbit altitude	_____ km
Natural orbital lifetime	_____ years
FCC 5-year compliant without propulsion?	Y / N
Orbit maintenance required?	Y / N
Constellation phasing required?	Y / N
Active deorbit required?	Y / N

Decision: Propulsion / No propulsion

If propulsion is selected, complete this section. Show all working.

Total ΔV required:

Manoeuvre	ΔV (m/s)
Orbit maintenance (_____ yr)	
Deorbit	
Collision avoidance margin	
Total	

Propulsion type selected: ___ $I_{sp} =$ _____ s

Propellant mass (Tsiolkovsky):

$m_{prop} = m_{dry} \times (e^{\Delta V/(I_{sp} \times g_0)} - 1)$

$= _ \times (e^{\ /(\ \times 9.81)} - 1) = \ \times (e^{\ } - 1) = __ \times _ = ___$ kg

---

System dry mass (thruster + tank + feed): _____ kg

Total propulsion system mass: _____ kg (propellant + dry)

Fraction of total spacecraft mass: _____ %

Part D: Equipment Selection Log

Using SpaceCDF's Equipment Browser, select components for each category:

Category	Component Selected	Manufacturer	Mass (kg)	Power (W)	Cost (kEUR)	Qty	Total Mass
EPS Board						1	
Battery						1	
Solar Panels							
OBC						1	
Reaction Wheel						x	
Magnetorquer						x	
Star Tracker							
Sun Sensor						x	
Transponder						1	
Antenna							
Payload						1	
Structure						1	
Harness						1	
Propulsion							
TOTAL			_____		_____		

Parametric estimate (from Session 2.4): _____ kg

Equipment-based total: _____ kg

Difference: _ **kg (** ___ %)

Any incompatibilities detected? (RF band mismatch, voltage mismatch, interface conflict)

Part E: Component Trade Study

Subsystem traded: _____ **Options compared:** 3

Criterion	Weight	Option A: ___	Score	Option B: ___	Score	Option C: ___	Score
Mass	0.20						
Power	0.15						
Cost	0.20						
TRL	0.15						
Heritage	0.15						
Performance	0.15						
Weighted	**1.00**		_____		_____		_____

Winner: ___ Score: ___

Rationale (beyond the numbers):

Part F: Final Budget Health Check

After equipment selection, record final budget status:

Budget	Value	Margin	Status
Mass (wet vs allocation)	_ / ___ kg	_____ %	G / A / R
Power (worst mode vs SA)	_ / ___ W	_____ %	G / A / R
Link margin	_____ dB	>= 3 dB?	G / A / R
Cost vs ceiling	_ / ___ MEUR	_____ %	G / A / R
Pointing (RSS vs req)	_ / ___ deg	_____ %	G / A / R
Data (DL vs gen)	_ / ___ GB/day	_____ %	G / A / R

Any budget that does not close? _____

Proposed fix: _____

Decision Justification

Explain WHY you made each key structural and propulsion decision.

Form factor selection rationale (why this size and not larger/smaller?):

Propulsion decision rationale (if no propulsion: why not? if propulsion: why this type?):

Equipment selection rationale (which component trade-offs were hardest? what compromises did you make?):

Notes & Reflections

Worksheet 4.1: Equipment Selection & Bill of Materials

Name: _____ Date: ___ Team: _____

Mission Name: _____ Form Factor: ____U

Part A: Make/Buy/Reuse Decisions

For each subsystem, document the procurement decision and rationale:

Subsystem	Decision (Buy/Reuse/Make)	Rationale	Lead Time
Structure			
EPS (solar array)			
EPS (battery)			
EPS (distribution)			
OBC			
AOCS (sensors)			
AOCS (actuators)			
TTC (transceiver)			
TTC (antenna)			
Thermal			
Payload			
Propulsion (if applicable)			

Part B: Bill of Materials (BOM)

Complete the BOM from the SpaceCDF Equipment Browser selections:

Item ID	Component	Manufacturer	Part Number	Qty	Mass (g)	Power (W)	Cost (kEUR)	TRL	ECCN
STR-001									
EPS-001									
EPS-002									
EPS-003									
OBC-001									
AOCS-001									

Item ID	Component	Manufacturer	Part Number	Qty	Mass (g)	Power (W)	Cost (kEUR)	TRL	ECCN
AOCS-002									
AOCS-003									
TTC-001									
TTC-002									
THM-001									
PLD-001									
HRN-001									

Part C: Budget Tracking

After all selections, record the live budget values from the SpaceCDF Dashboard:

Budget	BOM Total	Parametric Estimate	Allocation	Margin	Margin %
Dry mass	_____ g	_____ g	_____ g	_____ g	_____%
Peak power	_____ W	_____ W	_____ W	_____ W	_____%
Total cost	_____ kEUR	_____ kEUR	_____ kEUR	_____ kEUR	_____%
Volume	_% of _U		100%	_____%	

Does everything fit within allocation? Y / N

If not, what would you change? _____

Part D: Export Control Assessment

List any components that may require export licences:

Component	ECCN / USML Category	Controlled?	Licence Required?	Action Needed
		Y / N	Y / N	
		Y / N	Y / N	
		Y / N	Y / N	
		Y / N	Y / N	

Are there any ITAR-controlled items? Y / N

If yes, what TAA (Technical Assistance Agreement) or DSP-5 licence is needed?

Part E: Interface Compatibility Check

For each selected component, verify interface compatibility:

Component	Bus Voltage OK?	Data Protocol OK?	RF Band Match?	Connector OK?	Notes
	Y / N	Y / N	Y / N / NA	Y / N	
	Y / N	Y / N	Y / N / NA	Y / N	
	Y / N	Y / N	Y / N / NA	Y / N	
	Y / N	Y / N	Y / N / NA	Y / N	
	Y / N	Y / N	Y / N / NA	Y / N	
	Y / N	Y / N	Y / N / NA	Y / N	

Part F: Interface Incompatibility Resolution

Describe one interface incompatibility found during selection and how you resolved it:

Issue: _____

Components involved: _____

Root cause: _____

Resolution chosen: _____

Impact on budget: _____

Notes & Reflections

What was the most difficult equipment selection decision? Why?

Worksheet 4.2: Verification & Validation Planning

Name: _____ Date: ___ Team: _____

Mission Name: _____

Part A: IADT Method Assignment

For 10 key requirements from your SpaceCDF V&V Matrix, assign the verification method and justify:

#	Req ID	Requirement Text (abbreviated)	Method (I/A/D/T)	Phase (B/C/D)	Level (Unit/Sub/Sys)	Justification
1						
2						
3						
4						
5						
6						
7						
8						
9						
10						

Summary count: Analysis: ___ / Test: ___ / Demonstration: ___ / Inspection: ___

Part B: Dual-Method Requirements

Identify 3 requirements that need BOTH analysis (early confidence) AND test (final proof):

Requirement ID	Requirement Text	Analysis (Phase B)	Test (Phase C/D)	Why Both Needed?

Part C: Environmental Test Sequence

For your selected launch vehicle, specify the complete proto-flight test sequence:

Launch vehicle: ___ PUG reference: ___

Step	Test	Specification	Duration	Pass Criteria
1	Initial functional test	Full performance baseline	_____ hr	All parameters nominal

Step	Test	Specification	Duration	Pass Criteria
2	Sine vibration	_ g, - __ Hz	_____ min/axis x 3	No resonance shift > 5%
3	Random vibration	_____ gRMS overall	_____ min/axis x 3	No structural damage
4	Post-vibe functional test	Same as step 1	_____ hr	Compare to baseline
5	Thermal vacuum -- hot	+ _____ C (predicted max + 10 C)	_____ cycles	Functional at hot extreme
6	Thermal vacuum -- cold	- _____ C (predicted min - 10 C)	_____ cycles	Functional at cold extreme
7	Post-TVAC functional test	Same as step 1	_____ hr	Compare to baseline
8	EMC test	Required? Y / N	_____ day	No interference
9	Mass measurement	Calibrated scale (+/- 0.01 kg)	30 min	M <= _____ kg
10	Deployer fit check	_____ deployer	2 hr	Fits; slides freely

Total test campaign duration estimate: _____ weeks

Part D: Vibration Test Profile

Record the random vibration PSD profile from your launch vehicle PUG:

Frequency (Hz)	PSD Level (g^2/Hz)	Slope
20		Start
20 - _____		+ _____ dB/oct
_ - ___		Flat
_____ - 2000		- _____ dB/oct
2000		End

Overall gRMS (calculated): _____ gRMS

Show your calculation:

Part E: TVAC Cycle Profile

Sketch or describe one TVAC cycle:

Hot extreme: + _ **C Cold extreme: -** ___ C

Ramp rate: _ C/min Dwell time at extreme: ___ hr

Functional test at each extreme? Y / N **Duration of functional test:** _____ hr

Calculate time for one cycle:

t_cycle = _____

= _____ hours

Total TVAC campaign: _ cycles x *hr* + __ hr setup = _ **hr =** ___ days

Part F: Waiver Identification

Are there any requirements that cannot be verified by the standard method? Document potential waivers:

Req ID	Standard Method	Proposed Alternative	Justification	Risk

Notes & Reflections

Which environmental test do you consider most critical for your mission? Why?

Worksheet 4.3: Risk, Interfaces & FMECA

Name: _____ Date: ___ Team: _____

Mission Name: _____

Part A: Risk Register

Identify at least 5 technical risks for your mission design. Score each on the 5x5 matrix.

Scoring reminder: L x C = Score. Low (1-4), Medium (5-9), High (10-15), Critical (16-25).

#	Risk Description	L (1-5)	C (1-5)	Score	Category (L/M/H/C)	Mitigation Strategy	Owner (Position)	Post-Mitigation Score
1								
2								
3								
4								
5								
6								
7								

How many risks are High or Critical (before mitigation)? _____

How many remain High or Critical after mitigation? _____

Part B: 5x5 Risk Matrix Plot

Plot your risks on the matrix below. Write the risk number in the appropriate cell.

```
              Consequence ->
              1       2       3       4       5
  L   5 |     |       |       |       |       |
  i   4 |     |       |       |       |       |
  k   3 |     |       |       |       |       |
  e   2 |     |       |       |       |       |
  l   1 |     |       |       |       |       |
```

Part C: N-Squared (N^2) Interface Matrix

Fill in the interface matrix for your mission's subsystems. Use abbreviations: - **P** = Power (electrical) - **D** = Data (I2C, SPI, UART, CAN, RS-422) - **R** = RF (coaxial) - **M** = Mechanical (mounting, alignment) - **T** = Thermal (conductive/radiative)

	EPS	OBC	AOCS	TTC	Payload	Structure
EPS	---					
OBC		---				
AOCS			---			
TTC				---		
Payload					---	
Structure						---

Total number of interfaces identified: _____

Interface conflicts detected: (describe at least 2)

1. Conflict: _____

Resolution: _____

1. Conflict: _____

Resolution: _____

Part D: FMECA Table

Complete for 4 critical components:

Component	Function	Failure Mode	Cause	Local Effect	System Effect	Severity (1-5)	Detection Method	Compensating Provision

---

Part E: Single-Point Failure Analysis

List all single-point failures in your design:

#	Component	Failure Mode	Effect on Mission	SPF?	Acceptable?	Mitigation (if not acceptable)
1				Y/N	Y/N	
2				Y/N	Y/N	
3				Y/N	Y/N	

#	Component	Failure Mode	Effect on Mission	SPF?	Acceptable?	Mitigation (if not acceptable)
4				Y/N	Y/N	
5				Y/N	Y/N	

Total SPFs identified: _ Accepted: *Mitigated:* __

Part F: Reliability Calculation

Calculate the series reliability for your mission's critical chain (components that must all work for mission success):

Component	Individual Reliability (R_i)	Mission Duration Used

R_series = R_1 x R_2 x ... x R_n = _____

R_series = _____

If you added redundancy to one component (e.g., dual deployment mechanism), recalculate:

R_redundant = 1 - (1 - R)^2 = 1 - (1 - _)^2 = ___

New R_series (with redundancy) = _____

Improvement: from _ to (+ __ percentage points)

Part G: Risk Discussion

What is the highest-risk item in your design? What would it take (cost, mass, schedule) to bring the risk to an acceptable level?

Notes & Reflections

Which single-point failure concerns you the most? Would you accept it, or is mitigation essential?

Worksheet 4.4: Cost Estimation & Design Review

Name: _____ Date: ___ Team: _____

Mission Name: _____

Part A: Mission Cost Estimate (WBS)

Complete using both parametric estimates (from SpaceCDF Cost tab) and bottom-up estimates (from BOM vendor quotes):

WBS #	Element	Parametric (kEUR)	Bottom-Up (kEUR)	Difference	Notes
1.0	Programme Management				
2.0	Systems Engineering				
3.0	Mission Assurance				
4.0	Payload				
5.0	Spacecraft Bus Hardware				
	5.1 Structure & mechanisms				
	5.2 EPS				
	5.3 AOCS				
	5.4 TTC				
	5.5 OBC & data handling				
	5.6 Thermal				
	5.7 Propulsion				
	5.8 Harness				
6.0	Integration & Test				
7.0	Software (FSW + GSW)				
8.0	Launch Services				
9.0	Ground Segment				
10.0	Operations (_____ years)				
	TOTAL	_____	_____		

Part B: Confidence Levels

P50 estimate (your best estimate): _____ kEUR

P70 estimate (P50 x 1.2): _____ kEUR

P80 estimate (P50 x 1.3): _____ kEUR

Show your reasoning for the multiplier (is 1.3 appropriate, or should it be higher for your mission?):

Part C: Constellation Cost (if applicable)

Number of satellites: _ Learning rate: ___%

b = ln(learning_rate) / ln(2) = ln(_) / ln(2) = ___

Unit #	Unit Cost (kEUR) = C_1 x N^b	Cumulative (kEUR)
1		
2		
4		
8		
_____ (total)		

Show your calculation for the total constellation cost:

Hardware: _ **units x *kEUR avg* =** __ kEUR

Spares (_***%*):** __ units x _ **kEUR =** ___ kEUR

Launch: _ **sats x *kEUR* =** __ kEUR

Ground segment: _____ kEUR

Operations (_ **years):** ___ kEUR

PM/SE/MA: _____ kEUR

Total constellation cost: _ kEUR = ___ MEUR

P80 constellation cost (x 1.3): _____ MEUR

Part D: Cost Drivers

Top 3 cost drivers in your mission (from SpaceCDF Cost Breakdown):

Rank	Cost Element	Cost (kEUR)	% of Total	Why is it Expensive?
1				
2				
3				

What would you change to reduce total cost by 20%?

Part E: Design Review Readiness

From the SpaceCDF Gate Review tab, assess your readiness for SRR/PDR:

#	Gate Criterion	Status (Pass/Fail/ Manual)	Evidence Location	If Fail: Action Needed
1	Mission need justified			
2	Requirements complete & traceable			
3	All budgets close with margin			
4	Equipment selected (BOM complete)			
5	Interfaces defined, conflicts resolved			
6	V&V approach defined			
7	Risks identified with mitigations			
8	Cost estimate with WBS			
9	ConOps defined			
10	Sustainability compliance			

Must-pass criteria met: _ / ___

Overall readiness: READY / NOT READY

Action items to achieve readiness:

1.
2.
3.

Part F: Presentation Preparation

Outline your 12-minute design review presentation:

Section	Duration	Key Message	Evidence/Data to Show
1. Mission Need	2 min		
2. Why Space?	2 min		
3. System Design	3 min		
4. Equipment & V&V	2 min		

Section	Duration	Key Message	Evidence/Data to Show
5. Risk & Cost	2 min		
6. Operations & Lessons	1 min		

Notes & Reflections

What is the most uncertain element in your cost estimate? How would you reduce the uncertainty?

Worksheet 5.1: Ground Segment & Operations Architecture

Name: _____ Date: ___ Team: _____

Mission Name: _____ Orbit: *km*, __ deg inclination

Part A: Ground Station Contact Analysis

Calculate contact time and data volume per pass for your mission.

Input Parameters

Parameter	Value	Source
Orbital altitude (h)	_____ km	SpaceCDF orbit selection
Minimum elevation angle (epsilon)	_____ deg	Ground station spec (typically 5-10)
Downlink data rate (R)	_____ Mbps	SpaceCDF link budget
Protocol efficiency (eta)	_____ (typically 0.85-0.95)	CCSDS overhead estimate
Ground station latitude	_____ deg	Selected station
Earth radius (R_E)	6371 km	Constant

Calculations

Maximum slant range:

rho = R_E x (sqrt((h/R_E + 1)^2 - cos^2(epsilon)) - sin(epsilon))

rho = 6371 x (sqrt((_**/6371 + 1)^2 - cos^2(*)) - *sin(***__))

rho = _____ km

Average pass duration (approximate):

T_avg ~ _____ minutes (from STK/SpaceCDF or estimate: ~6 min for 500 km SSO)

Data volume per pass:

V_data = R x T x eta = **_ Mbps x *s* x** __ = **_ Mbit =** ___ MB

Part B: Data Budget Closure

Parameter	Value	Unit
Daily data generation (payload)		MB/day
Daily data generation (housekeeping)		MB/day
Total daily generation		**MB/day**

Parameter	Value	Unit
Passes per day (GS 1: _____)		passes
Data per pass (GS 1)		MB
Subtotal downlink (GS 1)		MB/day
Passes per day (GS 2: _____)		passes
Data per pass (GS 2)		MB
Subtotal downlink (GS 2)		MB/day
Total daily downlink capacity		**MB/day**
Data budget margin		**MB/day**
Margin percentage		**%**

Does the data budget close? Y / N

If not, what changes would close it? (check all that apply)

- [ ] Add another ground station at: ___
- [ ] Increase data rate to: _ **Mbps (requires:** _____)
- [ ] Reduce payload data generation to: _ **MB/day (impact:** _____)
- [ ] Add onboard compression (ratio: _____:1)
- [ ] Use SatNOGS network for additional passes
- [ ] Other: _____

Part C: Ground Station Network Design

Station #	Location	Latitude	Antenna	Band	Data Rate	Passes/Day	Role
1							Primary TTC
2							Payload DL
3							Backup/SatNOGS

Total ground segment cost estimate: _____ kEUR

Cost Element	Annual Cost (kEUR)	Notes
Antenna rental / ownership		
MCS software licence		
Network connectivity		
Operations staff (_____ FTE)		
Total annual		

Cost Element	Annual Cost (kEUR)	Notes
Mission lifetime (_____ yr)		

Part D: LEOP Timeline

Construct the LEOP timeline for the first 72 hours after separation:

Time (UTC)	Sim Time	Activity	Success Criterion	Status
T+0	Separation	Deployment switches release		[ ]
T+30 min	Timer expires	Antenna deployment commanded		[ ]
T+_____		Beacon acquisition	Carrier lock	[ ]
T+_____		First HK telemetry	Data decoded	[ ]
T+_____		First uplink command	ACK received	[ ]
T+_____		SA deployment (if separate)	Power generation confirmed	[ ]
T+_____		ADCS initialisation	Attitude determination active	[ ]
T+_____		ADCS calibration	Pointing < _____ deg	[ ]
T+_____		Full duplex validation	Uplink + downlink at ops rate	[ ]
T+_____		Health assessment complete	All subsystems nominal	[ ]

Part E: Mission Operations Timeline (Gantt Chart)

Sketch the operations timeline from launch to end of life:

```
Phase:       LEOP  Commission  Early Ops    Nominal Operations          EOL
Week:     1  2  3  4  5  6  7  8  9  ... 26 ... 52 ... 104 ... 156
          |  |  |  |  |  |  |  |  |       |      |       |       |
Staffing: ____________________________________________________
          24/7     16/7        8/5         8/5 or automated    8/5

Key milestones:
  L+___: First light
  L+___: Commissioning complete
  L+___: Orbit maintenance manoeuvre (if applicable)
  L+___: End of nominal mission
  L+___: Extended mission (if applicable)
  L+___: Passivation and deorbit
```

Notes & Reflections

What is the most challenging aspect of your ground segment design? What would you do differently with more budget?

Worksheet 5.2: Mission Operations Concepts

Name: _____ Date: ___ Team: _____

Mission Name: _____

Part A: Operational Modes

Define all operational modes for your spacecraft. Use the SpaceCDF ConOps Editor to enter these.

Mode	Description	Power (W)	Data Rate	Entry Condition	Exit Condition
Safe					
Detumble					
Standby					
Science					
Downlink					
Eclipse					
Manoeuvre					

Verify: Does the power budget close in every mode?

Mode	Power Consumption (W)	Power Available (W)	Margin (W)	Closes?
Safe				Y / N
Detumble				Y / N
Standby				Y / N
Science				Y / N
Downlink				Y / N
Eclipse (battery only)				Y / N

Part B: FDIR Rules

Define at least 5 FDIR rules for your spacecraft:

#	Fault Description	Detection Method	Threshold	FDIR Level (0-4)	Autonomous Response	Recovery Procedure
1						
2						

#	Fault Description	Detection Method	Threshold	FDIR Level (0-4)	Autonomous Response	Recovery Procedure
3						
4						
5						
6						

Can Safe Mode sustain the spacecraft indefinitely (power + thermal)? Y / N

If not, what is the maximum Safe Mode duration? _____ hours

Can every autonomous FDIR action be overridden from ground? Y / N

If not, which cannot? _____

Part C: Nominal Procedure

Write one nominal procedure for your mission (science observation or data downlink):

Procedure ID: ___ Title: _____

Preconditions:

Step	Action	Expected Result	If Not Achieved
1			
2			
3			
4			
5			
6			
7			
8			

Post-conditions:

Part D: Contingency Procedure

Write one contingency procedure (e.g., safe mode recovery, communication loss recovery):

Procedure ID: ___ Title: _____

Trigger condition:

Severity level: Green / Yellow / Orange / Red

Step	Action	Expected Result	If Not Achieved	Timeout
1				
2				
3				
4				
5				
6				

Escalation: If this procedure fails, escalate to: _____

Post-conditions:

Part E: Operations Staffing

Estimate the staffing requirements for each mission phase:

Phase	Duration	Shifts/Day	Staff/Shift	Total FTE	Annual Cost (kEUR)
LEOP	_____ days	3 (24/7)			
Commissioning	_____ weeks	2 (16/7)			
Early Operations	_____ months	1 (8/5)			
Nominal Operations	_____ years	1 (8/5) or auto			
End of Life	_____ weeks	1 (8/5)			
Total					

Part F: Anomaly Scenario

Hypothetical scenario: Your satellite enters Safe Mode during a weekend. The next ground contact is Monday morning (40 hours away). Walk through your response:

1. **What is the spacecraft doing in Safe Mode?**

1. **Will it survive 40 hours? (Check power and thermal)**

1. **What telemetry will you review first on Monday?**

1. **What is your recovery plan?**

Notes & Reflections

How much autonomy should your spacecraft have? Where is the line between onboard decision-making and ground control?

Worksheet 5.3: Mission Simulation Day

Name: _____ Date: ___ Team: _____

Mission Name: _____

Your Role: _____ (FD / SC / GSO / PO / FD-nav / Logger)

Simulation Event Log

Record ALL events, decisions, and actions in real-time during the simulation:

#	Sim Time	Real Time	Event / Observation	Action Taken	Result	Logged By
1	T+0		Separation			
2						
3						
4						
5						
6						
7						
8						
9						
10						
11						
12						
13						
14						
15						
16						
17						
18						
19						
20						

LEOP Checklist

Check off each LEOP milestone as it is achieved during the simulation:

#	Milestone	Time Achieved	Status	Notes
1	Separation confirmed		[ ]	
2	30-min timer expired		[ ]	
3	Antenna deployed		[ ]	
4	Beacon acquired		[ ]	
5	First HK telemetry decoded		[ ]	
6	First uplink command ACK'd		[ ]	
7	Solar array deployed (if sep.)		[ ]	
8	ADCS detumble complete		[ ]	
9	ADCS calibration complete		[ ]	
10	Full duplex validated		[ ]	
11	All subsystems nominal		[ ]	

Anomaly Response Form 1

Complete this form for each injected anomaly during the simulation.

Field	Entry
Anomaly ID	ANO-SIM-001
Sim Time	
Real Time	
Severity	Green / Yellow / Orange / Red
Affected Subsystem	

Observation (exact telemetry values and symptoms):

Initial Assessment (what do you think happened?):

Diagnosis (root cause after investigation):

Action Taken (commands sent, procedures executed):

Step	Time	Action	Expected Result	Actual Result
1				
2				
3				
4				
5				

Final Result (system state after response):

Follow-up Required:

Procedure used: ___ Did it work as written? Y / N / Partially

If not, what changes are needed? _____

Anomaly Response Form 2

Field	Entry
Anomaly ID	ANO-SIM-002
Sim Time	
Real Time	
Severity	Green / Yellow / Orange / Red
Affected Subsystem	

Observation (exact telemetry values and symptoms):

Initial Assessment (what do you think happened?):

Diagnosis (root cause after investigation):

Action Taken (commands sent, procedures executed):

Step	Time	Action	Expected Result	Actual Result
1				
2				
3				
4				
5				

Final Result (system state after response):

Follow-up Required:

Procedure used: ___ Did it work as written? Y / N / Partially

If not, what changes are needed? _____

Anomaly Response Form 3

Field	Entry
Anomaly ID	ANO-SIM-003
Sim Time	
Real Time	
Severity	Green / Yellow / Orange / Red
Affected Subsystem	

Observation (exact telemetry values and symptoms):

Initial Assessment:

Diagnosis:

Action Taken:

Step	Time	Action	Expected Result	Actual Result
1				
2				
3				
4				

Final Result:

Procedure used: ___ Did it work as written? Y / N / Partially

---

Commissioning Status

After the commissioning phase of the simulation, record subsystem checkout status:

Subsystem	Checkout Complete?	Performance vs Spec	Issues Found
EPS	[ ]	Nominal / Degraded / Failed	
OBC	[ ]	Nominal / Degraded / Failed	
AOCS	[ ]	Nominal / Degraded / Failed	
TTC	[ ]	Nominal / Degraded / Failed	
Thermal	[ ]	Nominal / Degraded / Failed	
Payload	[ ]	Nominal / Degraded / Failed	
Propulsion	[ ]	Nominal / Degraded / Failed / NA	

Nominal Operations Summary

After the nominal operations phase, record:

Metric	Planned	Actual (Simulated)	Variance
Science observations completed			
Data downlinked (MB)			
Anomalies encountered			
Ground contacts made			
Safe mode entries			

Simulation Debrief

What went well?

What could be improved?

Design weaknesses revealed by simulation:

Procedure changes needed:

FDIR gaps identified:

Notes & Reflections

What was the most stressful moment during the simulation? What did it teach you about operations?

Worksheet 5.4: Final Review & Presentations

Name: _____ Date: ___ Team: _____

Mission Name: _____

Part A: Final Design Summary

Record the final state of your design from the SpaceCDF Dashboard:

Parameter	Value	Unit	Margin	Margin %	Status (G/A/R)
Dry mass		kg			
Wet mass (if propulsion)		kg			
SA power (EOL)		W			
Battery capacity		Wh			
Link margin (worst case)		dB			
Pointing accuracy		deg			
Data balance		MB/day			
Total cost (P50)		kEUR			
Total cost (P80)		kEUR			
Debris compliance score		/100			
Mission lifetime		years			

Do all budgets close? Y / N **If not, which fail?** _____

Number of unresolved conflicts: _____

Part B: Documentation Checklist

Tick each document generated from SpaceCDF Exports tab:

Document	Generated?	Complete?	Notes
MRD (Mission Requirements Document)	[ ]	[ ]	
Technical Specification	[ ]	[ ]	
Verification Plan	[ ]	[ ]	
ConOps Document	[ ]	[ ]	
SEMP (Systems Engineering Mgmt Plan)	[ ]	[ ]	
IRD (Interface Requirements Document)	[ ]	[ ]	

Document	Generated?	Complete?	Notes
Risk Management Plan	[ ]	[ ]	
Test Plan	[ ]	[ ]	
Bill of Materials	[ ]	[ ]	
ITU Filing Template	[ ]	[ ]	
RSSSA Filing	[ ]	[ ]	
Export Control Assessment	[ ]	[ ]	
End-of-Life Report	[ ]	[ ]	
COPUOS Registration	[ ]	[ ]	

Total documents generated: _____ / 14

Part C: Presentation Outline

Prepare your 12-minute final design review presentation:

Section	Time	Key Points to Cover	Evidence / Data to Show
1. Mission Need	2 min		
2. Why Space?	2 min		
3. System Design	3 min		
4. Equipment & V&V	2 min		
5. Risk & Cost	2 min		
6. Operations & Lessons	1 min		

Presenter(s): _____

Backup slides prepared on: _____

Part D: Peer Review Evaluation

Complete this section while reviewing another team's presentation.

Team being reviewed: _____

Reviewers: _____

#	Criterion	Score (0-2)	Comments / Questions
1	Mission need clearly justified		
2	Requirements complete and traceable		
3	All budgets close with margin		

#	Criterion	Score (0-2)	Comments / Questions
4	Equipment selected with justified trades		
5	Interfaces defined, conflicts resolved		
6	V&V approach defined		
7	Risks identified with mitigations		
8	Cost estimate with WBS and P80		
9	ConOps and ground segment designed		
10	Sustainability compliance		
	TOTAL	**/ 20**	

Presentation quality:

Criterion	Score (1-5)	Comments
Clarity of communication		
Use of evidence (data, not assertions)		
Ability to answer questions		
Time management		

Board recommendation: GO / NO GO / GO WITH ACTIONS

Action items for the presenting team:

#	Action	Responsible	Deadline
1			
2			
3			

Strongest aspect of this team's design:

Biggest area for improvement:

Part E: Lessons Learned

1. **What was the most challenging design decision your team faced?**

1. **What would you do differently if starting the course over?**

1. **What does the design need next (Phase B activities)?**

1. **What was the most valuable SpaceCDF feature?**

1. **What feature was missing that you needed?**

1. **What concept from this course will you apply in your future work?**

Part F: Self-Assessment Rubric

Rate your own learning across the three weeks of the course. Be honest -- this is for your own reflection, not grading.

Competency	Week 1 Start (1-5)	Now (1-5)	Evidence of Growth
Define a space mission starting from the problem, not the solution			
Write verifiable, traceable requirements (SMART format)			
Conduct a structured trade study with weighted criteria			
Size subsystems using parametric models and engineering budgets			
Select COTS equipment and verify interface compatibility			
Construct a verification matrix with IADT methods			
Assess risk using a 5x5 matrix and define mitigations			
Estimate mission cost using parametric and bottom-up methods			
Design a ground segment and calculate data budget			
Define operational modes, FDIR rules, and procedures			
Respond to anomalies using a systematic process			

Competency	Week 1 Start (1-5)	Now (1-5)	Evidence of Growth
Present a design to a review board with evidence			

Scale: 1 = No familiarity, 2 = Basic awareness, 3 = Can apply with guidance, 4 = Can apply independently, 5 = Can teach others

Overall self-assessment:

At the start of this course, I would describe my systems engineering ability as:

Now, I would describe it as:

One thing I am most proud of from this course:

Notes & Reflections

Final thoughts on the course, the design process, and what comes next:

Final Review and Presentations

Moved from the textbook Part 2 wrap-up to the workbook so it sits with the worksheets it depends on.

> **Expected reading before this session.** SMAD4 Ch. 22 (optimization); Wertz et al. SME Ch. 9.

Prerequisites: All previous sessions (complete design through simulation) **References:** ECSS-M-ST-10C Rev.1 section 6 (Reviews), NPR 7123.1D Appendix G, NASA SEH Rev 2 section 3.7, ECSS-E-ST-10C section 4 (Technical Dossier)

Learning Objectives

By the end of this session, participants will be able to: 1. Prepare a complete design documentation package for review 2. Conduct a design review presentation with evidence-based arguments 3. Evaluate a peer team's design against gate criteria 4. Provide constructive technical feedback using a structured rubric 5. Identify lessons learned and articulate next steps for Phase B

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1. Design Documentation Review

Before presenting, each team must verify that their design documentation is complete and internally consistent.

Documentation Checklist

Document	Source in SpaceCDF	Status Check
Mission Requirements Document (MRD)	Exports -> ECSS Documents	All requirements baselined?
Technical Specification (TS)	Exports -> ECSS Documents	All budgets captured?
Concept of Operations (ConOps)	Exports -> ECSS Documents	All phases defined?
Verification Plan (VP)	V&V Matrix tab	All requirements have methods?
Bill of Materials (BOM)	Exports -> Design Data	All equipment listed with TRL?
Risk Register	Worksheet 4.3	All risks scored and mitigated?
Interface Matrix (N^2)	Worksheet 4.3	Interfaces identified? Conflicts resolved?
Cost Estimate	Worksheet 4.4	WBS complete? P80 computed?
Operations Concept	ConOps Editor	Modes, FDIR, procedures defined?
Sustainability Assessment	Sustainability Card	Debris compliance score?

Internal Consistency Checks

Before presenting, verify:

Check	How to Verify	Common Error
Mass budget closes	Dashboard mass KPI: margin > 0%	Equipment added without updating budget
Power budget closes	Dashboard power KPI: SA > loads in all modes	Eclipse mode power not checked
Link budget closes	Link Budget tool: margin >= 3 dB	Rain attenuation not included
Data budget closes	Data Budget: daily downlink >= daily generation	Assumed too many passes per day
Cost is within ceiling	Cost tab: total <= allocated budget	Forgot launch cost or operations
Requirements traceable	Every requirement maps to an objective	Orphan requirements (no parent objective)
V&V complete	Every requirement has a method assigned	"TBD" entries remaining
No unresolved conflicts	Dashboard: conflict count = 0	Interface mismatches not addressed

SpaceCDF Document Generation

Navigate to **Exports** tab and generate all documents: 1. ECSS Documents: MRD, TS, VP, ConOps, SEMP, IRD 2. Design Data: BOM, Parametric Model Data 3. Regulatory Filings: ITU API, RSSSA, Export Assessment, COPUOS, EOL Report 4. Presentation: Summary slide data (auto-populated)

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2. Presentation Structure

Each team presents a 12-minute design review followed by 5 minutes of Q&A from the peer review board.

[Source: ECSS-M-ST-10C Rev.1 section 6; NPR 7123.1D Appendix G (Review Process)]

Presentation Outline

Section	Duration	Content	Evidence
1. Mission Need	2 min	Problem statement; stakeholders; objectives with MoPs	Mission Need step
2. Why Space?	2 min	Trade study results; why space beats alternatives	Mission Trade results
3. System Design	3 min	Architecture; orbit; key parameters; budget status	Dashboard, budgets
4. Equipment & Verification	2 min	Key equipment; BOM summary; V&V approach; test plan	BOM, V&V Matrix
5. Risk & Cost	2 min	Top 5 risks; cost estimate (P50/P80); schedule	Risk register, cost tab

Section	Duration	Content	Evidence
6. Operations & Lessons	1 min	ConOps summary; FDIR; what would you change	ConOps, simulation debrief

Presentation Best Practices

Slide design: - One key message per slide - Data, not prose (budgets, trade tables, not paragraphs) - Include backup slides with detailed analyses for Q&A

Delivery: - Lead with conclusions: "Our mass budget closes with 22% margin" - Show evidence for every claim: "Link budget analysis shows 4.2 dB margin at worst case" - Acknowledge risks honestly: "Antenna deployment is our highest risk at 12 (L3 x C4)" - For questions you cannot answer: "Good question -- we will take that as an action item"

Common pitfalls to avoid: - Reading slides verbatim - Hiding known problems (review boards always find them) - Presenting analysis without stating assumptions - Spending too long on the mission description and rushing through technical results

---

3. Peer Review Process

Each team acts as a review board for another team. This mirrors the real design review process where an independent board evaluates the project.

Review Board Roles

Role	Responsibility	Key Questions to Ask
Chair	Manages review; ensures all criteria covered	"Let's systematically check each gate criterion"
Systems reviewer	Evaluates overall architecture and budgets	"Do all budgets close? What is the minimum margin?"
Technical reviewer	Evaluates subsystem design decisions	"Why did you choose this component over alternatives?"
Risk reviewer	Evaluates risk management and V&V	"What is your highest risk? Is the mitigation adequate?"
Operations reviewer	Evaluates ConOps, ground segment, regulatory	"What happens if you lose contact for 48 hours?"

Gate Criteria Evaluation

The review board evaluates the presenting team against PDR-level gate criteria:

#	Criterion	Score (0-2)	Notes
1	Mission need clearly justified	0 = missing, 1 = partial, 2 = strong	Is the problem real? Is space the right answer?
2	Requirements complete and traceable	0/1/2	All requirements linked to objectives? Verifiable?

#	Criterion	Score (0-2)	Notes
3	All budgets close with adequate margin	0/1/2	Mass, power, link, data, pointing -- all positive?
4	Equipment selected with justified trades	0/1/2	BOM complete? TRL adequate? Trade studies documented?
5	Interfaces defined and conflicts resolved	0/1/2	N^2 matrix? No unresolved conflicts?
6	V&V approach defined for all requirements	0/1/2	Methods assigned? Test plan outlined?
7	Risks identified with mitigations	0/1/2	Risk register with scores? Top risks mitigated?
8	Cost estimate with WBS and P80	0/1/2	Both parametric and bottom-up? Learning curve applied?
9	ConOps and ground segment designed	0/1/2	Modes defined? FDIR architecture? Ground stations?
10	Sustainability compliance	0/1/2	Debris compliance? FCC 5-year rule? Passivation plan?

Review Board Questions Bank

Standard questions for the review board to ask:

- "Why is space the right answer? What alternatives were considered?"
- "What is your mass margin? What drives it? What if the payload is 20% heavier than estimated?"
- "Walk me through your link budget at worst case. What is the minimum margin?"
- "What are your top 3 risks? What is the residual risk after mitigation?"
- "Show me your data budget. Does it close with a single ground station?"
- "What happens during a safe mode entry in eclipse? Does the thermal budget survive?"
- "Which component has the lowest TRL? What is your qualification plan?"
- "Have you started spectrum licensing? What is on the critical path?"
- "What is your single-point failure list? Which ones have you accepted?"
- "If you had 6 more months and 20% more budget, what would you change?"

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4. Team Presentations

Instructions

Presentation Schedule: 12 minutes presentation + 5 minutes Q&A per team.

For 4 teams: allocate 17 min x 4 = 68 min. Adjust timing based on actual number of teams. If fewer than 4 teams, allow 15 min presentation + 8 min Q&A.

For the presenting team: 1. Present the 6-section outline above 2. Use SpaceCDF live for demonstrations where helpful (show Dashboard, budgets) 3. Respond to board questions with evidence

For the review board: 1. Complete the gate criteria evaluation form on Worksheet 5.4 2. Ask at least 3 substantive questions 3. Provide a GO / NO GO / GO WITH ACTIONS recommendation 4. Document action items with responsible person and deadline

After each presentation: - Chair announces the board's recommendation - Action items are recorded - Brief applause and transition to next team

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5. Course Wrap-Up & Lessons Learned

Individual Reflection

Each participant completes the self-assessment rubric on Worksheet 5.4, reflecting on their learning across all three weeks.

Team Lessons Learned

Each team discusses and reports:

1. **Most challenging design decision:** What trade-off was hardest to resolve? Why?
2. **Biggest surprise:** What aspect of mission design was most different from expectations?
3. **Tool feedback:** What SpaceCDF feature was most valuable? What was missing?
4. **Process insight:** How did concurrent design change your approach to engineering?
5. **If starting over:** What would you do differently in Week 1?

Key Takeaways from the Course

Week	Theme	Core Lesson
Week 1	Requirements & Architecture	Start with the problem, not the solution. Requirements define WHAT, not HOW.
Week 2	Subsystem Design & Budgets	Engineering budgets are the language of systems engineering. Everything trades against everything.
Week 3	Integration, Verification & Operations	Building the thing right (verification) and building the right thing (validation) matter as much as designing it. Operations reveal what analysis cannot.

The Systems Engineering V-Model -- Completed

```
You started here:                              You finished here:

  Mission Need                                   Mission Validated
     |                                                  ^
     v                                                  |
  Requirements                                  Requirements Verified
     |                                                  ^
     v                                                  |
  System Design                                 System Integrated (sim)
     |                                                  ^
     v                                                  |
  Subsystem Design                              Subsystem V&V Planned
     |                                                  ^
```

```
    v                                                  |
Equipment Selected  ------>  Equipment Verified (BOM + compliance)

Week 1-2 (left side)         Week 3 (right side)
```

You have traversed the complete V-model: from mission need to verified, validated design.

Next Steps for Participants

- **SpaceCDF access** continues after the course
- **Apply the methodology** to your own missions or projects
- **Use the tool** for CDF-style studies with your team
- **Refer to the Course Textbook** for session plans and appendices
- **Share feedback** to improve the course for future cohorts

Course Evaluation

Distribute the course evaluation form. Key questions: - Which session was most valuable? Least valuable? - What content should be added? Removed? - How effective was the simulation day? - Would you recommend this course to colleagues? - What is one thing you will do differently in your engineering practice as a result?

Cited References

#	Reference	URL
1	ECSS-M-ST-10C Rev.1 (Project Planning and Implementation)	https://ecss.nl/standard/ecss-m-st-10c-rev-1-project-planning-and-implementation/
2	NPR 7123.1D Appendix G (Review Process)	https://nodis3.gsfc.nasa.gov/displayDir.cfm?t=NPR&c=7123&s=1D
3	NASA SEH Rev 2, section 3.7 (Technical Reviews)	https://www.nasa.gov/reference/systems-engineering-handbook/
4	ECSS-E-ST-10C (System Engineering)	https://ecss.nl/standard/ecss-e-st-10c-rev-1-system-engineering-general-requirements/

Session Summary

Topic	Key Takeaway
Documentation	Complete package: MRD, TS, ConOps, VP, BOM, Risk Register, Cost, Regulatory
Consistency	All budgets must close; no unresolved conflicts; requirements fully traceable
Presentation	6 sections: need, justification, design, V&V, risk+cost, operations+lessons
Peer review	Independent board evaluates against gate criteria; GO/NO GO/GO with actions
Best practices	Lead with conclusions, show evidence, acknowledge risks, answer directly

Topic	Key Takeaway
V-model	Course traverses full V: needs -> requirements -> design -> V&V -> validation
Lessons learned	Document what worked, what failed, and what to change -- this is the real output

Part 5 — OrbitNorth Week 3 Workbook

This part of the workbook is the operator's pad for the five-day **OrbitNorth Week 3** spacecraft operations training. It runs Monday 17 August to Friday 21 August 2026 at the uOttawa Cyberrange.

The structure mirrors Part 2 of the textbook. For every nominal and contingency procedure run during the week, there is a **confirm-each-step** worksheet: you tick each step when verified, write the actual telemetry value in the *note* column, and flag anything off-nominal. Anomaly tickets, handover templates, observer worksheets, and end-of-shift debrief pages are between the day worksheets.

How to use these worksheets.

- Bring the printed worksheet to the console at the start of every shift.
- Write in pen so corrections are visible (cross out, don't erase).
- Fill in every box, even if the answer is "nominal" or "n/a — not run".
- Hand the completed worksheet to the instructor at end of shift. The cohort lessons-learned register feeds off them.

The worksheet IDs follow the pattern `WB-5.X-NAME` so they cross-reference cleanly with textbook chapters.

Worksheet WB-5.0 — Orientation Check

Use: complete *before* arriving at the Cyberrange Monday morning. Brings you up to a common baseline with the rest of the cohort.

Name: _____ **Date:** _____

Section A — Mission facts

Fill in each value. Source: textbook §5.0.7.

Item	Your answer
EOSAT-1 platform (bus type, units)	
Orbit altitude (km)	
Orbit inclination (°)	
Local time of ascending node (LTAN)	
Mission band	
Number of imager bands and their centre wavelengths (nm)	
Ground stations — all five (name and latitude)	
Battery capacity (Wh)	
Battery chemistry and cell configuration	
Instrument swath (km) and GSD (m)	
Pointing requirement in NOMINAL, mode 4 (°)	
Pointing requirement in FINE_POINT, mode 5 (°)	
Propulsion: type and total Δv (m/s)	

Section B — Control room positions

For each position, write a one-sentence mandate in your own words.

Position	Mandate (one sentence)
Flight Director	
SPACON / FDIR & Systems	
AOCS Engineer	
EPS Engineer	
Thermal Engineer	
TT&C Engineer	
Payload Operations	

Position	Mandate (one sentence)
FDS / Orbit Analyst	
Ground Segment Engineer	
Mission Manager	

Section C — Pre-week reflection

What part of the operations work are you most looking forward to?

What part are you most nervous about?

Worksheet WB-5.1 — Position score cards (round-robin)

Use: Monday afternoon 13:00 – 15:00. One card per position. As each station's 12-minute slot ends, write a one-sentence observation under "Insight" and a one-sentence question under "Question to ask".

Name: _____

5.1.1 Flight Director

- **Task:** Run a GO/NO-GO round on the live dashboard. Poll three positions for *go for NOOP heartbeat*; record the responses.
- **Polled positions:** ___ , __ , ____
- **Responses recorded (verbatim):**
-
-
-
- **Time on FD seat:** _ : _ to _ : _
- **Insight:** _____
- **Question to ask:** _____

5.1.2 SPACON / FDIR & Systems

- **Task:** Walk through one FDIR rule on the FDIR panel; explain its trigger, threshold, severity, and the contingency procedure it points to.
- **Rule name selected:** _____
- **Trigger parameter:** _____
- **Threshold:** ___ **Persistence:** _
- **Severity:** _ ***Recommended procedure:*** _____
- **Insight:** _____
- **Question to ask:** _____

5.1.3 AOCS Engineer

- **Task:** Read `aocs.att_error` (0x0217) and verify it is inside the NOMINAL (mode 4, nadir-tracking) budget of < 1.0°. In FINE_POINT (mode 5) the budget is < 0.1°.
- **Reading:** _____ ° **In band?** ☐ yes ☐ no
- **AOCS mode (`aocs.mode` 0x020F):** ____ (0 OFF, 1 SAFE_BOOT, 2 DETUMBLE, 3 COARSE_SUN, 4 NOMINAL, 5 FINE_POINT, 6 SLEW, 7 DESAT, 8 ECLIPSE, 9 INERTIAL)
- **Star tracker status** (`aocs.st1_status` 0x0240 / `aocs.st2_status` 0x0243; 0=off, 1=boot, 2=tracking, 3=blind, 4=failed): ST1 _ ***ST2*** _ **Selected** (`aocs.st_selected` 0x0244): ____
- **Wheel speeds (RPM, read-only)** `aocs.rw1..rw4_speed` (0x0207–0x020A): 1=_ ***2=***_ 3=_ ***4=***_
- **Insight:** _____
- **Question to ask:** _____

5.1.4 EPS Engineer

- **Task:** Predict battery SoC at exit of the next eclipse.
- **SoC at eclipse entry:** _ ***% Eclipse duration:*** _ min
- **Predicted SoC at exit:** ____ % **(Method:** P_avg × t / E_batt, with E_batt = **100 Wh** and the nominal orbit-average load P_avg ≈ **28.7 W)**
- **Cross-check vs eclipse model:** ☐ matches ☐ off by ____ %
- **Insight:** _____
- **Question to ask:** _____

5.1.5 Thermal Engineer

- **Task:** Identify which heater is currently at the highest duty cycle. Predict whether it will saturate next eclipse.
- **Heater with highest duty:** ___ **Duty:** ____ %
- **Predicted next-eclipse duty:** ____ % **Will saturate?** ☐ yes ☐ no
- **Other thermal concerns:** _____
- **Insight:** _____
- **Question to ask:** _____

5.1.6 TT&C Engineer

- **Task:** Walk the TM rate ladder at AOS. The ladder has three rungs — **BEACON 1 kbps / LOW 16 kbps / HIGH 128 kbps** (`ttc.tm_data_rate` 0x0506 in bps, `ttc.tm_tier` 0x0539 = 0/1/2). LOW is the routine operations rung; HIGH is for dumps only and needs DOWNLINK mode on (func 79) plus elevation ≥ 10°. BEACON is not selectable with `TTC_SET_DATA_RATE` (S8 func 65).
- **Beacon rate (kbps):** __ **Routine LOW rate (kbps):** _ ***Dump HIGH rate (kbps):*** ___
- **Link margin post-transition (dB):** _____ (`ttc.link_margin` 0x0503)
- **BER reading (`ttc.ber` 0x050C, log10):** _____
- **Insight:** _____
- **Question to ask:** _____

5.1.7 Payload Operations

- **Task:** Schedule a single imaging session via MPS. Confirm FPA temperature is at setpoint before commanding.
- **Imaging target chosen:** _____
- **FPA temperature reading (`payload.fpa_temp` 0x0601, °C):** _____ **At setpoint?** ☐ yes ☐ no — the gate is `payload.fpa_ready` (0x0610) = 1, not a remembered temperature. Default setpoint **−15 °C** (`payload.fpa_setpoint` 0x0631; commandable envelope −20 °C to 0 °C).
- **Storage usage (`payload.mem_used_mb` 0x060B):** _____ / 20000 MB
- **Insight:** _____
- **Question to ask:** _____

5.1.8 FDS / Orbit Analyst

- **Task:** Compute AOS, TCA, LOS for the next two Iqaluit passes. Compare to the MPS contact schedule.
- **Pass 1:** AOS __ **TCA** _ ***LOS max el*** _ °

- **Pass 2:** AOS __ **TCA** _ ***LOS max el*** _ °
- **MPS agreement:** ☐ exact ☐ off by < 5 s ☐ off by > 5 s
- **Insight:** _____
- **Question to ask:** _____

5.1.9 Ground Segment Engineer

- **Task:** Verify the network is tracking the spacecraft for the next pass. EO-Sat-1 flies the **five-station** teaching baseline (`orbit.yaml`). Note the baseband configuration in use.
- **Iqaluit (63.75 N):** ☐ TRACKING ☐ NOT TRACKING
- **Troll (72.01 S):** ☐ TRACKING ☐ NOT TRACKING
- **Svalbard (78.23 N):** ☐ TRACKING ☐ NOT TRACKING
- **Inuvik (68.32 N):** ☐ TRACKING ☐ NOT TRACKING
- **O'Higgins (63.32 S):** ☐ TRACKING ☐ NOT TRACKING
- **Active station (`ttc.active_gs` 0x0523):** _____
- **Baseband config (rate, encoding):** ____
- **Insight:** _____
- **Question to ask:** _____

Worksheet WB-5.3-PP — Day 1 Pass-Plan Team Exercise

Team: ☐ Borealis ☐ Austral **Lead:** ___

Field	Entry
Target pass (station + AOS time)	
TCA time	
LOS time	
Pass duration (min)	
Procedures to run (IDs, in order)	
Imaging target ID	
Imaging window inside pass	
Bands selected and integration time	
Predicted ground swath (km)	
Downlink rate planned (kbps)	
Expected downlink duration (s)	
Plausible failure 1	
Recovery procedure for failure 1	
Plausible failure 2	
Recovery procedure for failure 2	

Read-aloud notes

What the other team challenged: _____

What we accepted: _____

What we defended: _____

Worksheet WB-5.4-VL — Day 2 Voice-Loop Callout Drill

Use: Tuesday 09:00 – 10:00. Pair drill. One plays FD, the other plays the named subsystem. Record what was said.

#	Scenario	Your callout (verbatim)	Pass/ Fail
1	FD asks AOCS for pointing report		☐ ☐
2	AOCS reports pointing 0.28°, nominal		☐ ☐
3	FD asks EPS for eclipse-entry predictor		☐ ☐
4	EPS reports eclipse entry T+04:12, SoC at exit 78%		☐ ☐
5	FD polls 3 positions for GO on payload power-on		☐ ☐
6	EPS responds: "no-go, payload line 3 already drawing 0.35 A with the line commanded OFF"		☐ ☐
7	FD calls HOLD on payload power-on		☐ ☐
8	TT&C calls anomaly: BER climbed to 1e-3		☐ ☐
9	FD requests rate fallback		☐ ☐
10	TT&C reports tier stepped down HIGH→LOW, rate 16 kbps, BER 1e-7 nominal		☐ ☐

Self-score (out of 10): _____

One callout I want to practice more: _ _____

Worksheet WB-5.4-N001 — NOM-001 Pass Startup (confirm each step)

Use: Tuesday afternoon dry-run, every nominal pass thereafter.

Operator: ___ **Pass:** _____

#	Owner	Action	Expected verification	☐ done	Note / actual value
1	Ground Seg	Confirm the scheduled station is tracking	station = TRACKING in MPS (five-station network)	☐	
2	TT&C	Verify the S-band chain is radiating	`ttc.pa_on` (0x0516) = 1 and `ttc.tx_active` (0x0525) = 1 before AOS-2 min; if 0, read `ttc.tx_gate_status` (0x052B)	☐	
3	TT&C	At AOS, verify link lock acquired	`ttc.link_status` (0x0501) = **2** (LOCKED) within 30 s of AOS; `ttc.carrier_lock` (0x0510) = 1	☐	
4	SPACON	Issue `CONNECTION_TEST` (S17.1)	S1.1 ack + S17.2 report received	☐	
5	SPACON	Verify full HK reporting	SIDs 1, 2, 3, 4, 5, 6 periodic (only 1 EPS and 11 Beacon are on at power-up — the rest need `HK_ENABLE` , S3.5); on-demand reads use `HK_REQUEST` , **S3.27**	☐	
6	AOCS	Confirm AOCS mode and rates	`aocs.mode` (0x020F) = **4** (NOMINAL, nadir-tracking); `aocs.rate_roll` / `rate_yaw` (0x0204/0x0206) < 0.05°/s, `aocs.rate_pitch` (0x0205) ≈ −0.063 ± 0.02°/s (nadir tracking rate)	☐	
7	EPS	Confirm battery and bus state	`eps.bat_soc` (0x0101) > 50 %, `eps.bus_voltage` (0x0105) > 27.0 V (nominal band 26–30 V; FDIR undervoltage at **24.5 V**)	☐	
8	Thermal	Confirm zones in band	All zones green	☐	
9	Payload	Confirm payload state matches plan	mode = per plan	☐	
10	SPACON	Log pass start in shift log	Log entry created	☐	

Off-nominal items observed: _____

FD signature: ____ Time complete: ___

Worksheet WB-5.4-C003 — CTG-003 TTC Link Loss Recovery (confirm each step)

Operator: ___ Trigger time: _

#	Owner	Action	Expected verification	☐ done	Note / actual value
1	TT&C	Confirm link loss on `ttc.link_status` (0x0501) and `ttc.frame_sync` (0x0512)	link_status ≠ 2, frame sync lost	☐	
2	Ground Seg	Verify ground station still tracking	Antenna pointing correct, baseband nominal	☐	
3	SPACON	Open FDIR panel; identify the rule that fired	rule + `fdir.level` (0x0700) recorded	☐	
4	TT&C	Blind `HK_REQUEST(sid=6)` (S3.27), then `CONNECTION_TEST` (S17.1)	If it responds, go to step 8	☐	
5	TT&C	If step 4 fails, blind `TTC_SWITCH_REDUNDANT` (S8 func 64), then `TTC_SWITCH_PRIMARY` (func 63)	`ttc.link_status` = 2 within 45 s	☐	
6	TT&C	If still silent, step the ladder down: `TTC_SET_DATA_RATE(rate=0)` (S8 func 65) = LOW	`ttc.tm_data_rate` (0x0506) = 16000 bps; `ttc.tm_tier` (0x0539) = 1	☐	
7	TT&C	If still no response, recommend TX power increase	FD GO required; `TTC_SET_TX_POWER(level=2)` (S8 func 68); readback `ttc.tx_fwd_power` (0x050D, dBm)	☐	
8	SPACON	Acknowledge FDIR rule, log recovery actions	FDIR cleared, log entry	☐	
9	TT&C	File anomaly ticket. If no link after two consecutive passes, escalate **EMG-004** Loss of Communication	Ticket logged	☐	

Step at which link was recovered (if any): _____

Anomaly ticket number: _____

FD signature: ____ Time complete: ___

Worksheet WB-5.4-N012 — NOM-012 Shift Handover Template

Use: every shift change. Outgoing FD reads each row; incoming FD reads back items 3 and 7. Target: under 6 minutes.

Outgoing FD: ___ Incoming FD: __ *Shift end*: __ Handover start: _

#	Item	Outgoing FD says	Incoming FD reads back?
1	Phase and time		
2	Last completed pass		
3	**Open anomalies** (rule name, status, recovery procedure step)		☐
4	Open work items (procedures in progress)		
5	Next pass (station, AOS, planned procedures, operator-of-record)		
6	Customer commitments active		
7	**GO authorisations outstanding**		☐
8	Voice-loop tone (team rested / tense / etc.)		

Handover end: ___ Duration: _ : _ (target < 6:00)

Both FDs sign below to acknowledge handover complete.

Outgoing: **____ Incoming: _____**

Worksheet WB-5.4-FD — Day 2 Flight-Dynamics Workshop

Operator: _ _____

Exercise 1 — Predict next three AOS times

Tool: `python3 tools/orbit_tools.py predict --hours 24`. Compare to MPS contact schedule.

Pass	Station	AOS (your prediction, UTC)	AOS (MPS, UTC)	Δ (s)
1				
2				
3				

Exercise 2 — Pass duration vs max elevation

Pass	Max el (°)	Predicted duration (min)	Actual duration from MPS (min)
1			
2			
3			

Formula used (write here): _____

Exercise 3 — Eclipse arithmetic

- **Longest eclipse in next 24 h:** __ **UTC, duration** __ min
- **Predicted SoC at exit** (using $\mathrm{SoC}_{\mathrm{exit}} = \mathrm{SoC}_{\mathrm{entry}} - \mathrm{P}_{\mathrm{avg}} \cdot \mathrm{t} / \mathrm{E}_{\mathrm{battery}}$): ____ %
- **Actual SoC at exit:** _ **% Δ:** _ %
- **Concern?** ☐ no ☐ yes — note: _____

Worksheet WB-5.4-Q — Day 2 Operations Training Quiz Response Sheet

Name: ___ Time started: _____

Pass mark: 11 of 15. Solutions on the back of the instructor pack.

Voice-loop discipline (3 Qs)

1.
2.
3.

MCS UI navigation (3 Qs)

1.
2.
3.

Telemetry interpretation — limits and SIDs (3 Qs)

1.
2.
3.

Procedure structure (2 Qs)

1.
2.

Anomaly response (2 Qs)

1.
2.

Pass timing arithmetic (2 Qs)

1.
2.

Self-mark when graded: ____ / 15 **Pass?** ☐ yes ☐ no

Worksheet WB-5.5-PB — Day 3 LEOP Pre-Pass Briefing Checklist

Team: ☐ Borealis ☐ Austral **Pass:** ☐ Pass 1 ☐ Pass 2

Item	☐
Position assignments confirmed and posted	☐
Comm check completed (all positions readable)	☐
BP-2 (`first_contact.yaml`) loaded; spacecraft in phase 3 (BOOTLOADER)	☐
`pass_override = 1` confirmed (param 0x05FF)	☐
FD has loaded LEOP procedure pack into procedure panel	☐
SPACON has reviewed the FDIR rule set on the FDIR panel (there is **no** FDIR table-load telecommand — the rule set is `subsystems/fdir.yaml + monitoring/`)	☐
AOCS has reviewed residual tumble rates (~1–2 °/s per axis from the separation impulse)	☐
EPS confirms SoC ~90 % (`eps.bat_soc` 0x0101); only the non-switchable `obc` (line 0) and `ttc_rx` (line 1) are ON — every switchable line defaults OFF	☐
TT&C has confirmed UHF beacon reception (`ttc.beacon_active` 0x0527 = 1); note the beacon is a separate radio from the S-band prime	☐
Ground Seg has confirmed Iqaluit antenna on-target, demodulator on the **BEACON** rung (1 000 bps, BPSK, conv(7,1/2)+RS, 1 kBd)	☐
FDS has produced 24-h pass schedule and handed to FD/TT&C/Ground Seg	☐
Anomaly ticket forms blank-printed and available at each console	☐
Voice-loop language declared (callouts in English)	☐

FD ready call (verbatim): _____

Pass start (UTC): _____

Worksheet WB-5.5-L001 — LEOP-001 First Acquisition of Signal & OBC Boot (confirm each step)

Operator (TT&C lead): ___ Pass start: _____

#	Owner	Action	Expected verification	☐ done	Actual note
1	Ground Seg	Configure GS receiver on the BEACON rung; force pass override in MCS	GS = TRACKING_READY; `pass_override = 1` (0x05FF)	☐	
2	TT&C	Send first uplink: `CONNECTION_TEST` (S17.1)	S1.1 ack < 10 s; S17.2 < 10 s; `eps.pl_ttc_tx` (0x0112) = 1 (auto-TX hold-down armed)	☐	
3	TT&C	Verify carrier lock and link quality	`ttc.link_status` (0x0501) = **2** (LOCKED) < 30 s; `ttc.rssi` (0x0502) > −110 dBm; `ttc.link_margin` (0x0503) > 3 dB	☐	
4	SPACON	Request beacon SID: `HK_REQUEST(sid=11)` (**S3.27**)	Beacon received; `obdh.sw_image` (0x0311) = 0 (BOOTLOADER) < 10 s	☐	
5	SPACON	Boot OBC into application: `OBC_BOOT_APP` (S8.1 func_id 55)	S1.1 ack + S1.7 < 5 s; `obdh.sw_image` = 1 < 15 s. Boot re-applies `default_enabled_sids: [1, 11]` — **only SIDs 1 and 11 stream**	☐	
5a	SPACON	Enable the platform SIDs: `HK_ENABLE(sid=3)`, `(sid=4)`, `(sid=6)` (**S3.5**)	SIDs 3 (60 s), 4 (8 s), 6 (8 s) begin within one structure interval	☐	
5b	TT&C	Clear the forced beacon tier, then select LOW: `TTC_SET_BEACON_MODE(on=0)` (S8 func 70), `TTC_SET_DATA_RATE(rate=0)` (S8 func 65)	`ttc.beacon_mode` (0x0521) = 0; `ttc.tm_data_rate` (0x0506) = 16000 bps; `ttc.tm_tier` (0x0539) = 1. Re-configure the ground demod to 16 kBd first	☐	
6	SPACON	Synchronise onboard time: `SET_TIME(cuc_seconds=T_corrected)` (S9.1), then `REQUEST_TIME` (S9.2)	Ack + complete < 5 s; |T_obc − T_gnd| < 1 s	☐	
7	SPACON	Request EPS HK: `HK_REQUEST(sid=1)` (S3.27)	`eps.bus_voltage` (0x0105) 26–30 V;	☐	

#	Owner	Action	Expected verification	☐ done	Actual note
			`eps.bat_soc` (0x0101) > 50 %		
8	TT&C	Log times — AOS, override, first accepted TC, app boot, time-sync residual, HK values	All times recorded	☐	

Recorded values: - AOS time (UTC): __ **- Override engaged:** _ **- *First accepted TC:*** ___ - App boot complete: __ **- Time-sync residual:** __ s - `eps.bus_voltage` : _ ***V*** ***`eps.bat_soc`*** **:** _ %

Common mistakes to avoid (instructor recap): - Forgetting `pass_override` before TCs. - Sending non-bootloader-allowlisted TCs before app boot (gets S1.8 failure code **0x0006**, "Not permitted in bootloader"). `SET_TIME` is one of them. - Letting the **5-minute** auto-TX hold-down (`auto_tx_hold_s` = 300 s, re-armed by every accepted TC) expire mid-procedure. - Trying `TTC_SET_DATA_RATE` before clearing beacon mode — refused with 0x0007; application boot does **not** clear it. - Expecting SIDs 2 and 5 to appear after step 5 (they don't — they are enabled in LEOP-007 once their lines are powered), or expecting any SID other than 1 and 11 without an explicit `HK_ENABLE` (S3.5).

Exit criterion met? ☐ yes ☐ no **FD signature:** ___

Worksheet WB-5.5-L006 — LEOP-006 Time Synchronisation (confirm each step)

Operator (SPACON): _ _ _ _ _ _

#	Action	Expected verification	☐ done	Note
1	`HK_REQUEST(sid=4)` (S3.27); read onboard time from `obdh.obc_time` (0x0309)	timestamp recorded	☐	
2	Issue `REQUEST_TIME` (**S9.2**) for the TM(9,2) report, **and** `HK_REQUEST(sid=6)` for `ttc.range_km` (0x0509) — SID 6 is the structure that carries the slant range	T_obc recorded; TM(9,2) agrees with 0x0309	☐	
3	Compute OWLT = `ttc.range_km` / 299792.458 × 1000 ms (expect **3–22 ms**), then Δt = T_obc − (T_gnd − OWLT)	OWLT (ms): _ _ **Δt (s):** _ _ _ _	☐	
4	Issue `SET_TIME(cuc_seconds=T_corrected)` (**S9.1**) — the argument goes *inside* the backticks; a bare `SET_TIME` packs cuc_seconds = 0, an epoch reset	ack + complete < 5 s, no S1.2	☐	
5	`REQUEST_TIME` (S9.2) again; verify residual < 1.0 s vs ground reference (LEOP threshold; NOM-008 tightens it in nominal ops)	residual (s): _ _ _ _ _ _	☐	

Exit criterion met? ☐ yes ☐ no

Worksheet WB-5.5-L002 — LEOP-002 Initial Health Check (confirm each step)

Operator (SPACON consolidates): _ _____

Subsystem	Operator	Tab read	Status reported	☐ done	Note
OBDH	SPACON	obdh	☐ OK ☐ NOT-OK	☐	
EPS	EPS	eps	☐ OK ☐ NOT-OK	☐	
TCS	Thermal	tcs	☐ OK ☐ NOT-OK	☐	
TT&C	TT&C	ttc	☐ OK ☐ NOT-OK	☐	
AOCS	AOCS	aocs	☐ OK ☐ NOT-OK	☐	
Payload	Payload	payload	☐ OK ☐ NOT-OK (typically OFF in LEOP)	☐	

Consolidated PASS / FAIL: ☐ PASS ☐ FAIL

If FAIL, which contingency procedure invoked: _ ____

Exit criterion met? ☐ yes ☐ no

Worksheet WB-5.5-L007 — LEOP-007 Sequential Power-On (confirm each step)

Operator (SPACON lead): _____

LEOP-007 is the long one — ~25 minutes of careful per-line power-on. Each new line is a separate commanding event with FD GO.

Power-line index map (`EPS_POWER_ON` / `EPS_POWER_OFF` / `EPS_RESET_OC_FLAG` all take `line_index` , range **0–8**): 0 `obc` , 1 `ttc_rx` (both non-switchable), 2 `ttc_tx` , 3 `payload` , 4 `fpa_cooler` , 5 `htr_bat` , 6 `htr_obc` , 7 `aocs_wheels` , 8 `prop` . Every switchable line defaults **OFF**. There is no IMU power line — the three gyros are S2 devices `S2_AOCS_GYROSCOPE_0/1/2` .

#	Owner	Action	Verification	☐ done	Note
1	EPS	Verify EPS baseline: `HK_REQUEST(sid=1)` (S3.27)	`eps.bus_voltage` (0x0105) 27.0–29.0 V; `eps.bat_soc` (0x0101) > 60 %; `eps.power_gen` (0x0107) > 0 W	☐	
2	EPS	Energise battery heater: `EPS_POWER_ON(line_index=5)` (S8.1 func_id **19**)	`eps.pl_htr_bat` (0x0115) = 1 < 5 s; consumption up ~6–8 W; `tcs.temp_battery` (0x0407) trending to the +5 °C setpoint	☐	
3	EPS	Energise OBC heater: `EPS_POWER_ON(line_index=6)` (func 19)	`eps.pl_htr_obc` (0x0116) = 1 < 5 s; `tcs.temp_obc` (0x0406) trending; consumption up ~3–4 W	☐	
4	EPS / AOCS	Energise wheels line and boot AOCS: `EPS_POWER_ON(line_index=7)` (func 19), then `HK_ENABLE(sid=2)` (**S3.5**), then `AOCS_SET_MODE(mode=2)` (S8.1 func_id **0**) — mode **2 = DETUMBLE**	`eps.pl_aocs_wheels` (0x0117) = 1; SID 2 emitting within one 4 s interval; `aocs.mode` (0x020F) = 2; `aocs.rate_roll/pitch/yaw` (0x0204–0x0206) decreasing; consumption up ~12 W	☐	
5	AOCS	Commission sensors, then sun-point: `HK_REQUEST(sid=2)` ; wait for all-axis rates < 0.5°/s; `AOCS_SET_MODE(mode=3)` (func 0) — mode **3 = COARSE_SUN**	`aocs.mag_enabled` (0x025C) = 1, field 25–65 µT; `aocs.mode` = 3; `aocs.att_error` (0x0217) decreasing to < 5°	☐	
6	EPS / Payload	Energise payload line and go to STANDBY: `EPS_POWER_ON(line_index=3)`	`eps.pl_payload` (0x0113) = 1;	☐	

#	Owner	Action	Verification	☐ done	Note
		(func 19), `PAYLOAD_SET_MODE(mode=1)` (S8.1 func_id **26**), `HK_ENABLE(sid=5)` (S3.5)	`payload.mode` (0x0600) = 1 (STANDBY); SID 5 emitting; `payload.fpa_temp` (0x0601) valid ≈ ambient		
7	EPS / Payload	Energise cooler line and start the cooler: `EPS_POWER_ON(line_index=4)` (func 19), `FPA_COOLER(on=1)` (S8.1 func_id **43**)	`eps.pl_fpa_cooler` (0x0114) = 1 **and** `tcs.cooler_fpa` (0x040C) = 1 — both, or the cooler is not really running; `payload.fpa_temp` falling toward the −15 °C default setpoint; consumption up ~15 W	☐	
8	FD / SPACON	GO/NO-GO poll across all positions. When GO is unanimous FD **declares COMMISSIONING by telecommand**: `SYS_SET_MISSION_PHASE(phase=5, flags=0)` (S8.1 func **120**) — nothing on board advances the phase past 4	`eps.spacecraft_phase` (0x0129) = 5 (COMMISSIONING) < 5 s	☐	
9	SPACON	Log all step times, rates and line currents	recorded	☐	

End-of-pass state: - Phase (0x0129): _ ***AOCS mode (0x020F):*** _ (expect 3 COARSE_SUN) - Body rates roll/pitch/yaw (°/s): // **`aocs.att_error`: ____° - Wheels enabled (`aocs.wheels_enabled` 0x0260): ____ of 4 Wheel speeds RW1–4 (RPM, 0x0207–0x020A):** ///__ - SoC: _ **% *`tcs.temp_battery`*** **:** _ °C `payload.fpa_temp` : ____ °C - SIDs periodic: 1 ☐ 2 ☐ 3 ☐ 4 ☐ 5 ☐ 6 ☐ 11 ☐

Exit criterion met? ☐ yes ☐ no **FD signature:** ___

Worksheet WB-5.5-L005 — LEOP-005 Sun Acquisition (optional, confirm each step)

Operator (AOCS): _____

#	Action	Verification	☐ done	Note
1	Confirm DETUMBLE active and rates damping: `HK_REQUEST(sid=2)` (S3.27)	`aocs.mode` (0x020F) = 2 (DETUMBLE); rates roll/pitch/yaw: // __ °/s	☐	
2	Wait for B-dot convergence. This is **slow**: the 0.5°/s DETUMBLE exit guard takes ~1 200 s, and a full 14°/s separation tumble ~3.5e3 s. Dipole saturates at 1.6 A·m² per axis on three torquers	all axes < 0.5°/s	☐	
3	Power a star tracker: `ST1_POWER(on=1)` (S8.1 func_id **4**) or `ST2_POWER(on=1)` (func **5**); select with `ST_SELECT` (func **6**)	`aocs.st1_status` (0x0240) / `aocs.st2_status` (0x0243) = 2 (tracking) within 30 s	☐	
4	Transition AOCS to COARSE_SUN (`AOCS_SET_MODE(mode=3)` , func 0) — mode **3 = COARSE_SUN**, the sun-safe hold	`aocs.mode` = 3 within 30 s	☐	
5	Wait for sun-acquisition settling	`aocs.att_error` (0x0217) < 5° and stable	☐	
6	Verify power balance improved: `HK_REQUEST(sid=1)`	`eps.power_gen` (0x0107) > `eps.power_cons` (0x0106); margin `eps.power_margin_w` (0x0135): ____ W	☐	
7	Log final attitude error	error: ____ °	☐	

Exit criterion met? ☐ yes ☐ no

Worksheet WB-5.5-AT — Day 3 Anomaly Ticket (3 copies)

Print three. Use one per anomaly observed (planned injections or unplanned). Each ticket signed by FD.

Ticket — copy 1

Field	Entry
Ticket ID (auto-assigned)	
Pass / shift	
Date / UTC time of first indicator	
Triggering parameter and value	
FDIR rule that fired (if any)	
Severity (0–4)	
First operator to call it (position)	
First-action time	
Procedure invoked (ID)	
Recovery actions taken (verbatim from loop)	
Outcome (recovered / safe / deferred)	
Root cause hypothesis	
Open items for next shift	
FD signature	

Ticket — copy 2

Field	Entry
Ticket ID	
Pass / shift	
Date / UTC time	
Parameter and value	
FDIR rule	
Severity	
First operator	
First-action time	

Field	Entry
Procedure invoked	
Actions taken	
Outcome	
Root cause hypothesis	
Open items	
FD signature	

Ticket — copy 3

Field	Entry
Ticket ID	
Pass / shift	
Date / UTC time	
Parameter and value	
FDIR rule	
Severity	
First operator	
First-action time	
Procedure invoked	
Actions taken	
Outcome	
Root cause hypothesis	
Open items	
FD signature	

Worksheet WB-5.5-GO — Day 3 End-of-Pass GO/NO-GO Scorecard

Assessor: __ Pass: ☐ Pass 1 (Team A) ☐ Pass 2 (Team B)

Criterion	☐ achieved
OBC application SW running (`obdh.sw_image` 0x0311 = 1)	☐
Onboard time set, residual < 1.0 s	☐
Beacon mode cleared, downlink on LOW (`ttc.tm_tier` 0x0539 = 1)	☐
Antenna confirmed deployed (`ttc.antenna_deployed` 0x0520 = 1)	☐
Battery heater line ON (`eps.pl_htr_bat` 0x0115 = 1)	☐
AOCS wheels line ON, AOCS through DETUMBLE (2) into COARSE_SUN (3)	☐
Body rates < 0.5°/s on all axes; `aocs.att_error` < 5°	☐
Anomaly ticket filed for any injection	☐
Phase advanced to COMMISSIONING (5) with `SYS_SET_MISSION_PHASE` (func 120) (Pass 2 only)	☐

Score: _ */ 9 (Pass 1) or* _ / 9 (Pass 2)

Notes for debrief: _____

Worksheet WB-5.5-OBS — Day 3 Observer Worksheets (axes used: Communications + Technical)

(Print both. Each observer fills one per pass they observe.)

Observer worksheet — Communications axis

Observer: ___ Pass observed: _____

Item	Note (timestamp + what + who)
Voice-loop discipline (callouts, hot-mike)	
GO/NO-GO protocol followed	
Cross-talk incidents (count + timestamps)	
MCS chat — used for what?	
Handover: timing, content, completeness	
Silent confusion (operator did wrong thing because no one said anything)?	

Worked well: _____

Could improve: _____

Specific recommendation: _____

Observer worksheet — Technical axis

Observer: ___ Pass observed: _____

Item	Note
Procedure adherence — any skipped or out-of-order steps?	
Anomaly detection time — first indicator to first action	
Telemetry interpretation — any misreads?	
Command discipline — any unauthorised commands?	
Use of FDIR panel — did anyone watch it?	
Outcome vs pass criteria	

Worked well: _____

Could improve: _____

Specific recommendation: _____

Worksheet WB-5.5-HO — Day 3 Inter-Pass Handover Turnover Form

Use: between Pass 1 and Pass 2 (Team A → Team B). One per position; outgoing operator signs, incoming operator signs.

Print 9 copies (one per console position).

Handover — Position: ____

Field	Entry
Outgoing operator	
Incoming operator	
Current parameter values (top 3 mnemonics)	
Current procedure state (file ID + step)	
Any pinned widget?	
Open question for next shift	
Outgoing signature	
Incoming signature	
Time	

Worksheet WB-5.5-DB — Day 3 Post-Shift Debrief Reflection

Name: ___

1. What is one decision you took during the shift that, in retrospect, you would not take again? Why?

1. What is one decision that surprised you with how well it worked?

1. Which voice callout did you most struggle to make in real time?

1. What is one thing the team did better as a unit than you expected?

1. What is one specific thing you will do differently tomorrow?

Worksheet WB-5.6-PB — Day 4 Commissioning Pre-Pass-Set Briefing

Item	☐
BP-3 `commissioning_day_start.yaml` scenario loaded (instructor-authored — see the Instructor Book; it is not in the shipped `configs/eosat1/scenarios/` set)	☐
Spacecraft in phase 4 (LEOP complete), `eps.spacecraft_phase` (0x0129) = 4	☐
Position assignments posted (different team from yesterday)	☐
Pass-set timeline understood (4 sim passes in 2.5 h real time)	☐
All 13 commissioning procedures (COM-001…COM-013) loaded in the procedure index	☐
EPS confirms SoC ~90 %, `eps.pl_payload` (0x0113) = 0	☐
Thermal confirms FPA at its uncooled equilibrium, `payload.fpa_temp` (0x0601) +8 to +14 °C	☐
Payload Ops has reviewed COM-009/-010/-011/-012 in detail	☐
First Light target candidate selected (FDS)	☐
Compressed-time speed multiplier set (instructor)	☐

Worksheet WB-5.6-C001 — COM-001 EPS Checkout (confirm each step)

#	Action	Expected	☐	Note
1	Baseline bus voltage — `HK_REQUEST(sid=1)` (S3.27)	`eps.bus_voltage` (0x0105) ∈ [27.5, 28.5] V	☐	V: ____
2	Battery V & SoC — `GET_PARAM(param_id=0x0100)`, `GET_PARAM(param_id=0x0101)` (S20.3)	`eps.bat_voltage` ∈ [26.5, 29.2] V (8S pack: 21.5 V at 0 %, 26.4 V nominal, 29.2 V at 100 %); `eps.bat_soc` > 70 %	☐	V: _ ***SoC:*** _
3	Solar array sunlit performance	`eps.power_gen` (0x0107) ∈ [40, 65] W (peak whole-spacecraft 88.6 W; wings 0.120 m² each at 267.9 W/m² EOL)	☐	____ W
4	Eclipse discharge monitoring	`power_gen` ~0 W < 30 s of entry; SoC drop < 1.5 %/min; `bus_voltage` > 27.0 V mid-eclipse	☐	
5	Eclipse exit charge recovery	`power_gen` > 35 W < 60 s; SoC rising < 120 s	☐	
6	Power budget verification — orbit-average gen > cons	nominal orbit-average load ≈ **28.7 W** against ≈ 38 W generation at worst beta (margin +9.3 W, ratio 1.33); end-of-eclipse SoC > 50 % on the **100 Wh** pack	☐	margin: ____ W
7	Bus regulation under load transient — `PAYLOAD_SET_MODE(mode=1)` then `(mode=0)` (S8.1 func_id **26**)	bus stays in [27.0, 29.0] V, returns to [27.5, 28.5] V	☐	min V: ____

Exit? ☐ yes ☐ no **GO COM-002:** ☐

Worksheet WB-5.6-C002 — COM-002 TCS Verification (confirm each step)

#	Action	Expected	☐	Note
1	Thermal HK — `HK_REQUEST(sid=3)` (S3.27; SID 3 = TCS, 60 s cadence)	`tcs.temp_obc` (0x0406) ∈ [−5, +45] °C; `tcs.temp_battery` (0x0407) ∈ [+5, +35] °C; `tcs.temp_fpa` (0x0408) reported	☐	obc: **bat:** fpa:__
2	Exercise each heater individually for 5 min — `HEATER_BATTERY(on=1)` (S8.1 func **40**), `HEATER_OBC(on=1)` (func **41**)	duty `tcs.htr_duty_battery` (0x040E) / `tcs.htr_duty_obc` (0x040F) < 80 %; zone reaches setpoint	☐	
3	Verify thermostatic control of battery zone (setpoint `tcs.setpoint_bat` 0x0330; restore with `HEATER_AUTO_MODE`, func **45**)	hysteresis between on/off observed on `tcs.htr_battery` (0x040A)	☐	
4	Confirm radiator performance via TCS zone response in eclipse	zones stable, no runaway	☐	

Exit? ☐ yes ☐ no **GO COM-005:** ☐

Worksheet WB-5.6-C005 — COM-005 AOCS Mode Transitions (confirm each step)

Mode codes (`aocs/aocs_mode_codes.yaml` , canonical): 0 OFF, 1 SAFE_BOOT, 2 DETUMBLE, 3 COARSE_SUN, 4 NOMINAL, 5 FINE_POINT, 6 SLEW, 7 DESAT, 8 ECLIPSE, 9 INERTIAL. Every transition is `AOCS_SET_MODE(mode=N)` (S8.1 **func_id 0**).

#	Action	Expected	☐	Note
1	Confirm starting mode = COARSE_SUN (**3**) after LEOP-007	`aocs.mode` (0x020F) = 3; `aocs.att_error` (0x0217) < 5°; `aocs.rate_roll` (0x0204) < 0.1°/s	☐	rates roll/ pitch/ yaw: // __
2	Transition to NOMINAL (`AOCS_SET_MODE(mode=4)`)	mode = 4 within 30 s; att_error < 1.0° within 120 s; all three rates < 0.05°/s	☐	
3	Read the FINE precondition mask: `AOCS_FINE_READY_CHECK` (S8.1 func **89**)	`aocs.fine_ready` (0x02C5) = 0. Bit 1 = *fewer than three* wheels — a 3-of-4 set still flies FINE, flagged at `aocs.fine_degraded` (0x0299)	☐	mask: ____
4	Transition to FINE_POINT (`AOCS_SET_MODE(mode=5)`)	mode = 5 within 30 s; att_error < 0.1° within 300 s; all rates < 0.01°/s	☐	error: ____ °
5	Step back down: 5 → 4 → 3 → 2 (DETUMBLE) → 3	each `aocs.mode` readback matches the command; cross-check `aocs.commanded_mode` (0x0297) and `aocs.mode_demoted` (0x0298)	☐	

Exit? ☐ yes ☐ no

Worksheet WB-5.6-C006 — COM-006 TTC Link Verification (confirm each step)

The TM rate ladder has exactly three rungs: **BEACON 1 kbps / LOW 16 kbps / HIGH 128 kbps**. BEACON is not selectable with `TTC_SET_DATA_RATE` — its baseline comes from the LEOP-001 pass.

#	Action	Expected	☐	Note
0	Configure the S-band chain: `EPS_POWER_ON(line_index=2)` (func 19), `TTC_PA_ON` (func **66**), `TTC_SET_BEACON_MODE(on=0)` (func **70**), `TTC_DEPLOY_ANTENNA` (func **69**)	`eps.pl_ttc_tx` (0x0112) = 1; `ttc.pa_on` (0x0516) = 1; `ttc.beacon_mode` (0x0521) = 0; `ttc.antenna_deployed` (0x0520) = 1; `ttc.tx_active` (0x0525) = 1 (else read `ttc.tx_gate_status` 0x052B)	☐	
1	Baseline the primary transponder at LOW: `TTC_SET_DATA_RATE(rate=0)` (S8.1 func **65**)	`ttc.link_status` (0x0501) = **2** (LOCKED); `ttc.tm_data_rate` (0x0506) = 16000 bps; `ttc.tm_tier` (0x0539) = 1; `ttc.link_margin` (0x0503) > 6 dB (expect **55–59 dB** at 30° — this link is not power-limited)	☐	____ dB
2	Step to HIGH: `TTC_SET_DOWNLINK_MODE(on=1)` (func **79**) then `TTC_SET_DATA_RATE(rate=1)` (func 65)	`tm_data_rate` = 128000 bps; `tm_tier` = 2; `ttc.tm_symbol_rate` (0x053B) = 64000 Bd (QPSK); margin expect **45–48 dB** at 30°	☐	____ dB
3	Compute the rung step: margin(LOW) − margin(HIGH)	**10.5 ± 0.5 dB**. Anything else is a modelling error, not a link problem	☐	step: ____ dB
4	Below 10° elevation (`ttc.contact_elevation` 0x050A), attempt HIGH again	Command **rejected**, S1.8 error 0x0007 naming the elevation gate; rate stays 16000 bps. This is correct behaviour, not an anomaly	☐	
5	Safe the link: `TTC_SET_DATA_RATE(rate=0)`, `TTC_SET_DOWNLINK_MODE(on=0)`, `TTC_SWITCH_PRIMARY` (func **63**)	`tm_tier` = 1, `ttc.tm_tier_cmd` (0x053A) = 1, `ttc.downlink_mode` (0x053C) = 0. Antenna stays deployed and the PA stays ON	☐	
6	Record margins at both operational rungs in the shift log	recorded	☐	

Exit? ☐ yes ☐ no

Worksheet WB-5.6-C007 — COM-007 OBDH Checkout (confirm each step)

#	Action	Expected	☐	Note
1	Exercise the watchdog: `OBC_SET_WATCHDOG_PERIOD(period_ticks=N)` (S8.1 func **58**), `OBC_WATCHDOG_ENABLE` (func **59**) / `OBC_WATCHDOG_DISABLE` (func **60**). (There is no `OBC_WATCHDOG_KICK` command)	no reboot; `obdh.reboot_count` (0x030A) unchanged, `obdh.uptime` (0x0308) still incrementing	☐	
2	Read and verify mass-memory usage — `HK_REQUEST(sid=4)` (S3.27)	`obdh.mmm_used` (0x0303, %) as expected; payload store `payload.mem_used_mb` (0x060B) of `payload.mem_total_mb` (0x060A)	☐	used: ____ MB
3	Test an S6 memory read on a known region: `MEM_DUMP(memory_id, address, length)` (**S6.5**); optionally `MEM_CHECK` (S6.9)	data matches the memory map	☐	
4	Verify bus A and bus B health	`obdh.bus_a_status` (0x030F) = 0 and `obdh.bus_b_status` (0x0310) = 0 (0 = OK); active bus at `obdh.active_bus` (0x030E)	☐	

Exit? ☐ yes ☐ no

Worksheet WB-5.6-C008 — COM-008 FDIR Configuration (confirm each step)

There is **no** FDIR table-load telecommand. The rule set is static configuration (`subsystems/fdir.yaml` plus the S12 limit definitions and S19 event-action rules); commissioning verifies and exercises it, it does not upload it.

Keep the two ladders distinct: **FDIR severity 0–4** (how bad the event is) and **autonomy level 0–3** (who is allowed to act — 0 hardware, 1 subsystem SW, 2 OBC FDIR, 3 ground). They are not the same scale and must never be conflated on the loop.

#	Action	Expected	☐	Note
1	Read the FDIR state — `GET_PARAM(param_id=0x0700)` (S20.3)	`fdir.level` = 0 (0 nominal, 1 caution, 2 warning, 3 critical); `fdir.active_alarms` (0x0701) = 0	☐	
2	Review the rule set on the FDIR panel against `subsystems/fdir.yaml`	key thresholds confirmed: `eps.bus_voltage` < **24.5 V**, `eps.bat_soc` < 20 % / < 15 %, `aocs.att_error` > 5° sustained 420 s, `obdh.temp` > 65 °C	☐	count: ____ rules
3	Enable/verify the S12 monitoring service — `MON_ENABLE` (**S12.1**), `MON_REPORT_DEFS` (S12.12)	definition list returned; alarms visible in the MCS	☐	
4	Inject a benign test condition by narrowing one limit — `MON_ADD_DEF` (S12.6) or `SET_PARAM` (S20.1) — then restore it	the relevant rule triggers and clears; `fdir.level` returns to 0	☐	rule: ____
5	Verify no autonomous action fired (test condition kept below the action threshold)	no mode change, no line opened, no safe-mode entry recorded	☐	

Exit? ☐ yes ☐ no

Worksheet WB-5.6-C009 — COM-009 Payload Power-On (confirm each step)

#	Action	Expected	☐	Note
1	FD GO confirmed	recorded	☐	
2	Energise the payload line: `EPS_POWER_ON(line_index=3)` (S8.1 func_id **19**)	`eps.pl_payload` (0x0113) = 1 < 5 s; `eps.line_current_3` (0x011B) within envelope — standby is 2.4 W, i.e. ≈ **0.09 A** at 28 V (the overcurrent trip on this line is 2.5 A)	☐	I: ____ A
3	Command STANDBY: `PAYLOAD_SET_MODE(mode=1)` (S8.1 func_id **26**)	`payload.mode` (0x0600) = 1 (STANDBY). Modes: 0 OFF, 1 STANDBY, 2 IMAGING, **3 CALIBRATION**	☐	
4	Enable payload HK: `HK_ENABLE(sid=5)` (**S3.5**), then `HK_REQUEST(sid=5)` (S3.27)	SID 5 periodic within one 8 s interval; `payload.imager_status` (0x0611) no fault bits; `payload.fpa_temp` (0x0601) valid, +8 to +14 °C	☐	
5	Verify storage initialised	`payload.mem_used_mb` (0x060B) of `payload.mem_total_mb` (0x060A) = 20000 MB; `payload.product_count` (0x0648)	☐	____ MB

Exit? ☐ yes ☐ no **GO COM-010:** ☐

Worksheet WB-5.6-C010 — COM-010 FPA Cooler Activation (confirm each step; spans eclipse)

The FPA cooler is a **miniature Stirling-class cryocooler** running at about 23 % of Carnot (`fpa_cooler_carnot_fraction` 0.23) — not a Peltier stack. Its authority falls as the lift grows, so the achievable floor moves with the sink temperature and the imaging load. Default setpoint **−15 °C**; commandable envelope **−20 °C to 0 °C** (`PAYLOAD_COOLER_SETPOINT`, S8.1 func **36**, argument is a uint16 in **centi-kelvin** — −15 °C is 0x64D7). The loop settles in about 10 minutes and never overshoots the setpoint.

#	Action	Expected	☐	Note
1	Record the pre-activation state; cooler OFF	`payload.fpa_temp` (0x0601) at the uncooled equilibrium +8 to +14 °C; `payload.mode` (0x0600) = 1	☐	T_fpa: ____ °C
2	Enable the cooler: `FPA_COOLER(on=1)` (S8.1 func_id **43**)	`tcs.cooler_fpa` (0x040C) = 1 **and** `eps.pl_fpa_cooler` (0x0114) = 1 — both, or the cooler is not actually running	☐	
3	Verify the power draw — `HK_REQUEST(sid=1)` (S3.27)	`eps.power_cons` (0x0106) up **14–16 W**; `payload.cooler_pwr` (0x0602) ≈ 15 W; `eps.bus_voltage` (0x0105) > 27.0 V	☐	____ W
4	Monitor the ramp (exponential, not linear — a slower reading at T+5 than at T+2 is expected)	T+2 min < +4 °C; T+5 min < −8 °C	☐	T at 5 min: __ °C
5	Monitor settling	T+10 min ∈ [−16.0, −14.0] °C; T+15 min ∈ [−16.0, −15.0] °C. A reading below −16.5 °C is a **control fault**, not an over-performing cooler	☐	T at 15 min: __ °C
6	Verify readiness and margin	`payload.fpa_ready` (0x0610) = 1 (60 s hysteresis); `payload.fpa_setpoint` (0x0631) = −15.0; `payload.cooler_margin_k` (0x064B) > 0 (expect ≈ +5 K); floor at `payload.cooler_floor_c` (0x064A)	☐	margin: ____ K
7	Hold through eclipse	T_fpa stays in [−16.5, −14.5] °C; bus > 27.0 V	☐	

If `cooler_margin_k` ≤ 0: the cooler physically cannot reach the commanded setpoint at the present sink and load. The answer is a **warmer setpoint**, not more waiting.

Exit? ☐ yes ☐ no **GO COM-011:** ☐

Worksheet WB-5.6-FPA — FPA Cool-Down Monitor Sheet

Use: during COM-010. The loop closes in about 10 minutes, so record `payload.fpa_temp` (0x0601) every minute for the first 15 minutes, then every 5 minutes through the stability hold.

Elapsed	T_fpa (°C)	`payload.cooler_pwr` 0x0602 (W)	Eclipse?	Note
0:00			☐	start, +8 to +14 °C
0:01			☐	
0:02			☐	expect < +4 °C
0:03			☐	
0:04			☐	
0:05			☐	expect < −8 °C
0:06			☐	
0:07			☐	
0:08			☐	
0:09			☐	
0:10			☐	expect −16.0 to −14.0 °C
0:11			☐	
0:12			☐	
0:13			☐	
0:14			☐	
0:15			☐	setpoint band expected by here
0:20			☐	stability hold
0:25			☐	
0:30			☐	
0:35			☐	
0:40			☐	
0:45			☐	p-p variation < 1.0 °C over the hold

`payload.fpa_ready` (0x0610) set at elapsed: __ min `payload.cooler_margin_k` (0x064B) at setpoint: _____ K

Worksheet WB-5.6-C011 — COM-011 Payload Calibration (confirm each step)

Radiometric calibration is its own payload mode: `payload.mode` (0x0600) = **3 (CALIBRATION)**. Dark calibration no longer requires IMAGING mode. If a calibration is refused, read the reason code — 0x0024 refused because IMAGING, 0x0025 FPA not ready, 0x0026 shutter in the wrong position, 0x0027 lamp in the wrong state, 0x0028 capture with the shutter closed.

#	Action	Expected	☐	Note
1	Close the shutter: `PAYLOAD_SET_SHUTTER(position=0)` (S8.1 func **119**)	`payload.shutter_pos` (0x063C) = 0 (closed)	☐	
2	Dark set: `PAYLOAD_SET_MODE(mode=1)` (func 26), `PAYLOAD_CAL_LAMP(on=0)` (func **95**), `PAYLOAD_START_CALIBRATION(mode=0)` (func **37**, 0 = DARK)	`payload.mode` = 3 (CALIBRATION); `payload.calibration_active` (0x062A) = 1; on completion S5 event **0x060B CALIBRATION_COMPLETE**; `payload.dark_frame_cnt` (0x0642) > 0	☐	frames: ____
3	Flat set: `PAYLOAD_SET_SHUTTER(position=1)` (func 119), `PAYLOAD_CAL_LAMP(on=1)` (func 95), `PAYLOAD_START_CALIBRATION(mode=1)` (func 37, 1 = FLAT)	`payload.cal_lamp_on` (0x0615) = 1; `payload.bias_frame_cnt` (0x0643) > 0; lamp back OFF afterwards	☐	frames: ____
4	Verify the correction set is valid across the four bands	`payload.cal_valid_mask` (0x0646) bits 0–3 (blue/green/red/nir) all set; `payload.snr` (0x0616) recorded	☐	mask: ____
5	Calibration-site overpass: `AOCS_FINE_READY_CHECK` (func 89), then `AOCS_SET_MODE(mode=5)` (func 0) — FINE_POINT is mode **5**	`aocs.fine_ready` (0x02C5) = 0; `aocs.mode` (0x020F) = 5	☐	
6	Return: `PAYLOAD_SET_MODE(mode=1)` STANDBY, shutter OPEN, `AOCS_SET_MODE(mode=4)` NOMINAL	`payload.mode` = 1; `payload.shutter_pos` = 1; `aocs.mode` = 4	☐	

Exit? ☐ yes ☐ no **GO COM-012:** ☐

Worksheet WB-5.6-C012 — COM-012 First Light (confirm each step)

#	Action	Expected	☐	Note
1	FD GO for First Light	recorded	☐	
2	AOCS to FINE_POINT: `AOCS_FINE_READY_CHECK` (func 89), `AOCS_SET_MODE(mode=5)` (func 0)	`aocs.fine_ready` (0x02C5) = 0; `aocs.mode` (0x020F) = 5	☐	
3	Verify the FPA is at temperature	`payload.fpa_ready` (0x0610) = 1; `payload.fpa_temp` (0x0601) ∈ [−16.5, −14.5] °C	☐	T: ____ °C
4	Set the four band integration times: `PAYLOAD_SET_INTEGRATION_TIME(blue_ms=, green_ms=, red_ms=, nir_ms=)` (S8.1 func **34**)	readback `payload.int_time_blue/green/red/nir` (0x0638–0x063B)	☐	ms: ///
5	Command IMAGING and capture: `PAYLOAD_SET_MODE(mode=2)` (func **26**), then `PAYLOAD_CAPTURE(scene_id=1, lines=2048)` (S8.1 func **28**)	S1.1 accept + S1.7 complete; S5 event 0x0613 IMAGE_CAPTURED. 2048 lines at 244 Hz = 8.4 s = 60 km along-track	☐	scene: ____
6	Verify the product stored	`payload.mem_used_mb` (0x060B) increased; `payload.product_count` (0x0648) +1; size ≈ **4 MB** (`image_size_mb` , the courseware abstraction)	☐	____ MB
7	Return to STANDBY and start the transfer: `PAYLOAD_SET_MODE(mode=1)` , `PAYLOAD_INITIATE_TRANSFER(scene_id=1)` (S8.1 func **94**)	`payload.transfer_active` (0x062D) = 1; S5 event 0x0610 TRANSFER_START, then 0x0611 TRANSFER_COMPLETE	☐	rx time: _____
8	Quality assessment (next pass if needed)	mean DN, contrast, geo-registration; `payload.last_scene_quality` (0x060D); `aocs.geoloc_err_m` (0x02C7)	☐	quality: ____

Exit? ☐ yes ☐ no **First Light declared?** ☐ yes ☐ formally deferred

If deferred, justification: _____

Worksheet WB-5.6-FL — First Light Image Assessment

Field	Entry
Image ID	
Target name	
Acquisition UTC	
AOCS mode at acquisition	
FPA temperature at acquisition (°C)	
Integration time per band (ms), bands at 443/560/665/865 nm	/ / /
Image size (MB) — expect ~4	
Ground swath (km) — `payload.swath_width_km` (0x0619)	240
GSD (m) — `payload.gsd_m` (0x0628)	29.3
Mean DN per band	
Cloud cover (%)	
Geo-registration error (m) — `aocs.geoloc_err_m` (0x02C7)	
Quality verdict	☐ science usable ☐ marginal ☐ retake
Customer bulletin drafted? (Mission Mgr)	☐ yes

Worksheet WB-5.6-MA — Day 4 Multi-Anomaly Response Form

Use: if the instructor injects more than one anomaly in a pass-set. One row per anomaly; signed by FD at end of recovery.

Anomaly #	First indicator time	Position to flag it	Procedure invoked	Recovery time (s)	Outcome	FD sig
1						
2						
3						

Order of priority taken: _____

Conflicts encountered (procedure A requires X, procedure B requires not-X):

How conflicts resolved: _____

Worksheet WB-5.6-GO — Day 4 Pass-Set Criteria Scorecard

Assessor: __ Pass-set: ☐ Pass-set 1 (Team A) ☐ Pass-set 2 (Team B)

Criterion	☐ achieved
Phase advanced to COMMISSIONING (5) — `SYS_SET_MISSION_PHASE` (func 120) sent and verified	☐
EPS checkout complete, no faults	☐
AOCS NOMINAL (mode 4) achieved, `aocs.att_error` < 1.0°; FINE_POINT (5) demonstrated at < 0.1°	☐
FDIR rule set verified and exercised on the FDIR panel	☐
Payload powered ON (`eps.pl_payload` 0x0113 = 1, `payload.mode` = 1)	☐
FPA cooler at its commanded setpoint (−15 °C default), `payload.fpa_ready` (0x0610) = 1, `cooler_margin_k` > 0	☐
First Light image acquired or formally deferred	☐
Anomaly tickets filed for all injections	☐

Score: ____ / 8

Worksheet WB-5.6-OBS — Day 4 Observer Worksheets (axes: Team + Space usage)

Observer worksheet — Team axis

Observer: _____

Item	Note
Did each position know its role?	
Was the FD authoritative?	
Anyone over-loaded or under-utilised?	
Did anyone do another position's job?	
Was fatigue handled (breaks, water)?	
How did the team handle disagreement?	

Worked well: ___ ***Could improve:*** _____

Specific recommendation: _____

Observer worksheet — Space-usage axis

Observer: _____

Item	Note
Console layout — could every position see what they needed?	
Line of sight — could FD see all positions?	
Distractions (phones, side talk, music)?	
Lighting, temperature, noise — issues?	
Movement around the room (procedures, water)	
Physical comfort — chairs, monitors, posture	

Worked well: ___ ***Could improve:*** _____

Specific recommendation: _____

Worksheet WB-5.6-HO — Day 4 Inter-Pass Handover Turnover Form

(Same format as WB-5.5-HO; print 9 copies, one per console position.)

Handover — Position: ____

Field	Entry
Outgoing operator	
Incoming operator	
Current parameter values (top 3)	
Current procedure state (file ID + step)	
Any pinned widget?	
FDIR rules currently active (acknowledged)	
Open question for next shift	
Outgoing signature	
Incoming signature	
Time	

Worksheet WB-5.6-DB — Day 4 Post-Shift Debrief Reflection

(Same five-question prompt as WB-5.5-DB.)

1. What is one decision you took during the shift that, in retrospect, you would not take again? Why?

1. What is one decision that surprised you with how well it worked?

1. Which procedure step did you most struggle to execute in real time?

1. What is one thing the team did better as a unit than you expected?

1. What is one specific thing you will take into your future engineering work?

Worksheet WB-5.7-SA — Day 5 Personal Self-Assessment

Name: ___

For each row, score yourself 1–3 (1 = developing, 2 = competent, 3 = strong). Be honest — the cohort average is 2.0.

Row	Competency	Score (1–3)	Evidence
1	Voice-loop discipline		
2	Procedure compliance		
3	Telemetry interpretation		
4	Anomaly response timing		
5	Handover quality		
6	Position rotation completion		
7	Pass-plan quality (Day 1 exercise)		
8	Quiz score / 15		
9	Debrief contribution		

Total: ____ / 27 **Honours threshold:** 22

Three things I am proudest of this week:

1.
2.
3.

Three things I want to keep working on:

1.
2.
3.

Worksheet WB-5.7-LL — Cohort Lessons-Learned Contribution

Name: ___

Each cohort member adds at least three lessons to the cohort register. The register is read by next year's cohort.

Lesson 1

Title: _____

What we observed: _____

Why it matters: _____

Recommendation: _____

Lesson 2

Title: _____

What we observed: _____

Why it matters: _____

Recommendation: _____

Lesson 3

Title: _____

What we observed: _____

Why it matters: _____

Recommendation: _____

Worksheet WB-5.7-WI — "What If" Branch Reflection

If the instructor ran a branch-replay, fill this in.

Original decision: _____

Alternative decision tried in replay: _____

What changed in the next 15 simulated minutes:

Which decision would you take next time? Why?

Worksheet WB-5.7-FR — Final Mission Review Prep

The cohort produces a 4–6 page CFMR (final mission review) document on Day 5 afternoon. Use this worksheet to collect material.

Section 1 — Mission objectives

Were the Week 3 objectives met? (Day 5, list each with PASS/FAIL.)

Objective	PASS / FAIL	Evidence
Operated all 9 positions for at least one full pass	☐ / ☐	
Ran a planning cycle end-to-end	☐ / ☐	
Executed one nominal procedure as lead	☐ / ☐	
Executed one contingency procedure as lead	☐ / ☐	
Critiqued a peer team's shift	☐ / ☐	
Produced a shift log and anomaly ticket	☐ / ☐	

Section 2 — Telemetry summary across the week

Three most-watched parameters and their typical values:

1. _____
2. _____
3. _____

Section 3 — Anomalies log

Date	Anomaly	Procedure	Outcome	Root cause
Wed				
Wed				
Thu				
Thu				

Section 4 — Compliance with ECSS-E-ST-70-11C

Tick each clause you can demonstrate compliance with from the cohort's shift logs and worksheets.

Clause area	☐
5.1 Operability principles	☐
5.2 Operations preparation	☐
5.3 Procedure design	☐
5.4 Anomaly management	☐

Clause area	☐
5.5 Reporting and review	☐

Section 5 — Crew performance self-assessment

(Aggregated from WB-5.7-SA across the cohort.)

Position	Cohort mean score (1–3)
FD	
SPACON	
AOCS	
EPS	
Thermal	
TT&C	
Payload	
FDS	
Ground Seg	
Mission Manager	

Section 6 — Recommendations for next cycle

1.
2.
3.

Worksheet WB-5.7-CE — Course Evaluation

Anonymous. Drop in the box by the door.

Item	1 (poor)	2	3	4	5 (excellent)
Lesson-plan structure (Mon/Tue)	☐	☐	☐	☐	☐
Simulator realism (Wed/Thu)	☐	☐	☐	☐	☐
Instructor coaching	☐	☐	☐	☐	☐
Cyberrange facility	☐	☐	☐	☐	☐
Workbook usefulness	☐	☐	☐	☐	☐
Pace (too slow / right / too fast)	slow ☐	☐	right ☐	☐	fast ☐
Confidence to operate a real shift now	low ☐	☐	medium ☐	☐	high ☐

One thing you would change for next year:

One thing you would keep exactly as it is:

Appendix WB-A — Contingency confirm-each-step (Week-3-likely)

The contingency confirm pages below are reference: they sit in this workbook so that if the instructor injects a scenario, the team has a ready-printed page to follow on the console.

CTG-001 Under-Voltage Load Shed

#	Owner	Action	Verification	☐	Note
1	EPS	Confirm `eps.bus_voltage` (0x0105) below the FDIR undervoltage threshold **24.5 V** (in-model shed starts at 25.0 V; bus warning 27.0 V, bus critical 25.0 V)	recorded	☐	V: ____
2	EPS	Report to FD; recommend priority-1 shed	recommendation logged	☐	
3	FD	GO for load shed	granted	☐	
4	EPS	Priority 1 — payload OFF: `PAYLOAD_SET_MODE(mode=0)` (S8.1 func **26**)	`payload.mode` (0x0600) = 0	☐	
5	EPS	Priority 2 — cooler OFF: `FPA_COOLER(on=0)` (func **43**)	`tcs.cooler_fpa` (0x040C) = 0	☐	
6	EPS	Priority 3 — PA off: `TTC_PA_OFF` (func **67**). Note the S-band TM stops but the **UHF beacon survives** — it is a separate radio on the non-switchable `ttc_rx` line, so HK SID 11 keeps arriving	`ttc.pa_on` (0x0516) = 0, `ttc.beacon_active` (0x0527) still 1	☐	
7	EPS	Priority 4 — wheels line off: `EPS_POWER_OFF(line_index=7)` (func **20**)	`eps.pl_aocs_wheels` (0x0117) = 0	☐	
8	EPS	(Last resort) staged shed `EPS_EMERGENCY_LOAD_SHED(stage=N)` (S8.1 func **25**, stages 1–3) or `OBC_SET_MODE(mode=2)` EMERGENCY (func **50**)	`eps.load_shed_stage` (0x0134)	☐	
9	EPS	Verify SoC stops dropping; bus recovers above 25.5 V (shed threshold + 0.5 V hysteresis; auto-restore then walks the lines back one per tick, in reverse shed order)	recovery time: ____ s	☐	
10	SPACON	Acknowledge FDIR rule; log	logged	☐	
11	EPS	File anomaly ticket	ticket #: ____	☐	

CTG-002 AOCS Anomaly Recovery

#	Action	Verification	☐	Note
1	Read which sensor / actuator is faulted from the FDIR panel; `HK_REQUEST(sid=2)` (S3.27)	identified: _____	☐	
2	Command the sun-safe hold: `AOCS_SET_MODE(mode=3)` (S8.1 func **0**) — mode **3 = COARSE_SUN**. (There is no SAFE_POINT mode; mode 1 is SAFE_BOOT, a post-boot propagate-only state, not a recovery target)	`aocs.mode` (0x020F) = 3	☐	
3	Verify the spacecraft is stable	rates (0x0204–0x0206) < 0.05°/s; `aocs.att_error` (0x0217) < 5°; no further alarm	☐	
4	If momentum is the problem, dump it: `AOCS_DESATURATE` (S8.1 func **1**) — DESAT is mode 7. Budget **20–50 min** for a full dump; imaging is inhibited and the spacecraft returns to its pre-desat mode	`aocs.h_body_mag` (0x02B9) falling; `aocs.desat_ineffective` (0x029C) = 0	☐	
5	Follow the specific sub-procedure for the failed component (CTG-007 wheel / CTG-008 star tracker)	sub-procedure ID: ____	☐	
6	Restore NOMINAL: `AOCS_SET_MODE(mode=4)` ; escalate to DETUMBLE (`mode=2`) only on multi-wheel failure	`aocs.mode` = 4	☐	
7	File anomaly ticket	ticket #: ____	☐	

CTG-007 Reaction Wheel Anomaly

EO-Sat-1 flies **four** wheels in a pyramid. Speeds are READ-ONLY at `aocs.rw1..rw4_speed` (0x0207–0x020A); temps 0x0218–0x021B; currents 0x0250–0x0253; bearing health 0x0284–0x0287. Speed is commanded with `RW_SET_SPEED_BIAS` (S8.1 func **8** — the bias is **added** to the current speed) and `AOCS_RW_RAMP_DOWN` (func **14**, `wheel_idx` 255 = all).

#	Action	Verification	☐	Note
1	Identify the affected wheel (index 0–3) from speed, current and bearing health	wheel: ____	☐	
2	Disable it: `AOCS_DISABLE_WHEEL(wheel_idx=N)` (S8.1 func **2**)	`aocs.rwN_enabled` (0x0254–0x0257) = 0; `aocs.wheels_enabled` (0x0260) = 3	☐	
3	Desaturate the remaining set: `AOCS_DESATURATE` (func **1**)	`aocs.h_body_mag` (0x02B9) falling; allow **20–50 min**	☐	
4		`aocs.att_error` (0x0217): ____°; `aocs.fine_degraded`	☐	

#	Action	Verification	☐	Note
	Verify pointing on three wheels. A 3-of-4 subset still spans R^3, so **FINE_POINT still works** — flagged, not refused	(0x0299) = 1; `aocs.fine_ready` (0x02C5) bit 1 clear		
5	If pointing acceptable, stay in NOMINAL (`AOCS_SET_MODE(mode=4)`); else fall back to COARSE_SUN (**3**)	mode: ____	☐	
6	File anomaly ticket	ticket #: ____	☐	

CTG-008 Star Tracker Failure

EO-Sat-1 carries **two** star trackers, ST1 and ST2 — this is a redundancy switch, not a single-head loss. There is no `AOCS_EST_SET` command and no `ss_imu` source name: the estimator source is the read-only bitmask `aocs.est_source` (**0x02C1**; bit0 ST, bit1 CSS, bit2 MAG, bit3 gyro), and the only estimator command is `AOCS_EST_RESET` (S8.1 func **91**, `mode` 0 = cold, 1 = warm).

#	Action	Verification	☐	Note
1	Confirm the failure: `HK_REQUEST(sid=2)` (S3.27)	`aocs.st1_status` (0x0240) = 3 (blind) or 4 (failed); rising `aocs.est_innov_rej_count` (0x02C2)	☐	ST1: __ ST2: __
2	Bring up the other head: `ST2_POWER(on=1)` (S8.1 func **5**)	`aocs.st2_status` (0x0243) = 2 (tracking) within its boot time	☐	
3	Select it: `ST_SELECT(unit=2)` (S8.1 func **6**)	`aocs.st_selected` (0x0244) updated; `aocs.est_source` (0x02C1) bit 0 still set	☐	
4	Isolate the failed head: `ST1_POWER(on=0)` (S8.1 func **4**)	`aocs.st1_status` = 0 (off)	☐	
5	If the attitude solution has diverged, re-initialise it: `AOCS_EST_RESET(mode=1)` (S8.1 func **91**, warm) — or `mode=0` for a cold reset after a time jump	`aocs.knowledge_err_3sig` (0x02A7) settling; innovation rejections stop climbing	☐	
6	If **both** heads are lost, fall back to the sun-safe hold: `AOCS_SET_MODE(mode=3)` COARSE_SUN, and `PAYLOAD_SET_MODE(mode=0)`	`aocs.mode` (0x020F) = 3; `aocs.est_source` shows CSS+MAG+gyro, ST bit clear	☐	mode: ____
7	If a head recovers, restore NOMINAL (`AOCS_SET_MODE(mode=4)`); else hold COARSE_SUN	mode: ____	☐	
8	File anomaly ticket	ticket #: ____	☐	

CTG-010 OBDH Watchdog Recovery

#	Action	Verification	☐	Note
1	Confirm the reboot: `HK_REQUEST(sid=4)` (S3.27)	`obdh.reboot_count` (0x030A) incremented; `obdh.last_reboot_cause` (0x0316) = 1 (watchdog); `obdh.uptime` (0x0308) near zero	☐	timestamp: _____
2	If the spacecraft came back in the bootloader (`obdh.sw_image` 0x0311 = 0), re-boot the application: `OBC_BOOT_APP` (S8.1 func **55**)	`obdh.sw_image` = 1 within 15 s	☐	
3	Re-enable housekeeping — application boot re-applies `default_enabled_sids: [1, 11]`, so **every other SID is off again**: `HK_ENABLE(sid=2/3/4/5/6)` (**S3.5**)	all required SIDs periodic again	☐	
4	Re-synchronise time: `SET_TIME(cuc_seconds=T_corrected)` (S9.1), verify with `REQUEST_TIME` (S9.2)	residual < 1.0 s	☐	
5	Verify OBC mode; restore NOMINAL if in SAFE: `OBC_SET_MODE(mode=0)` (S8.1 func **50**)	`obdh.mode` (0x0300) = 0	☐	mode: ____
6	Re-apply settings lost in the reset (heater setpoints, S12 limit overrides, TM tier, beacon mode)	settings restored	☐	
7	File anomaly ticket	ticket #: ____	☐	

CTG-012 Overcurrent Response

Line indices run **0–8**: 0 `obc`, 1 `ttc_rx`, 2 `ttc_tx`, 3 `payload`, 4 `fpa_cooler`, 5 `htr_bat`, 6 `htr_obc`, 7 `aocs_wheels`, 8 `prop`.

#	Action	Verification	☐	Note
1	Identify the tripped line from `eps.oc_trip_flags` (0x010D) and the per-line currents `eps.line_current_0..7` (0x0118–0x011F; line 8 is 0x014B)	line: _____	☐	I: ____ A
2	Report to FD; note that the FDIR autonomous action has usually already opened the line	recorded	☐	
3	FD GO	granted	☐	
4	Ensure the line is open: `EPS_POWER_OFF(line_index=N)` (S8.1 func **20**)	`eps.pl_*` status bit = 0	☐	
5	Verify the line current is 0	I: ____ A	☐	
6			☐	

#	Action	Verification	☐	Note
	Only after the fault is understood, clear the latch and re-enable: `EPS_RESET_OC_FLAG(line_index=N)` (S8.1 func **21**), then `EPS_POWER_ON(line_index=N)` (func **19**)	`eps.oc_trip_flags` bit clear; line current back within envelope		
7	File anomaly ticket; root cause for next shift	ticket #: ____	☐	

CTG-015 Corrupted Image Recovery

#	Action	Verification	☐	Note
1	Confirm the S5 event **0x0607 IMAGE_CHECKSUM_ERROR** and read `payload.checksum_errors` (0x0609)	event recorded	☐	count: ____
2	Check the FPA: `HK_REQUEST(sid=5)` and `(sid=3)` (S3.27)	`payload.fpa_temp` (0x0601) in band and `payload.fpa_ready` (0x0610) = 1 at capture time?	☐	T: ____ °C
3	If the FPA was warm, abort to STANDBY (`PAYLOAD_SET_MODE(mode=1)`, func 26), re-enable the cooler (`FPA_COOLER(on=1)`, func **43**) and wait for `fpa_ready`	ready = 1	☐	
4	Check memory: `payload.mem_segments_bad` (0x0612); mark a bad segment with `PAYLOAD_MARK_BAD_SEGMENT(segment_id=N)` (S8.1 func **31**)	segments: ____	☐	
5	Delete and re-capture: `PAYLOAD_DELETE_IMAGE(scene_id=N)` (func **30**), then `PAYLOAD_CAPTURE(scene_id=N, lines=L)` (func **28**)	new scene: _; *S5 0x0613 IMAGECAPTURED*	☐	
6	If recurring, investigate FPA temperature or pointing	finding: _____	☐	
7	File anomaly ticket	ticket #: ____	☐	

CTG-017 Bus Failure Switchover

Switching the **bus** and switching the **OBC unit** are different commands. `OBC_SELECT_BUS` (func 54) changes the CAN bus; `OBC_SWITCH_UNIT` (func 53) fails over to the redundant OBC and is CTG-011, not this card.

#	Action	Verification	☐	Note
1	Confirm the bus fault: `HK_REQUEST(sid=4)` (S3.27)	`obdh.bus_a_status` (0x030F) = 1 (ERROR); `obdh.active_bus` (0x030E) = 0 (A)	☐	
2	FD GO for bus switchover	granted	☐	

#	Action	Verification	☐	Note
3	Issue `OBC_SELECT_BUS(bus=1)` (S8.1 func_id **54**) to move to bus B (use `bus=0` to go back to A)	S1.1 accept + S1.7 complete	☐	
4	Verify bus B in use and all housekeeping resumes	`obdh.active_bus` = 1; `obdh.bus_b_status` (0x0310) = 0; SIDs 1–6 flowing again	☐	
5	File anomaly ticket. Note there is now **no bus redundancy left**	ticket #: ____	☐	

CTG-019 No Telemetry at Pass Start

#	Action	Verification	☐	Note
1	Verify the scheduled station and baseband config (five-station network)	OK	☐	
2	Verify orbit / pass prediction (could be geometry)	prediction OK	☐	
3	Blind-command the TX chain: `HK_REQUEST(sid=1)` (S3.27), `EPS_POWER_ON(line_index=2)` (func **19**), `TTC_PA_ON` (func **66**)	`eps.pl_ttc_tx` (0x0112) = 1; `ttc.pa_on` (0x0516) = 1 once TM returns	☐	
4	Blind transponder swap: `TTC_SWITCH_REDUNDANT` (func **64**), then `TTC_SWITCH_PRIMARY` (func **63**)	`ttc.link_status` (0x0501) = **2** (LOCKED)	☐	
5	If the antenna was never deployed, `TTC_DEPLOY_ANTENNA` (func **69**) — irreversible burn-wire	`ttc.antenna_deployed` (0x0520) = 1	☐	
6	Remember the UHF beacon is a **separate radio**: if HK SID 11 is still arriving while the S-band is silent, the fault is in the S-band chain, not the spacecraft	`ttc.beacon_active` (0x0527): ____	☐	
7	If link established, run NOM-001; if no link after two consecutive passes, escalate **EMG-004** (Loss of Communication)	path: ____	☐	
8	File anomaly ticket	ticket #: ____	☐	

CTG-020 Ground Station Antenna Failure

CTG-020 is an **advisory** card (`advisory: true` in `procedure_index.yaml`) — it is not machine-runnable, and it contains **no spacecraft telecommands**. This is a ground-segment failure: everything is done on the ground side. Do not command the spacecraft for it.

#	Action	Verification	☐	Note
1		confirmed	☐	

#	Action	Verification	☐	Note
	Confirm the failure is ground-side, not spacecraft-side (`ttc.gs_equipment_status` 0x0524 bitmask; a healthy spacecraft still radiates)			
2	Attempt local station recovery with the station operator	result: _____	☐	
3	Notify FD and Mission Manager; assess the impact on the shift plan	logged	☐	
4	Re-plan the remaining activities onto the next station in the five-station network; verify its baseband and tracking	station: _____	☐	
5	File anomaly ticket	ticket #: ____	☐	

Appendix WB-B — End-of-Week Shift Log (cohort copy)

Use: completed by the FD at end of every shift. One sheet per shift. Goes into cohort archive.

Field	Entry
Shift date	
Shift FD	
Team	☐ Borealis ☐ Austral
Pass(es) covered	
Procedures executed	
Anomalies (count + ticket IDs)	
FDIR rules triggered	
GO/NO-GO outcome	
Customer commitments met / deferred	
One-line summary for cohort archive	
FD signature	

Appendix WB-C — Workbook self-check

At end of Week 3, the workbook should contain:

- ☐ WB-5.0 orientation check (filled before Monday)
- ☐ WB-5.1 nine position score cards (one per position)
- ☐ WB-5.3-PP pass-plan exercise output
- ☐ WB-5.4-VL voice-loop drill responses
- ☐ WB-5.4-N001 NOM-001 confirm-each-step (one per pass)
- ☐ WB-5.4-C003 CTG-003 confirm-each-step (if invoked)
- ☐ WB-5.4-N012 NOM-012 handover (one per handover)
- ☐ WB-5.4-FD flight-dynamics workshop output
- ☐ WB-5.4-Q quiz response sheet (graded)
- ☐ WB-5.5-PB LEOP pre-pass briefing checklist (one per pass)
- ☐ WB-5.5-L001 / -L006 / -L002 / -L007 / -L005 confirm-each-step (one each per pass operated)
- ☐ WB-5.5-AT anomaly tickets (≥ 1 per pass operated)
- ☐ WB-5.5-GO end-of-pass scorecard
- ☐ WB-5.5-OBS observer worksheet (one per pass observed)
- ☐ WB-5.5-HO inter-pass handover (one per position)
- ☐ WB-5.5-DB debrief reflection
- ☐ WB-5.6-PB pre-pass-set briefing
- ☐ WB-5.6-C001 through WB-5.6-C012 confirm-each-step (one each)
- ☐ WB-5.6-FPA cool-down monitor sheet
- ☐ WB-5.6-FL First Light assessment
- ☐ WB-5.6-MA multi-anomaly form (if applicable)
- ☐ WB-5.6-GO pass-set scorecard
- ☐ WB-5.6-OBS observer worksheet
- ☐ WB-5.6-HO inter-pass handover
- ☐ WB-5.6-DB debrief reflection
- ☐ WB-5.7-SA personal self-assessment
- ☐ WB-5.7-LL ≥ 3 cohort lessons-learned contributions
- ☐ WB-5.7-WI what-if reflection (if a branch replay ran)
- ☐ WB-5.7-FR final mission review prep (cohort copy)
- ☐ WB-5.7-CE course evaluation (anonymous; in the box)
- ☐ WB-A contingency confirm-each-step pages used as needed
- ☐ WB-B shift log per shift

FD signature at end of Week 3: _____

Workbook handed to instructor: _ : _ (UTC)